

# Matrix Compositions: Exact Spectra, Hankel Products, and Uniform Asymptotics

OEIS A261781, its empty-row transition,  
and refined asymptotics for A261784

Part II: finite spectra, uniform asymptotics, and two Poisson limits

Part III: Cauchy limits, arithmetic atoms, and edge laws for the Hankel zeros

Part IV: packed matrices on the total-weight  $2N$  diagonal

Part I: OpenAI ChatGPT

Research prepared for Vladimir Reshetnikov

Part II: a research report prepared for Vladimir Reshetnikov

Part III: OpenAI ChatGPT

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## Abstract

Let  $T(n, k)$  count nonnegative integer matrices of total sum  $n$ , with  $k$  ordered rows, an arbitrary number of ordered columns, and no zero row or column. These are the entries of OEIS A261781. We establish a collection of exact and asymptotic results for this family. First, its minimal eventual constant-coefficient recurrence has order  $k(k+1)/2$ , and its full-rank shifted Hankel determinants admit an explicit positive product formula, including a polynomial refinement marking the number of columns. Second, we give an exponential relative error bound for the unrestricted count that is uniform in the number of rows. This yields a Poisson empty-row law at  $n = k(\log k + s)$ , an explicit first correction with a summable error of order  $(\log k)^2/k^2$ , and the sharp total-variation rate with its leading constant. We also prove joint Gaussian–Poisson independence for column count and empty-row count in this window. Third, for  $n/k$  in compact subsets of  $(1, \infty)$ , a positive Stirling-number transform gives a uniform saddle expansion to every fixed order. For A261784, this identifies the exact amplitude in the recorded leading equivalent and gives the first relative correction. Finally, fixing the number of columns produces a different explicit coverage threshold.

Attribution matters here: the generating functions, a recurrence of the stated order, and fixed-row asymptotics were already published by Munarini, Poneti, and Rinaldi. Thus the two conjecture labels still present in A261781 lead in part to a reconciliation with prior literature. The main extensions of this article are the factored determinant and the uniform distributional and asymptotic results. Full proofs, reproducible computations, a bounded source audit, and further research questions are included. These are unrefereed mathematical proofs; no proof-assistant verification or historical-priority guarantee is claimed.

**Editorial note (3 October 2026, batch 85B).** This report is built from two manuscripts of ProveIt’s incoming-reports intake, written independently of each other on the same subject: manuscript 06 of batch 85 (Part I, the base, whose abstract stands above) and manuscript 01 of batch 85 (Part II). It is AI-assisted, unrefereed and not formalized. Where the two manuscripts prove the same statement, it is printed once, in Part I, in manuscript 06’s form, with a bracketed note marked “[Added 3 October 2026, batch 85B]” that names manuscript 01’s version and its route; Part II (Section 11 onwards) prints everything that manuscript 01 adds. *All of Part II assumes the column weight  $u = 1$* : manuscript 01 never weights columns. The two manuscripts permute their Greek letters; everything here is printed in manuscript 06’s letters, and the dictionary of Section 1.5 discloses every rename. Manuscript 01 presented its recurrence theorem as a proof of the OEIS order conjecture; Munarini, Poneti and Rinaldi had already proved a recurrence of that order with the same denominator [4, Proposition 25 and Remark 26, p. 16], so only its minimality is new (Section 11.2). Part I’s labels carry the prefix `mx`:, Part II’s the prefix `mx:tp`;; Appendix A records the provenance and every editorial change.

**Editorial note (5 October 2026, batch 98).** Since batch 98 the report has three Parts. Part III (Section 23 onwards) is built from manuscript 08 of batch 98, which answers Part I’s Research question 7, on the limiting distribution of the zeros of the Hankel determinant of Theorem 3.1: after the factors at  $u = 0$  and  $u = -1$  are removed, the zeros converge to the image on  $\operatorname{Re} u = -\frac{1}{2}$  of a Cauchy law of scale  $\frac{1}{2}$ , with explicit cyclotomic multiplicities, the sharp Kolmogorov constant  $9/\pi^2$ , an arithmetic edge law, the transition of the absolute moments and the limit under the full-degree normalization. Its re-derivation of Theorem 3.1 is not reprinted. Part III’s labels carry the prefix `mx:hz`:, and it renames manuscript 08’s letters that clash with Parts I and II (Table 10). Dated notes in Sections 1, 1.5, 3 and 10 and in Appendix A record what has changed. Part III is AI-assisted, unrefereed and not formalized.

**Editorial note (6 October 2026, batch 108).** Since batch 108 the report has four Parts. Part IV (Section 34 onwards) is built from Report 169 of a session bundle, written without sight of this report, on the diagonal  $a_N = T(2N, N)$  (A261784). It proves again, by a different route, the expansion of  $a_N$  to every order (Corollaries 7.4 and 17.1), the column mean and variance at  $n/k = 2$  (Corollary 7.5) and the first step of the inversion with its integer brackets (Section 18); these are printed as marked second routes. It adds two further terms of the inversion, an expansion uniform for complex markers of the columns and of the cells with entry at least 2, and the Poisson limit, of mean  $Lr$ , of the number of such cells, independent of the Gaussian column count. It answers no question of this report; it bears on one question of Part II and answers, on one ray, an analogue of another (Section 34). Part IV’s labels carry the prefix `mx:pm`:, and it renames Report 169’s letters into those of Parts I and II (Table 17). Dated notes in Sections 1, 1.5, 7, 10, 17, 18 and 20 and in Appendix A record what has changed. Part IV is AI-assisted, unrefereed and not formalized.

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## Part I

# Exact spectra, Hankel products, and uniform asymptotics

## 1 The problem, the sources, and the contribution

### 1.1 Why this family is worth studying

One generating function can behave quite differently when a parameter is fixed or grows with the coefficient index. Matrix compositions provide an especially explicit setting in which to study this difference. For fixed  $k$ , the sequence is a finite sum of exponential terms and has an exact linear recurrence. At  $n \asymp k$ , requiring every row to be used is a rare event, and a saddle point governs its count. At  $n = k(\log k + O(1))$ , the same requirement has a nondegenerate limiting probability. These regimes cannot be interchanged by simply substituting a growing  $k$  into a fixed- $k$  asymptotic.

The current A261781 entry attributes to Václav Kotěšovec two conjectures dated October 14, 2017: a column recurrence order  $k(k+1)/2$ , and the limit  $d(k+1) - d(k) \rightarrow 1/\log 2$  for its exponential growth constants [2]. Its companion A261780 counts the same matrices while allowing zero rows [1]. The diagonal  $T(2N, N)$  is A261784 [3].

### 1.2 What was already known

The primary enumerative source is *Matrix Compositions* by Munarini, Poneti, and Rinaldi [4], preceded by a 2006 conference paper [5]. In the 2009 article, Proposition 1 gives the generating function, Propositions 12–13 give a root-of-unity formula and fixed-row asymptotics, Propositions 22–24 give the no-zero-row inclusion–exclusion formula, Proposition 25 gives a recurrence of order  $\binom{k+1}{2}$ , and Proposition 29 gives the positive Stirling/Fubini transform used below. Our elementary derivations of these identities make the article self-contained; the identities themselves are attributed to those authors.

Louchard [6] studies fixed-row probabilistic asymptotics, including the number of columns and column-size statistics. An empty *column size* in that work is distinct from a zero *row* here. The A261784 entry already records a leading equivalent with numerical amplitude and an explicit exponential constant, credited to Kotěšovec in 2017 and updated in 2024 [3].

Accordingly, this article does not claim the discovery of the fixed-row growth formula, recurrence existence, or the leading A261784 exponential rate. The first OEIS statement receives an explicit minimality proof and a careful indexing convention; the second follows from an already known closed form. The stronger developments concern exact determinants and uniform estimates as the number of rows grows.

### 1.3 Relation to ProveIt and the audit boundary

The target was selected after inspecting the OEIS research in ProveIt [7]. Searches for A261780, A261781, and A261784 found no indexed matching report. The relevant repository hit for matrix compositions was a citation in the report on A260700 and parabolic double cosets, which studies a different enumeration. The repository search exposed indexed commit `ce37e13f4aa16819c87c3ccc611362d15758f2ac`; the search result is not an exhaustive statement about every branch or historical archive. No unformalized repository theorem is assumed in our proofs.

| Topic                 | Existing foundation                          | Result established here                                             |
|-----------------------|----------------------------------------------|---------------------------------------------------------------------|
| Recurrences           | Order- $\binom{k+1}{2}$ recurrence           | Minimal eventual order; complete spectrum; initial impulse          |
| Hankel determinants   | Size- $k$ identity for unrestricted matrices | Factored size- $\binom{k+1}{2}$ determinant with column weights     |
| Fixed-row asymptotics | Dominant pole and amplitude                  | Explicit error uniform in all $k \geq 1$                            |
| Empty rows            | Exact inclusion–exclusion                    | Poisson window, first correction, sharp total variation             |
| Column count          | Fixed-row Gaussian law                       | Joint Gaussian–Poisson limit; conditional limits                    |
| Diagonal A261784      | Recorded leading equivalent                  | Exact amplitude, first correction, uniform all-orders ray expansion |
| Fixed column count    | Stars-and-bars enumeration                   | Explicit shifted coverage threshold when columns grow slowly        |

Table 1: Attribution and the scope of the extensions. “Established here” describes the proofs in this article, not a claim of priority over all literature.

The source audit compared current OEIS records with the primary 2006, 2008, and 2009 works, and checked targeted searches for growing-row, empty-row, and coupon-collection results. It found no matching statement of the specific uniform expansions below. This is a bounded audit, so the responsible claim is a set of fully specified results and proofs, rather than a declaration of a historical breakthrough.

[Added 3 October 2026, batch 85B: the search statement describes the pinned index `ce37e13f4`. At that commit manuscript 01, the source of Part II, had not yet arrived; it now forms Part II of this report, which is filed at `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a261781-matrix-compositions`. Three repository neighbours bear on the method, and none is a dependency. (i) The report on A260700 [10] uses the same two devices as Section 7, a uniform extraction of the Fubini pole at  $L$  (its Section `pd:sec:poles`) followed by an outer Stirling transform (its Section `pd:sec:outer`), for a different sequence; it cites [4] as related work only. (ii) The canonical transseries volume proves the exact pole-lattice formula for the Fubini numbers, its Theorem `q2:thm:fubini` [11], of which the estimate (7.5) is the one-pole truncation. (iii) The only ingredient of this report that is formalized anywhere in the repository is the Fubini exponential generating function used in Section 7: the Lean declarations `Fabius.fubini` (the definition  $F_q = \sum_b b! \left\{ \begin{smallmatrix} q \\ b \end{smallmatrix} \right\}$ ), `Fabius.egfA_fubini` and `Fabius.two_sub_exp_mul_egfA_fubini`, the identity  $(2 - e^t) \sum_q F_q t^q / q! = 1$ , in `Analysis/FabiusFunction/Lean/FabiusFunction/OrderedBell.lean` [12]. Nothing else stated in either Part is formalized, and placement beside the repository’s formal projects confers no formal status.] [Added 5 October 2026, batch 98: nor is anything stated in Part III.] [Added 6 October 2026, batch 108: nor anything stated in Part IV.]

## 1.4 Notation and conventions

Throughout,  $n$  is total matrix mass,  $k$  is the number of rows, and  $C$  is the number of columns. Unless stated otherwise,  $M$  denotes the number of zero rows in an unrestricted uniformly chosen matrix. Write

$$L = \log 2, \quad a_k = 2^{-1/k}, \quad r_k = 1 - a_k, \quad d_k = r_k^{-1}, \quad c_k = \frac{a_k}{2kr_k}.$$

The empty-row window uses  $h = n/k$  and  $\eta = ke^{-h}$ . The dense-ray regime uses  $\lambda = n/k$  and a saddle  $r > 0$ ; these are separate local notations. The falling factorial is  $(x)_j = x(x-1) \cdots (x-j+1)$ . All logarithms are natural. Constants in uniform big- $O$  notation may depend on the fixed compact sets specified in each theorem.

We include one empty matrix at  $n = 0$  for each fixed height when zero rows are allowed. Thus  $A(0, k) = 1$ ,  $A(n, 0) = 0$  for  $n > 0$ ,  $T(0, 0) = 1$ , and  $T(0, k) = 0$  for  $k > 0$ . These conventions matter for recurrences and Hankel rank at the initial index.

## 1.5 Notation dictionary for the two manuscripts

[Added 3 October 2026, batch 85B.] Manuscripts 06 (Part I) and 01 (Part II) use the same objects but different letters, and three Greek letters are cyclically permuted between them: manuscript 06's  $\eta$ ,  $\lambda$ ,  $\rho$  are manuscript 01's  $\lambda$ ,  $\rho$ ,  $\eta$ . Copying a formula from one manuscript into the other without translation therefore gives a formula that is wrong, not merely differently named. This report prints *everything* in manuscript 06's letters. Table 2 lists every symbol of manuscript 01 that was renamed for Part II and Part I's notes, with the tempting false reading where one exists; manuscript 01's other symbols are unchanged. Normalizations are unchanged throughout: every rename is a change of letter only. All of Part II is at column weight  $u = 1$ , so its  $A$ ,  $T$ ,  $c_k$ ,  $d_k$  are the unweighted ones of Part I. [Added 5 October 2026, batch 98: the renames of manuscript 08, the source of Part III, are in its own Table 10; this table is unchanged.] [Added 6 October 2026, batch 108: those of Report 169, the source of Part IV, are in Table 17.]

Table 2: Manuscript 01's symbols and the letters used for them in this report. “False reading” gives what the manuscript-01 letter means in this report.

| Notion                                          | This report<br>(ms. 06)             | Ms. 01                         | False reading if copied                                                                       |
|-------------------------------------------------|-------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------|
| Matrices allowing zero rows                     | $A(n, k)$                           | $U(n, k)$                      | —                                                                                             |
| Its generating function                         | $F_k(z; 1)$                         | $U_k(z)$                       | —                                                                                             |
| Zero rows of an unrestricted matrix             | $M$                                 | $Z_{n,k}$                      | —                                                                                             |
| Poisson mean $ke^{-n/k}$                        | $\eta$                              | $\lambda$                      | $\lambda$ is $n/k$ on dense rays                                                              |
| Ratio $n/k$ on compact rays                     | $\lambda$                           | $\rho$                         | $\rho$ is the pole-bound ratio                                                                |
| Pole-bound ratio $L/\sqrt{L^2 + 8}$             | $\rho$                              | $\eta$                         | $\eta$ is the Poisson mean                                                                    |
| Dense-ray saddle,<br>$r/(1 - e^{-r}) = \lambda$ | $r$                                 | $\tau$                         | $\tau$ is a formal variable in (7.16);<br>ms. 01's $r$ is $n - k$ or a moment index           |
| Half-saddle mass $Lr/2$                         | $v$                                 | $\mu$                          | $\mu_k(u)$ is an amplitude in<br>Theorem 4.1 and a column mean in (6.1)                       |
| Fixed- $k$ constants                            | $c_k, d_k; d(x)$ at<br>real $x$     | $c(k), d(k),$<br>$d(x)$        | —                                                                                             |
| Diagonal index                                  | $N,$<br>$a_N = T(2N, N)$            | $m, a(m)$                      | $m$ indexes $d_m$ and counts columns<br>in Section 8                                          |
| Diagonal constants                              | $c_*, d_*$                          | $c, d$                         | $c_k, d_k$ are the fixed- $k$ constants                                                       |
| Diagonal corrections                            | $\beta_1 = \beta, \beta_2, \beta_3$ | $\delta_1, \delta_2, \delta_3$ | ms. 01's “ $\beta_1$ ” is $\tilde{\beta}_1 = \beta_1 + \frac{1}{6}$ , a<br>different constant |
| Saddle correction coefficients                  | $E_j(\lambda)$                      | $A_m(\rho)$                    | $A(n, k)$ is a count                                                                          |
| Leading saddle approximation                    | $\mathcal{B}(n, k)$                 | $B(n, k)$                      | $B, B(u)$ are pole-bound constants                                                            |
| Truncation order of an<br>expansion             | $J$                                 | $M$                            | $M$ is the zero-row count                                                                     |

| Notion                                   | This report<br>(ms. 06)                                                                                                                  | Ms. 01                                                                                | False reading if copied                                               |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Coverage correction polynomials          | $Q_j(h, \eta; L)$                                                                                                                        | $Q_j(h, \lambda; L)$                                                                  | $Q_q(d)$ , $Q_{k,u}(z)$ , $Q$ are other objects                       |
| Coverage exponent                        | $\Psi(j, h, \epsilon)$ ,<br>$\epsilon = 1/k$                                                                                             | $G(r, h, \epsilon)$                                                                   | —                                                                     |
| Moment index; point mass                 | $j; m$                                                                                                                                   | $r; t$                                                                                | $r$ is the saddle                                                     |
| Reduced denominator; its degree          | $Q_{k,1}(z); D$                                                                                                                          | $D_k(z); N_k$                                                                         | $D$ is also $n - Q$ in Section 7.6                                    |
| Denominator coefficients                 | $q_i$                                                                                                                                    | $d_r$                                                                                 | $d_k$ is a growth constant                                            |
| Roots and roots of unity                 | $a_j = 2^{-1/j}$ , $z_{j,\ell}$ ,<br>$\omega_j$                                                                                          | $q_j$ , $r_{j,h}$ , $\zeta_j$                                                         | $q_i$ are denominator coefficients                                    |
| Fubini numbers                           | $F_q$                                                                                                                                    | $F_j$                                                                                 | $F_k(z; u)$ is a generating function                                  |
| Entire Stirling kernel; its coefficients | $W_n(w); \chi_{n,\ell}$                                                                                                                  | $W_n(v); w_{n,\ell}$                                                                  | $W_N$ is a diagonal diagnostic, $w_{j,\lambda}$ a Hankel weight       |
| Kernel variable; Euler operator          | $w = Lz; D_w$                                                                                                                            | $v; D_v$                                                                              | $v = Lr/2$                                                            |
| Kernel correction polynomials            | $P_r(w)$                                                                                                                                 | $p_r(v)$                                                                              | $P_j(t)$ in Section 3 is another polynomial                           |
| Kernel-proof variable $1/n$              | $\epsilon$                                                                                                                               | $h$                                                                                   | $h = n/k$                                                             |
| Kernel-proof auxiliaries                 | $\Sigma_j(n)$ , $\mathbf{a}_j$ , $\mathbf{e}_\ell$ , $\mathbf{r}_\ell$ ,<br>$\mathbf{b}_r$ , $\Pi_r(\ell)$ , $\mathbf{B}_a$ , $\delta_0$ | $S_j(n)$ , $A_j$ ,<br>$E_\ell$ , $R_\ell$ , $B_r$ ,<br>$P_r(\ell)$ , $B_a$ , $\delta$ | $E_j(\lambda)$ , $R_p(w)$ , $B$ , $\delta$ of Lemma 7.2               |
| Gaussian functional                      | $\mathcal{G}_b$                                                                                                                          | $\mathfrak{G}_b$                                                                      | —                                                                     |
| Cumulant recursion                       | $\mathcal{K}_j(z, a)$                                                                                                                    | $K_j(z, a)$                                                                           | $K$ is a compact set of rays                                          |
| Stirling index; Stirling defect          | $Q_{n,k}$ ;<br>$\Delta_{n,k} = n - Q_{n,k}$                                                                                              | $J_{n,k}$ ; $\Delta_{n,k}$                                                            | $J_q$ is a block count; $\Delta_{n,k}$ is Section 7.6's $D$           |
| Defect generating variable               | $\xi$                                                                                                                                    | $u$                                                                                   | $u$ is the column weight                                              |
| Smooth comparator; its root              | $\mathcal{A}^{[J]}(x); x_J(Y)$                                                                                                           | $A^{[M]}(x);$<br>$x_M(Y)$                                                             | —                                                                     |
| Inverse core                             | $\Lambda$ , $\varpi = W_0(\alpha\Lambda)$                                                                                                | $X$ , $w$                                                                             | $X$ is a Poisson or normalized variable, $w$ a PGF or kernel variable |
| Integer threshold; its bracket           | $\mathcal{N}(Y); \epsilon_J, C_J$                                                                                                        | $N(Y); \epsilon_M$ ,<br>$C_M$                                                         | $N$ is the diagonal index                                             |
| $\log 2$                                 | $L$                                                                                                                                      | $L$ , macro<br>$\backslash \text{Li}$                                                 | —                                                                     |

## 2 Exact enumeration and the complete spectrum

**Definition 2.1.** A matrix composition of mass  $n$  and height  $k$  is a  $k \times C$  matrix of nonnegative integers with positive column sums and total entry sum  $n$ . Its columns and rows are ordered. Let  $A(n, k)$  count these matrices, and let  $T(n, k)$  count those with every row nonzero.

The word interpretation in A261781 is equivalent: a column of sum  $i$  specifies the multiplicities of the  $k$  letters in an alphabetically ordered word of length  $i$ . Every letter occurs somewhere precisely when every row is nonzero. For example,  $T(2, 2) = 3$ : the one-column matrix with entries 1, 1, and the two  $2 \times 2$  permutation matrices.

### 2.1 Generating functions and a positive transform

Let  $u > 0$  mark columns. Define

$$A(n, k; u) = \sum_B u^{C(B)}, \quad T(n, k; u) = \sum_{B: M(B)=0} u^{C(B)}.$$

At  $u = 1$  these reduce to the unweighted counts. A nonzero column has generating function  $(1 - z)^{-k} - 1$ , since a column of sum  $i$  has  $\binom{i+k-1}{i}$  possibilities. Taking an ordered sequence of columns gives the classical identity

$$F_k(z; u) := \sum_{n \geq 0} A(n, k; u) z^n = \frac{1}{1 - u((1 - z)^{-k} - 1)} = \frac{(1 - z)^k}{(1 + u)(1 - z)^k - u}. \quad (2.1)$$

This also gives the positive recurrence

$$A(n, k; u) = u \sum_{i=1}^n \binom{i+k-1}{i} A(n-i, k; u) \quad (n \geq 1, k \geq 1). \quad (2.2)$$

Deleting any designated set of zero rows proves inclusion–exclusion:

$$T(n, k; u) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} A(n, j; u). \quad (2.3)$$

Equations (2.1)–(2.3) are established enumeration, not new formulas [4, 1, 2].

For later use let  $F_q = \sum_{b=0}^q b! \left\{ \begin{smallmatrix} q \\ b \end{smallmatrix} \right\}$  be the ordered Bell or Fubini number, with  $F_0 = 1$ . The two Stirling triangles satisfy

$$x(x+1) \cdots (x+n-1) = \sum_q \begin{bmatrix} n \\ q \end{bmatrix} x^q, \quad (e^z - 1)^k = k! \sum_{q \geq k} \left\{ \begin{smallmatrix} q \\ k \end{smallmatrix} \right\} \frac{z^q}{q!}.$$

Expanding (2.1) at  $u = 1$  in powers of  $(1 - z)^{-k}$  gives

$$A(n, k) = \frac{1}{2} \sum_{b \geq 0} 2^{-b} \binom{n+kb-1}{n} = \frac{1}{n!} \sum_{q=0}^n \begin{bmatrix} n \\ q \end{bmatrix} F_q k^q.$$

For  $n > 0$  the  $b = 0$  term is zero. The identity  $F_q = \frac{1}{2} \sum_{b \geq 0} b^q 2^{-b}$  follows by comparing the exponential generating functions, with the usual  $0^0 = 1$  convention. Applying the  $k$ th binomial difference to  $j^q$  yields

$$\boxed{T(n, k) = \frac{k!}{n!} \sum_{q=k}^n \begin{bmatrix} n \\ q \end{bmatrix} \left\{ \begin{smallmatrix} q \\ k \end{smallmatrix} \right\} F_q.} \quad (2.4)$$

This is Proposition 29 of [4]. Its nonnegative summands will allow controlled asymptotic replacement without alternating-sum cancellation. In particular  $T(n, n) = F_n$ .

## 2.2 Every pole survives

For fixed  $u > 0$  put

$$b = \frac{u}{1+u}, \quad L_u = \log(1 + 1/u), \quad a_{j,u} = b^{1/j}, \quad \omega_j = e^{2\pi i/j}.$$

The denominator in  $F_j(z; u)$  has simple zeros  $z_{j,\ell} = 1 - a_{j,u} \omega_j^\ell$ ,  $0 \leq \ell < j$ . Their residues are  $-a_{j,u} \omega_j^\ell / ((1+u)j)$ . Consequently, for  $n \geq 1$ ,

$$A(n, j; u) = \frac{a_{j,u}}{(1+u)j} \sum_{\ell=0}^{j-1} \omega_j^\ell (1 - a_{j,u} \omega_j^\ell)^{-n-1}. \quad (2.5)$$

The polynomial part of  $F_j$  is the constant  $1/(1+u)$ , so (2.5) is deliberately stated only for  $n \geq 1$ . Its unweighted specialization is the published Binet formula [4, Proposition 12].

**Theorem 2.2** (Minimal recurrence and spectrum). *Fix  $k \geq 1$  and  $u > 0$ , and put  $D = \binom{k+1}{2}$ . The reduced denominator of  $\sum_{n \geq 0} T(n, k; u) z^n$  is*

$$Q_{k,u}(z) = \prod_{j=1}^k ((1+u)(1-z)^j - u). \quad (2.6)$$

*Thus the minimal eventual homogeneous constant-coefficient recurrence has order exactly  $D$ , over  $\mathbb{C}$  as well as over  $\mathbb{R}$ . Its characteristic polynomial is*

$$P_{k,u}(x) = \prod_{j=1}^k ((1+u)(x-1)^j - ux^j). \quad (2.7)$$

*All  $D$  characteristic roots are simple and nonzero. At  $u = 1$  the polynomial is monic with integer coefficients.*

*Proof.* Roots belonging to different factors in (2.6) cannot coincide: their distances from 1 are  $b^{1/j}$ , strictly increasing with  $j$ . Within each factor the roots are simple. At a root of the  $j$ th factor, only the  $j$ th term in (2.3) has a pole, and its coefficient  $(-1)^{k-j} \binom{k}{j}$  and residue are nonzero. Therefore no pole cancels. This proves the exact reduced denominator and its degree.

If an eventual recurrence of smaller order existed, multiplying the generating function by its recurrence polynomial would give a polynomial; that recurrence polynomial would have to vanish at every one of these  $D$  poles. Its degree would be at least  $D$ , a contradiction. Reversing the factors in (2.6) gives (2.7); its leading coefficient is  $(1+u) - u = 1$  in each factor. No pole is zero, since  $b < 1$ , and every corresponding characteristic root is finite and nonzero.  $\square$

*Remark 2.3* (Both manuscripts; the published part). [Added 3 October 2026, batch 85B.] Manuscript 01 proves Theorem 2.2 at  $u = 1$  by the same argument: the roots of  $2(1-z)^j - 1$  satisfy  $|1-z| = 2^{-1/j}$ , which determines  $j$ , so the root sets are disjoint; at a root only the  $j$ th inclusion–exclusion summand is singular, with a nonzero coefficient; and an eventual recurrence of order  $D'$  would give a denominator of degree at most  $D'$ . It writes the recurrence as  $T(n, k) = -\sum_{i=1}^D q_i T(n-i, k)$  for  $n > D$ , with  $Q_{k,1}(z) = \sum_{i=0}^D q_i z^i$  normalized by  $q_0 = 1$ . The weighted statement (every  $u > 0$ ) and Remark 2.4 are manuscript 06's alone.

What is new in either manuscript is the *minimality* (no cancellation, so no shorter recurrence even over  $\mathbb{C}$ ) and the indexing of Remark 2.4. The existence of a recurrence of order  $\binom{k+1}{2}$  is Proposition 25 of [4] (printed page 16, with  $m$  rows there): its proof displays the generating function of the no-zero-row column as  $x^m F_m(x)$  over  $(1-2x)(1-4x+2x^2) \cdots (2(1-x)^m - 1)$  with  $\deg F_m \leq \binom{m}{2}$ , which is  $Q_{k,1}$  in our letters. Munarini, Poneti and Rinaldi do not claim minimality. Their Remark 26 gives the cases of two and three rows; for  $k = 2$  it is the example that manuscript 01 prints,

$$\sum_{n \geq 0} T(n, 1) z^n = \frac{z}{1-2z}, \quad \sum_{n \geq 0} T(n, 2) z^n = \frac{z^2(3-2z)}{1-6z+10z^2-4z^3}, \quad (2.8)$$

whence  $T(n, 2) = 6T(n-1, 2) - 10T(n-2, 2) + 4T(n-3, 2)$  for  $n \geq 4$ . The recurrence fails at  $n = 3$  with the zero initial terms: there  $T(3, 2) - 6T(2, 2) = 16 - 18 = -2 = (-1)^{k+D} 2^{k-1}$ , the impulse of Remark 2.4 at  $k = 2$ ,  $D = 3$ . Manuscript 01 presented its theorem as a proof of the OEIS order conjecture without citing Proposition 25; Section 11.2 records the correction.

*Remark 2.4* (The initial impulse). The rational function for  $T(n, k; u)$  tends to  $(-1)^k u / (1+u)$  as  $z \rightarrow \infty$ : the  $j = 0$  term of inclusion–exclusion is  $(-1)^k$ , and every other  $F_j$  tends to  $1/(1+u)$ . Hence its numerator has degree  $D$ , not less than  $D$ . The denominator recurrence begins at  $n = D + 1$  for

the full column indexed from zero; it is an order- $D$  recurrence on the tail  $T(1, k; u), T(2, k; u), \dots$ . At  $u = 1$ , writing  $Q_{k,1}(z) = \sum_{i=0}^D q_i z^i$ , one has

$$\sum_{i=0}^D q_i T(D-i, k) = (-1)^{k+D} 2^{k-1} \neq 0.$$

If a convention insists on an order- $r$  recurrence starting already at index  $r$  on the unshifted column, its minimum is  $D+1$ , obtained by adjoining a zero characteristic root. This explains why the word “eventual” is part of the theorem.

### 2.3 The growth-constant conjecture

The smallest-modulus pole of  $F_k(z; 1)$  is  $r_k = 1 - 2^{-1/k}$ , uniquely. It is also the smallest-modulus pole across all terms of (2.3), since  $r_j > r_k$  for  $j < k$ . Thus, for each fixed  $k$ ,

$$T(n, k) \sim c_k d_k^n, \quad d_k = \frac{1}{1 - 2^{-1/k}}, \quad c_k = \frac{1}{2k(2^{1/k} - 1)}. \quad (2.9)$$

The underlying pole and unrestricted asymptotic are classical [4, Proposition 13]; inclusion–exclusion transfers them to  $T$ . These constants agree with the numerical columns in A261781.

**Corollary 2.5.** *The second conjecture recorded in A261781 holds:*

$$d_{k+1} - d_k \longrightarrow \frac{1}{L}.$$

*More precisely,*

$$d_k = \frac{k}{L} + \frac{1}{2} + \frac{L}{12k} - \frac{L^3}{720k^3} + O(k^{-5}), \quad (2.10)$$

$$d_{k+1} - d_k = \frac{1}{L} - \frac{L}{12k(k+1)} + O(k^{-4}), \quad (2.11)$$

$$c_k = \frac{1}{2L} - \frac{1}{4k} + \frac{L}{24k^2} - \frac{L^3}{1440k^4} + O(k^{-6}). \quad (2.12)$$

*Proof.* Use the Taylor expansion at zero of  $1/(1 - e^{-x}) - 1/x$  with  $x = L/k$ . It is analytic in a neighborhood of zero; Taylor’s theorem supplies the stated remainders. Then subtract adjacent values and use  $c_k = (d_k - 1)/(2k)$ .  $\square$

[Added 3 October 2026, batch 85B: manuscript 01 proves (2.9) and the expansions (2.10) and (2.12) in the same way, by expanding  $x/(e^x - 1)$ , and states, like manuscript 06, that the limit is immediate from the closed form of [4, Proposition 13]. The increment expansion (2.11) is manuscript 06’s alone. Manuscript 01 adds that  $d_{k+1} - d_k$  *increases strictly* to  $1/L$  and stays below it (Theorem 12.1).]

## 3 A factored Hankel determinant and its polynomial refinement

Let  $T_u(n, k)$  be the polynomial that counts  $k$ -row matrix compositions of mass  $n$ , with no zero row or zero column, assigning weight  $u$  to each column. Thus  $T_1(n, k) = T(n, k)$  is the triangle A261781. For  $k \geq 1$  we set  $T_u(0, k) = 0$ . In the proofs we initially take  $u > 0$ ; the resulting determinant identity is a polynomial identity and hence extends to every complex  $u$ .

Put

$$a = 1 + u, \quad b = u, \quad d = \binom{k+1}{2}, \quad c = \binom{k+1}{3}.$$

For  $s \geq 1$  define the  $d$ -by- $d$  Hankel determinant

$$H_{k,s}(u) = \det[T_u(s+p+q, k)]_{p,q=0}^{d-1}.$$

**Theorem 3.1** (Maximal Hankel determinant). *For all integers  $k, s \geq 1$ , the following identity holds in  $\mathbb{Z}[u]$ :*

$$H_{k,s}(u) = u^{d+2c} (1+u)^{k(s-1)+2c} \prod_{j=1}^k \binom{k}{j}^j \prod_{1 \leq i < j \leq k} \left( (1+u)^{(j-i)/g_{ij}} - u^{(j-i)/g_{ij}} \right)^{2g_{ij}}, \quad (3.1)$$

$$g_{ij} = \gcd(i, j).$$

Here  $d + 2c = \sum_{j=1}^k j^2 = k(k+1)(2k+1)/6$ . In particular,  $H_{k,s}(u) > 0$  for every real  $u > 0$ , and

$$H_{k,s+1}(u) = (1+u)^k H_{k,s}(u).$$

The determinant is maximal for the positive-index tail: the infinite matrix  $[T_u(1+p+q, k)]_{p,q \geq 0}$  has rank exactly  $d$  for  $u > 0$ . Every minor of order greater than  $d$  in that infinite matrix vanishes. The initial term at  $n = 0$  requires separate treatment, given below.

### 3.1 Exact spectrum and residue weights

Use the column-weight notation  $T_u(n, k) = T(n, k; u)$  from Section 2. The generating function and inclusion–exclusion have already been proved in (2.1)–(2.3).

For  $j \geq 1$ , set

$$P_j(t) = a(t-1)^j - bt^j.$$

Since  $a - b = 1$ , this is a monic polynomial of degree  $j$ . Its roots can be written

$$\lambda = \frac{1}{1-r}, \quad r^j = \frac{b}{a}.$$

For  $u > 0$  all such roots are finite and nonzero. They are simple, and the root sets for distinct  $j$  are disjoint: a common root would give an  $r$  with two distinct prescribed moduli  $(b/a)^{1/j}$ . There are therefore exactly  $d$  roots in the union of the root sets.

Partial fractions yield, for  $n \geq 1$ ,

$$T_u(n, k) = \sum_{j=1}^k \sum_{P_j(\lambda)=0} w_{j,\lambda} \lambda^n, \quad w_{j,\lambda} = (-1)^{k-j} \binom{k}{j} \frac{\lambda - 1}{aj}. \quad (3.2)$$

All the weights are nonzero. For completeness, at a pole  $z_0 = 1/\lambda$ , with  $r = 1 - z_0$ , the numerator of the rational generating function is  $b/a$  and the derivative of its denominator is  $-ajr^{j-1}$ . Its coefficient of  $(1 - \lambda z)^{-1}$  is thus

$$\frac{-\lambda(b/a)}{-ajr^{j-1}} = \frac{\lambda r}{aj} = \frac{\lambda - 1}{aj}.$$

The polynomial part of the unrestricted generating function is the constant  $1/a$ , so it does not affect any  $n \geq 1$  coefficient.

### 3.2 Root products, discriminants, and resultants

The following identities are enough to evaluate the determinant:

$$\prod_{P_j(\lambda)=0} \lambda = a, \quad (3.3)$$

$$\prod_{P_j(\lambda)=0} (\lambda - 1) = (-1)^{j+1} b, \quad (3.4)$$

$$\text{Disc}(P_j) = (-1)^{(j-1)(j-2)/2} j^j (ab)^{j-1}, \quad (3.5)$$

$$\text{Res}(P_i, P_j) = (-1)^i (ab)^i \left( a^{(j-i)/g} - b^{(j-i)/g} \right)^g, \quad 1 \leq i < j, \quad g = \gcd(i, j). \quad (3.6)$$

The first two identities follow from the constant coefficient of  $P_j$  and from  $P_j(1) = -b$ . At a root  $\lambda$ ,

$$P'_j(\lambda) = j \left( a(\lambda - 1)^{j-1} - b\lambda^{j-1} \right) = \frac{jb\lambda^{j-1}}{\lambda - 1}.$$

Taking products and using the root products gives

$$\prod_{P_j(\lambda)=0} P'_j(\lambda) = (-1)^{j+1} j^j (ab)^{j-1}.$$

For a monic degree- $j$  polynomial, multiplication by  $(-1)^{j(j-1)/2}$  converts this product into its discriminant. Since

$$\frac{j(j-1)}{2} + j + 1 \equiv \frac{(j-1)(j-2)}{2} \pmod{2},$$

this proves (3.5).

To prove the resultant formula, put  $h = j - i$  and  $g = \gcd(i, h)$ . For a root of  $P_i$  write  $r = 1 - 1/\lambda$ , so that  $r^i = b/a$ . Then

$$P_j(\lambda) = \lambda^j (ar^j - b) = b\lambda^j (r^h - 1).$$

As  $r$  ranges through the  $i$  roots of  $r^i = b/a$ , its  $h$ th powers are the  $i/g$  roots of

$$v^{i/g} = (b/a)^{h/g},$$

each occurring with multiplicity  $g$ . Consequently

$$\begin{aligned} \prod_{r^i=b/a} (r^h - 1) &= (-1)^i \left( 1 - (b/a)^{h/g} \right)^g \\ &= (-1)^i a^{-h} \left( a^{h/g} - b^{h/g} \right)^g. \end{aligned}$$

Because  $P_i$  is monic, its resultant with  $P_j$  is the product of  $P_j(\lambda)$  over its roots. Using  $\prod \lambda = a$  gives

$$\begin{aligned} \text{Res}(P_i, P_j) &= b^i a^j (-1)^i a^{-h} \left( a^{h/g} - b^{h/g} \right)^g \\ &= (-1)^i (ab)^i \left( a^{h/g} - b^{h/g} \right)^g, \end{aligned}$$

as asserted.

### 3.3 Vandermonde evaluation

Enumerate all  $d$  roots in (3.2) as  $\lambda_1, \dots, \lambda_d$ , keeping the  $j$ -blocks together, and write  $w_\nu$  for the corresponding weights. The Hankel matrix factors as

$$[T_u(s + p + q, k)]_{p,q=0}^{d-1} = V \operatorname{diag}(w_\nu \lambda_\nu^s)_{\nu=1}^d V^\top, \quad V_{p,\nu} = \lambda_\nu^p.$$

Hence

$$H_{k,s}(u) = \left( \prod_{\nu=1}^d w_\nu \lambda_\nu^s \right) \prod_{\mu < \nu} (\lambda_\nu - \lambda_\mu)^2. \quad (3.7)$$

Splitting the squared Vandermonde product into within-block and between-block factors gives

$$\prod_{\mu < \nu} (\lambda_\nu - \lambda_\mu)^2 = \prod_{j=1}^k \operatorname{Disc}(P_j) \prod_{1 \leq i < j \leq k} \operatorname{Res}(P_i, P_j)^2.$$

For one block, the weights, the root power, and the discriminant combine to

$$\begin{aligned} & \left( \prod_{P_j(\lambda)=0} w_{j,\lambda} \lambda^s \right) \operatorname{Disc}(P_j) \\ &= (-1)^{j(k-j)+\binom{j}{2}} \binom{k}{j}^j b^j a^{s-1}. \end{aligned}$$

There is no residual sign after multiplying over  $j$ , because

$$\sum_{j=1}^k \left( j(k-j) + \binom{j}{2} \right) = 2 \binom{k+1}{3} = 2c.$$

The squared resultants supply

$$(ab)^{2 \sum_{1 \leq i < j \leq k} i} \prod_{i < j} \left( a^{(j-i)/g_{ij}} - b^{(j-i)/g_{ij}} \right)^{2g_{ij}} = (ab)^{2c} \prod_{i < j} \left( a^{(j-i)/g_{ij}} - b^{(j-i)/g_{ij}} \right)^{2g_{ij}}.$$

Substitution in (3.7) proves (3.1) for  $u > 0$ . Both sides are polynomials in  $u$ , so agreement on this interval proves the identity in  $\mathbb{Z}[u]$ .

### 3.4 The unweighted specialization

At  $u = 1$  the formula becomes

$$\det[T(s + p + q, k)]_{p,q=0}^{d-1} = 2^{k(s-1)+2\binom{k+1}{3}} \prod_{j=1}^k \binom{k}{j}^j \prod_{1 \leq i < j \leq k} \left( 2^{(j-i)/\gcd(i,j)} - 1 \right)^{2\gcd(i,j)}. \quad (3.8)$$

The first few values of  $H_{k,1}(1)$  are

$$H_{1,1}(1) = 1, \quad H_{2,1}(1) = 8, \quad H_{3,1}(1) = 62208.$$

For example, the complete  $k = 3$  polynomial is

$$H_{3,s}(u) = 27u^{14}(1+u)^{3s+5}(1+2u)^2.$$

This extends the Cassini-type phenomenon already observed for unrestricted matrix compositions. In [4, Proposition 21], the unrestricted height- $k$  sequence has a  $k$ -by- $k$  determinant  $(-1)^{\lfloor k/2 \rfloor} 2^{s-1}$ . Enforcing all rows to be used mixes the root sets for heights  $1, \dots, k$ . The determinant above has order  $d = k(k+1)/2$  and includes the explicit pairwise resultant factors caused by that mixture. The primary paper does not state this order- $d$  determinant; the proof here is independent of any priority claim.

### 3.5 Recurrence minimality and the initial-term correction

The nonzero determinant is an independent proof of the exact tail rank  $d$  and the minimality in Theorem 2.2. It also makes the initial impulse of Remark 2.4 quantitative.

At  $n = 0$  the spectral sum has the value

$$\sum_{\nu=1}^d w_{\nu} = (-1)^{k+1} \frac{u}{1+u},$$

whereas  $T_u(0, k) = 0$ . Thus the all-index identity is

$$T_u(n, k) = \sum_{\nu=1}^d w_{\nu} \lambda_{\nu}^n + (-1)^k \frac{u}{1+u} \mathbf{1}_{\{n=0\}}. \quad (3.9)$$

In particular, the minimal forward-shift annihilator valid from  $n = 0$  is  $tP_{k,u}(t)$ ; the extra root is zero. In the conventional eventual-recurrence language the order is still  $d$ . This distinction also accounts for the rank of the unshifted infinite Hankel matrix when  $u > 0$ :

$$\text{rank}[T_u(p+q, k)]_{p,q \geq 0} = d + 1.$$

Indeed, adjoining the zero root to the Vandermonde factorization gives the exact exceptional determinant

$$\det[T_u(p+q, k)]_{p,q=0}^d = (-1)^k \frac{u}{1+u} H_{k,2}(u). \quad (3.10)$$

The extra squared Vandermonde factors are  $\prod_{\nu} \lambda_{\nu}^2$ , which change  $H_{k,0}$  for the pure spectral extension into  $H_{k,2}$ . At  $u = 1$ ,

$$\det[T(p+q, k)]_{p,q=0}^d = (-1)^k 2^{k-1} H_{k,1}(1).$$

### 3.6 Coefficient positivity and the location of the zeros

Every nonconstant difference factor on the right of (3.1) is

$$(1+u)^m - u^m = \sum_{r=0}^{m-1} \binom{m}{r} u^r,$$

which has positive coefficients. Therefore  $H_{k,s}(u) \in \mathbb{Z}_{\geq 0}[u]$ . This statement concerns these maximal determinants. It does not assert total positivity of the Hankel matrix: for  $k = 2$ , for example,

$$\det \begin{pmatrix} T_u(1, 2) & T_u(2, 2) \\ T_u(2, 2) & T_u(3, 2) \end{pmatrix} = -u^2(1+2u)^2 < 0 \quad (u > 0).$$

The factorization also determines the geometry of all complex zeros. Apart from  $u = 0$  and  $u = -1$ , every zero lies on the vertical line  $\text{Re } u = -1/2$ . Indeed, a zero of  $(1+u)^m - u^m$  has the form

$$u = \frac{1}{\zeta - 1}, \quad \zeta^m = 1, \quad \zeta \neq 1,$$

and

$$\frac{1}{e^{i\theta} - 1} = -\frac{1}{2} - \frac{i}{2} \cot \frac{\theta}{2}.$$

The multiplicity at  $u = 0$  is exactly  $\sum_{j=1}^k j^2$ . When its displayed exponent is positive, the multiplicity at  $u = -1$  is exactly  $k(s-1) + 2\binom{k+1}{3}$ . All other multiplicities follow by collecting

the cyclotomic difference factors in the product. [Added 5 October 2026, batch 98: Part III collects them. Each zero coming from a primitive  $m$ -th root of unity,  $2 \leq m \leq k-1$ , has the multiplicity  $E_m(k) = 2 \sum_{i < j, m|q_{ij}} g_{ij}$ , with  $q_{ij} = (j-i)/g_{ij}$ ; every such order occurs ((25.1)–(25.3)), and  $E_m(k) = k^2 \log k / (\zeta(2)\psi(m)) + O(k^2)$  uniformly in  $m$  (Proposition 25.2).]

## 4 An exponential error bound uniform in the row count

The essential analytic input for the growing-row results is stronger than a fixed-parameter pole asymptotic. The bound below does not deteriorate when arbitrarily many poles are present.

**Theorem 4.1** (Uniform dominant-pole estimate). *For every  $n, k \geq 1$ ,*

$$A(n, k) = c_k d_k^n (1 + \varepsilon_{n,k}), \quad |\varepsilon_{n,k}| \leq B \rho^{n-1}, \quad (4.1)$$

where

$$B = \frac{\pi^2 L^2}{24}, \quad \rho = \left(1 + \frac{8}{L^2}\right)^{-1/2} < 0.239. \quad (4.2)$$

More generally, with  $b = u/(1+u)$ ,  $L_u = \log(1+1/u)$ ,  $a_{k,u} = e^{-L_u/k}$ ,  $r_{k,u} = 1 - a_{k,u}$ , and

$$d_k(u) = r_{k,u}^{-1}, \quad \mu_k(u) = \frac{a_{k,u}}{(1+u)kr_{k,u}},$$

one has

$$A(n, k; u) = \mu_k(u) d_k(u)^n (1 + \varepsilon_{n,k}(u)), \quad |\varepsilon_{n,k}(u)| \leq B(u) \rho(u)^{n-1}, \quad (4.3)$$

where

$$B(u) = \frac{\pi^2 L_u^2}{48b}, \quad \rho(u) = \left(1 + \frac{16b}{L_u^2}\right)^{-1/2}.$$

For  $u$  in a fixed compact subset of  $(0, \infty)$  the constants can be replaced by common  $B_I < \infty$  and  $\rho_I < 1$ .

*Proof.* In (2.5), divide by the positive  $\ell = 0$  term. Its coefficient is exactly  $\mu_k(u) d_k(u)^n$ . If  $j = \min(\ell, k - \ell)$ , then  $\sin(\pi\ell/k) \geq 2j/k$ ,  $a_{k,u} \geq b$ , and  $r_{k,u} \leq L_u/k$ . Therefore

$$\begin{aligned} |1 - a_{k,u} \omega_k^\ell|^2 &= r_{k,u}^2 + 4a_{k,u} \sin^2(\pi\ell/k) \\ &\geq r_{k,u}^2 \left(1 + \frac{16b}{L_u^2} j^2\right). \end{aligned}$$

Put  $D_u = 16b/L_u^2$ . There are at most two roots for each  $j$ , so

$$\begin{aligned} |\varepsilon_{n,k}(u)| &\leq 2 \sum_{j \geq 1} (1 + D_u j^2)^{-(n+1)/2} \\ &\leq 2(1 + D_u)^{-(n-1)/2} \sum_{j \geq 1} \frac{1}{D_u j^2} = \frac{\pi^2}{3D_u} \rho(u)^{n-1}. \end{aligned}$$

For  $k = 1$  the omitted sum is empty and the error is exactly zero. On a compact interval use its minimum  $b$  and maximum  $L_u$  in the same inequalities. At  $u = 1$ ,  $b = 1/2$ , giving (4.2).  $\square$

*Remark 4.2* (A second route with a  $k$ -dependent constant). [Added 3 October 2026, batch 85B.] At  $u = 1$  manuscript 01 proves the same one-pole dominance with a different constant:

$$|\varepsilon_{n,k}| \leq (k-1)\rho^{n+1} \quad (n, k \geq 1), \quad (4.4)$$

with the same  $\rho = L/\sqrt{L^2 + 8}$  (manuscript 01 calls it  $\eta$ ; see Table 2). Its proof bounds each of the  $k-1$  nonprincipal terms of (2.5) separately: with  $a_k \geq \frac{1}{2}$ ,  $r_k \leq L/k$  and  $\sin(\pi/k) \geq 2/k$  one gets  $r_k/|1 - a_k\omega_k^\ell| \leq \rho$  for every  $\ell \neq 0$ , and each ratio is at most  $\rho^{n+1}$ . Since  $\rho^2 = L^2/(L^2 + 8) < 0.057$  and  $B = \pi^2 L^2/24 > 0.197$ , the bound (4.4) is the sharper one for  $k \leq 4$  and Theorem 4.1 for  $k \geq 5$ ; only Theorem 4.1 is uniform in  $k$ , which the growing-row results need. Manuscript 01 says so itself: its relative bound is useful only when  $k\rho^n$  is small, and is not claimed to be small for large  $k$  and fixed  $n$ ; both of its growing regimes satisfy the condition by a wide margin.]

*Remark 4.3.* The theorem controls the unrestricted count  $A(n, k)$  uniformly. It does not say that  $T(n, k) \sim c_k d_k^n$  uniformly: the subtraction enforcing all rows nonzero becomes significant when  $n \approx k \log k$  and exponentially significant when  $n \asymp k$ . The next sections quantify these effects.

## 5 The empty-row transition and its first correction

Put  $L = \log 2$  and suppose that  $A(n, k)$  counts unrestricted objects on  $k$  labelled rows, with

$$\sum_{n \geq 0} A(n, k) z^n = \frac{(1-z)^k}{2(1-z)^k - 1}.$$

For  $m \geq 1$  write

$$d_m = (1 - e^{-L/m})^{-1}, \quad c_m = \frac{d_m - 1}{2m}.$$

Theorem 4.1 supplies the estimate

$$A(n, m) = c_m d_m^n (1 + \varepsilon_{n,m}), \quad |\varepsilon_{n,m}| \leq C_0 \rho^{n-1}, \quad 0 < \rho < 1, \quad (5.1)$$

uniformly for  $m \geq 1$  and  $n \geq 2$ .

Choose an unrestricted matrix of mass  $n$  and height  $k$  uniformly and let  $M$  be its number of unused rows. Its factorial moments are exactly

$$b_j := \mathbb{E}[(M)_j] = (k)_j \frac{A(n, k-j)}{A(n, k)} \quad (0 \leq j \leq k). \quad (5.2)$$

Here  $(x)_j = x(x-1)\cdots(x-j+1)$ ,  $b_0 = 1$ , and  $b_k = 0$  when  $n > 0$ .

**Theorem 5.1** (Coverage-window expansion). *Fix  $S, R < \infty$ . Set*

$$h = \frac{n}{k}, \quad s = h - \log k, \quad \eta = ke^{-h} = e^{-s}.$$

*As  $k \rightarrow \infty$ , uniformly for integers  $n$  satisfying  $|s| \leq S$  and complex  $w$  satisfying  $|w| \leq R$ ,*

$$\mathbb{E}(1+w)^M = e^{\eta w} \left\{ 1 - \frac{(1+h)\eta^2 w^2 + (1-L)h\eta w}{2k} + O_{S,R}\left(\frac{h^2}{k^2}\right) \right\}. \quad (5.3)$$

*In particular,*

$$\mathbb{P}(M=0) = e^{-\eta} \left\{ 1 + \frac{(1-L)h\eta - (1+h)\eta^2}{2k} + O_S\left(\frac{(\log k)^2}{k^2}\right) \right\}. \quad (5.4)$$

*The definition  $s = n/k - \log k$  includes the integer rounding of  $n$ ; there is no unstated choice of rounding convention.*

*Proof.* We first obtain a bound valid for all inclusion–exclusion indices, not only fixed  $j$ . The elementary inequalities

$$\frac{m}{L} \leq d_m \leq \frac{m}{L} + 1$$

follow respectively from  $1 - e^{-x} \leq x$  and  $e^x \geq 1 + x$ . Also,  $c_m$  is bounded above and bounded away from zero uniformly for  $m \geq 1$ . Equation (5.1) therefore gives, for all sufficiently large  $k$  in the indicated window and  $0 \leq j \leq k - 1$ ,

$$\begin{aligned} b_j &\leq C(k)_j \left( \frac{d_{k-j}}{d_k} \right)^n \\ &\leq Ck^j \left( 1 - \frac{j-L}{k} \right)^n \\ &\leq Ce^{Lh} (ke^{-h})^j = Ce^{Lh} \eta^j. \end{aligned} \quad (5.5)$$

The same bound holds for  $j = k$  since  $b_k = 0$ .

Let  $J = \lceil 4 \log k \rceil$ . Since  $\eta \leq e^S$  and  $e^{Lh} \leq e^{LS} k^L$ , the factorial tail bound gives

$$\sum_{j>J} \frac{|w|^j b_j}{j!} \leq Ck^L \sum_{j>J} \frac{(Re^S)^j}{j!} = O_{S,R,B}(k^{-B}) \quad (5.6)$$

for every fixed  $B > 0$ . Indeed the logarithm of the last factorial tail is at most  $-J \log J + O_{S,R}(J)$ . This bound directly justifies absolute truncation of the exact factorial-moment polynomial.

For the remaining indices  $1 \leq j \leq J$ , the needed expansion is uniform. Define

$$g_0(x) = \log \frac{Lx/2}{\sinh(Lx/2)}, \quad g_0(0) = 0.$$

Near zero,  $g_0$  is analytic,  $g_0(x) = O(x^2)$  and  $g'_0(x) = O(x)$ . The exact identities

$$\log d_m = \log \frac{m}{L} + \frac{L}{2m} + g_0(1/m), \quad \log c_m = -\log(2L) - \frac{L}{2m} + g_0(1/m)$$

yield

$$\begin{aligned} \log \frac{d_{k-j}}{d_k} &= -\frac{j}{k} - \frac{j^2}{2k^2} + \frac{Lj}{2k^2} + O\left(\frac{j^3}{k^3}\right), \\ \log \frac{c_{k-j}}{c_k} &= O\left(\frac{j}{k^2}\right), \\ \log(k)_j &= j \log k - \frac{j(j-1)}{2k} + O\left(\frac{j^3}{k^2}\right). \end{aligned}$$

All constants here are absolute, once  $J \leq k/2$ . Consequently

$$\log b_j = j \log \eta + \frac{-j(j-1) + h(Lj - j^2)}{2k} + O\left(\frac{(1+h)j^3}{k^2} + \rho^{n-1}\right). \quad (5.7)$$

Since  $hJ^2/k \rightarrow 0$ , exponentiation is uniform and gives

$$b_j = \eta^j \left\{ 1 + \frac{-j(j-1) + h(Lj - j^2)}{2k} + O\left(\frac{h^2 j^4}{k^2} + \rho^{n-1}\right) \right\}. \quad (5.8)$$

The index  $j = 0$  is exact and is treated separately.

Now use the exact identity

$$\mathbb{E}(1+w)^M = \sum_{j=0}^k \frac{b_j w^j}{j!}.$$

The error in (5.8) sums to  $O_{S,R}(h^2/k^2)$ , because  $\sum_{j \geq 0} j^4 (Re^S)^j / j! < \infty$  and  $\rho^{n-1}$  is smaller than every fixed power of  $k^{-1}$  in this window. Equation (5.6) permits extension of all the main sums to infinity. With  $x = \eta w$ ,

$$\sum_{j \geq 0} \frac{[-j(j-1) + h(Lj - j^2)]x^j}{j!} = -[(1+h)x^2 + (1-L)hx]e^x.$$

This proves (5.3); the additive and relative forms of its error are equivalent on each fixed disk because  $|e^{\eta w}|$  is bounded away from zero there. Set  $w = -1$ , with the polynomial interpretation  $0^0 = 1$ , to obtain (5.4).  $\square$

[Added 3 October 2026, batch 85B: manuscript 01 proves the case  $w = -1$ , that is (5.4) with error  $O((1+h)^2/k^2)$ , which is the same order, and the Poisson limit of  $M$ , by a second route. It expands the factorial moments exactly as in (5.8) (its form is  $\log(\mathbb{E}(M)_j/\eta^j) = -(h+1)j^2 + (Lh+1)j/(2k) + O((1+h)(1+j)^4/k^2)$  for  $j \leq C \log k$ , the same polynomial), then truncates the inclusion–exclusion series for  $\mathbb{P}(M=0)$  by the Bonferroni inequalities after  $R \asymp \log k$  terms, the next term being at most  $C'^{R+1}/(R+1)!$ , and sums with  $\sum_j (-\eta)^j j / j! = -\eta e^{-\eta}$  and  $\sum_j (-\eta)^j j^2 / j! = (\eta^2 - \eta)e^{-\eta}$ . The same device gives every fixed order (Theorem 13.1). Manuscript 06 proves the whole generating function in complex  $w$ , and from it the probability-mass and total-variation statements below, which manuscript 01 does not have.]

## 5.1 A summable expansion of the entire missing-row law

For  $\eta > 0$  put  $\pi_\eta(r) = e^{-\eta} \eta^r / r!$  for  $r \geq 0$ , and set  $\pi_\eta(r) = 0$  for negative integers  $r$ . Define

$$C_2(r; \eta) = r(r-1) - 2\eta r + \eta^2 = (r-\eta)^2 - r.$$

**Corollary 5.2** (Probability-mass and total-variation expansion). *Under the assumptions of Theorem 5.1, uniformly for  $r \geq 0$ ,*

$$\mathbb{P}(M=r) = \pi_\eta(r) \left\{ 1 + \frac{(1-L)h(\eta-r) - (1+h)C_2(r; \eta)}{2k} \right\} + \mathcal{R}_{k,r}, \quad (5.9)$$

where

$$|\mathcal{R}_{k,r}| \leq C_S 2^{-r} \frac{h^2}{k^2}, \quad \sum_{r \geq 0} |\mathcal{R}_{k,r}| = O_S \left( \frac{h^2}{k^2} \right).$$

In particular, if  $X \sim \text{Poisson}(\eta)$ , then

$$d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(\eta)) = \frac{1}{4k} \mathbb{E} |(1-L)h(\eta-X) - (1+h)C_2(X; \eta)| + O_S \left( \frac{h^2}{k^2} \right). \quad (5.10)$$

The total-variation distance is  $\Theta_S((\log k)/k)$ , uniformly for  $|s| \leq S$ . If  $s \rightarrow c$  and  $\eta_0 = e^{-c}$ , its leading constant is given by

$$\frac{k}{\log k} d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(ke^{-n/k})) \rightarrow \frac{1}{4} \mathbb{E} |(1-L)(\eta_0 - X_0) - C_2(X_0; \eta_0)| > 0, \quad (5.11)$$

where  $X_0 \sim \text{Poisson}(\eta_0)$ .

*Proof.* Substitute  $w = z - 1$  in (5.3). Its error is uniformly  $O_S(h^2/k^2)$  on  $|z| \leq 2$ , using  $R = 3$ . Cauchy's coefficient bound gives  $|\mathcal{R}_{k,r}| \leq C_S 2^{-r} h^2/k^2$ . The coefficients of  $(z - 1)e^{\eta(z-1)}$  and  $(z - 1)^2 e^{\eta(z-1)}$  are respectively

$$\pi_\eta(r - 1) - \pi_\eta(r), \quad \pi_\eta(r - 2) - 2\pi_\eta(r - 1) + \pi_\eta(r).$$

Writing these as multiples of  $\pi_\eta(r)$  proves (5.9). The reverse triangle inequality and summability of  $\mathcal{R}_{k,r}$  prove (5.10).

The function

$$D(\eta) = \mathbb{E} |(1 - L)(\eta - X) - C_2(X; \eta)|, \quad X \sim \text{Poisson}(\eta),$$

is continuous for  $\eta > 0$  and strictly positive: its integrand is the absolute value of a nonzero quadratic polynomial and a Poisson law with positive mean assigns positive mass to every nonnegative integer. Therefore  $D$  has a positive minimum and a finite maximum on  $[e^{-S}, e^S]$ . The difference between the expectation in (5.10) and  $hD(\eta)$  is bounded by  $\mathbb{E}|C_2(X; \eta)| = O_S(1)$ . This proves the uniform two-sided rate and the claimed limit.  $\square$

## 5.2 Mean and variance interpretation

The same expansion gives

$$\nu_k := \mathbb{E}M = \eta - \frac{(1 - L)h\eta}{2k} + O_S\left(\frac{h^2}{k^2}\right),$$

and

$$\text{Var}(M) - \mathbb{E}M = -\frac{(1 + h)\eta^2}{k} + O_S\left(\frac{h^2}{k^2}\right).$$

Thus the correction includes both a shift of the Poisson mean and a negative second factorial cumulant. Adjusting only the mean does not remove the leading dependence correction. More precisely,

$$d_{\text{TV}}(\mathcal{L}(M), \text{Poisson}(\nu_k)) = \frac{1 + h}{4k} \mathbb{E}|C_2(X; \eta)| + O_S\left(\frac{h^2}{k^2}\right), \quad X \sim \text{Poisson}(\eta).$$

To see this, replace the linear term in (5.3) by  $e^{(\nu_k - \eta)w}$ , with uniform error  $O_S(h^2/k^2)$ , and apply the same coefficient argument. Replacing  $\eta$  by  $\nu_k$  in the expectation changes the displayed answer by only  $O_S(h^2/k^2)$ : the expectation is locally Lipschitz as a function of its positive Poisson parameter, by dominated summation of the polynomially weighted Poisson series on a compact parameter interval.

**Corollary 5.3** (The coverage threshold). *If  $n/k - \log k \rightarrow s \in \mathbb{R}$ , then*

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \frac{T(n, k)}{A(n, k)} \longrightarrow \exp(-e^{-s}).$$

For bounded  $s_k = n/k - \log k$ , with  $\eta = ke^{-n/k}$  and  $h = n/k$ ,

$$T(n, k) = c_k d_k^n e^{-\eta} \left[ 1 + \frac{(1 - L)h\eta - (1 + h)\eta^2}{2k} + O\left(\frac{(\log k)^2}{k^2}\right) \right]. \quad (5.12)$$

*Proof.* The generating-function expansion implies the Poisson distributional limit; alternatively use the summable probability-mass expansion. The event  $M = 0$  is precisely membership in the class counted by  $T$ . Multiply its probability by Theorem 4.1; the exponential error is negligible at the displayed scale.  $\square$

[Added 3 October 2026, batch 85B: the limit statements are also manuscript 01's (its Theorem on the Poisson transition), proved there by marked inclusion–exclusion with  $\mathbb{E}(M)_{j+m}/(m!j!)$  for each fixed point mass  $m$ , or equivalently by the factorial-moment method. Table 8 in Part II gives manuscript 01's coverage probabilities as ratios of exact integers for  $k = 10, \dots, 120$ ; at  $k = 20$ ,  $n = 60$  they agree with Table 3 in every printed digit.]

*Remark 5.4* (Moving versus fixed Poisson mean). The sharp rate in (5.11) compares the actual law with  $\text{Poisson}(ke^{-n/k})$ . If only  $n/k - \log k \rightarrow s$  is assumed, replacing this moving mean by  $e^{-s}$  incurs an additional error governed by the rate of that convergence. For  $n = \text{round}(k(\log k + s))$ , the additional error is  $O(1/k)$  and does not change the leading  $(\log k)/k$  constant. No such rate is asserted for an arbitrary unspecified  $o(1)$  perturbation.

## 6 Column counts and joint Gaussian–Poisson independence

The row-coverage law survives a fixed positive weight on the number of columns. This permits a useful independence theorem by exponential tilting. Under the probability measure proportional to  $u^{C(B)}$ , denote expectation and probability by  $\mathbb{E}_u$  and  $\mathbb{P}_u$ .

**Lemma 6.1** (Uniformity under column weights). *If  $n/k - \log k \rightarrow s \in \mathbb{R}$ , then, uniformly for  $u$  in each fixed compact interval  $I \subset (0, \infty)$ ,*

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \mathbb{P}_u(M = m) \rightarrow e^{-e^{-s}} \frac{e^{-ms}}{m!} \quad (m \geq 0 \text{ fixed}).$$

*The same conclusion holds along any sequence of weights in  $I$ .*

*Proof.* Deleting designated zero rows gives the exact identity

$$\mathbb{E}_u[(M)_j] = (k)_j \frac{A(n, k-j; u)}{A(n, k; u)}.$$

Use Theorem 4.1. Uniformly for  $u \in I$ , with  $L_u = \log(1 + 1/u)$  and  $q = 1/(1 + u)$ ,

$$\log d_k(u) = \log k - \log L_u + \frac{L_u}{2k} + O_I(k^{-2}), \quad \log \mu_k(u) = \log \frac{q}{L_u} - \frac{L_u}{2k} + O_I(k^{-2}).$$

For each fixed  $j$  this gives

$$\log \mathbb{E}_u[(M)_j] = -j(n/k - \log k) + O_{I,j}((\log k)/k).$$

Thus all fixed factorial moments tend uniformly to  $e^{-js}$ . To make the distributional step explicit, expand  $\mathbb{E}_u z^M$  at  $z = 1$  for  $0 \leq z \leq 1$ . Taylor's remainder after  $J - 1$  terms is bounded by  $\mathbb{E}_u[(M)_J]/J!$ . First pass to the limit with  $J$  fixed, then let  $J \rightarrow \infty$ ; the limiting error  $e^{-Js}/J!$  tends to zero. This identifies the Poisson probability generating function. The first moment bounds give tightness. Every subsequential limit is therefore Poisson, uniformly in the allowed weights. On the nonnegative integers, weak convergence gives convergence of each point probability by taking an interval with half-integer endpoints.  $\square$

Fix a baseline  $u_0 > 0$ . In the formulas below evaluate

$$b = \frac{u_0}{1 + u_0}, \quad q = \frac{1}{1 + u_0}, \quad L_{u_0} = \log(1 + 1/u_0), \quad a = e^{-L_{u_0}/k}, \quad r = 1 - a$$

at this baseline. Define

$$\mu_k(u_0) = \frac{qa}{kr}, \quad \sigma_k^2(u_0) = \frac{\mu_k(u_0)^2}{a} - b\mu_k(u_0) = \frac{qa}{k^2r^2}(q - bkr). \quad (6.1)$$

These are the first two derivatives of  $\ell_k(v) = \log d_k(u_0 e^v)$  at  $v = 0$ . Indeed logarithmic differentiation gives  $q' = -bq$ ,  $L'_u = -q$ ,  $a' = aq/k$ , and  $r' = -aq/k$ . The variance constant is strictly positive, since  $q/b = e^{L_{u_0}} - 1 > L_{u_0} \geq kr$ .

**Theorem 6.2** (Joint limit and conditional Gaussian laws). *Suppose  $n/k - \log k \rightarrow s \in \mathbb{R}$ , and choose matrix compositions of mass  $n$  and height  $k$  with probabilities proportional to  $u_0^C$ , for fixed  $u_0 > 0$ . Then*

$$\left( \frac{C - n\mu_k(u_0)}{\sqrt{n\sigma_k^2(u_0)}}, M \right) \xrightarrow{d} (Z, P), \quad (6.2)$$

where  $Z$  is standard normal,  $P$  has law  $\text{Poisson}(e^{-s})$ , and  $Z, P$  are independent. For each fixed  $m \geq 0$ , the same Gaussian limit holds conditional on  $M = m$ . In particular it holds for the A261781 class, which requires  $M = 0$ .

*Proof.* Put  $s_k = \sqrt{n\sigma_k^2(u_0)}$ ,  $X = (C - n\mu_k(u_0))/s_k$ , and  $u_v = u_0 e^{v/s_k}$  for a fixed real  $v$ . The exact tilting identity is

$$\mathbb{E}_{u_0} e^{vX} = e^{-vn\mu_k(u_0)/s_k} \frac{A(n, k; u_v)}{A(n, k; u_0)}.$$

The two weights lie in a common compact interval. Applying (4.3) and Taylor's theorem gives

$$\begin{aligned} \log \mathbb{E}_{u_0} e^{vX} &= n [\ell_k(v/s_k) - \ell_k(0) - \mu_k(u_0)v/s_k] + \log \frac{\mu_k(u_v)}{\mu_k(u_0)} + O(\rho_I^{n-1}) \\ &= \frac{v^2}{2} + O_v(n^{-1/2}) + O(\rho_I^{n-1}). \end{aligned}$$

The derivatives used here are bounded uniformly in  $k$ : write

$$\log d_k(u) = \log k - \log L_u + g(L_u/k), \quad g(x) = \log \frac{x}{1 - e^{-x}},$$

where  $g$  is smooth at zero. The variance constants have a positive limit and remain bounded away from zero and infinity. The logarithmic derivatives of  $\mu_k$  are likewise uniformly bounded. Convergence of moment generating functions on a neighborhood of zero proves the marginal normal limit.

For every fixed  $m$ , the conditional tilting identity is also exact:

$$\mathbb{E}_{u_0} [e^{vX} \mid M = m] = \mathbb{E}_{u_0} e^{vX} \frac{\mathbb{P}_{u_v}(M = m)}{\mathbb{P}_{u_0}(M = m)}.$$

Both probabilities tend to the same strictly positive Poisson probability by Lemma 6.1. Their ratio tends to one, proving the conditional Gaussian limit. Summing the conditional limits over finitely many  $m$  and controlling the remaining mass by tightness of  $M$  proves the joint limit and independence.  $\square$

For the unweighted model these constants simplify to

$$\mu_k = c_k, \quad \sigma_k^2 = \frac{a_k(1 - kr_k)}{4k^2 r_k^2}, \quad \mu_\infty = \frac{1}{2L}, \quad \sigma_\infty^2 = \frac{1 - L}{4L^2}. \quad (6.3)$$

In the coverage window, using  $n\mu_\infty$  instead of  $n\mu_k$  also gives the same limit: their difference is  $O(n/k) = O(\log k)$ , which is  $o(\sqrt{n})$ . The finite- $k$  centering keeps this correction visible. Louchard's fixed-row Gaussian theorem is an antecedent [6]; the new conclusion here is uniformity with growing rows and its joint and conditional form.

## 7 Uniform asymptotics on dense rays

We now require every row to be nonzero and let  $n/k$  stay bounded above and away from 1. The Poisson-window expansion does not apply here. The positive transform (2.4) gives a different route.

### 7.1 The saddle and the first correction

Put  $f(z) = e^z - 1$ . For  $\lambda > 1$ , let  $r = r(\lambda) > 0$  solve

$$\frac{r}{1 - e^{-r}} = \lambda. \quad (7.1)$$

With  $\mathcal{D} = r \, d/dr$ , define

$$\kappa_j = \mathcal{D}^j \log f(r), \quad b = \kappa_2 = \lambda(1 + r - \lambda), \quad v = \frac{Lr}{2}. \quad (7.2)$$

In particular  $\kappa_1 = \lambda$  and  $b > 0$ .

**Theorem 7.1** (Uniform ray expansion). *Let  $K$  be any fixed compact subset of  $(1, \infty)$ . Uniformly over integers  $n, k \rightarrow \infty$  with  $\lambda = n/k \in K$ ,*

$$T(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}} \left[ 1 + \frac{E_1(\lambda)}{k} + O_K(k^{-2}) \right], \quad (7.3)$$

where

$$E_1(\lambda) = \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b} + \frac{v^2/3 - v}{\lambda}. \quad (7.4)$$

The expansion continues to every fixed order in integral powers of  $1/k$ , uniformly on  $K$ . Section 7.4 gives an explicit finite coefficient-generation rule at each order.

The derivative definitions in (7.2) are exact finite formulas, not additional implicitly determined constants. If expanded,

$$\begin{aligned} \kappa_3 &= \lambda(2\lambda^2 - 3\lambda r - 3\lambda + r^2 + 3r + 1), \\ \kappa_4 &= \lambda(-6\lambda^3 + 12\lambda^2 r + 12\lambda^2 - 7\lambda r^2 - 18\lambda r - 7\lambda \\ &\quad + r^3 + 6r^2 + 7r + 1). \end{aligned}$$

## 7.2 Reducing the positive sum to an analytic amplitude

The Fubini generating function is

$$\sum_{q \geq 0} F_q \frac{z^q}{q!} = \frac{1}{2 - e^z}.$$

Its poles are  $L + 2\pi i j$ ,  $j \in \mathbb{Z}$ , with residue  $-1/2$ . Subtracting the pole at  $L$  and applying Cauchy's coefficient estimate on a circle between it and the next poles proves

$$F_q = \frac{q!}{2L^{q+1}}(1 + O(\vartheta^q)) \quad (7.5)$$

for any fixed  $\vartheta$  strictly between  $L/\sqrt{L^2 + 4\pi^2}$  and 1. Every term in (2.4) is nonnegative and  $q \geq k$ , so this replacement incurs a relative error  $O(\vartheta^k)$  in the entire sum.

[Added 3 October 2026, batch 85B: manuscript 01 proves the same estimate (its Lemma on the Fubini approximation) by subtracting  $1/(2(L - z))$  from  $1/(2 - e^z)$  and applying Cauchy's bound on a circle of radius between  $L/\vartheta$  and  $\sqrt{L^2 + 4\pi^2}$ . Both are classical. The repository's transseries volume has the exact convergent pole-lattice formula  $F_q = \frac{q!}{2} \sum_{j \in \mathbb{Z}} (L + 2\pi i j)^{-q-1}$ ,  $q \geq 1$  (Theorem `q2:thm:fubini` of [11]), of which (7.5) keeps the term  $j = 0$ ; and the generating function  $1/(2 - e^z)$  is formalized in Lean as `Fabius.two_sub_exp_mul_egfA_fubini` [12].]

Define the polynomial

$$H_{n,k}(z) = \sum_{d=0}^{n-k} L^d \begin{bmatrix} n \\ n-d \end{bmatrix} \left( \frac{(n-d)!}{n!} \right)^2 z^d. \quad (7.6)$$

Using  $[z^q]f(z)^k = k! \{^q_k\} / q!$ , including both factorials from (2.4) and (7.5), gives

$$T(n, k) = \frac{n!}{2L^{n+1}} [z^n] f(z)^k H_{n,k}(z) (1 + O(\vartheta^k)). \quad (7.7)$$

Terms with  $d > n - k$  would have zero coefficient contribution since  $f(z)^k$  begins at degree  $k$ . Notice that the numerator in this formula is  $n!$ , not  $1/n!$ ; the square of the factorial ratio in (7.6) is essential.

**Lemma 7.2** (Uniform analytic amplitude). *Fix  $R > 0$  and  $\delta > 0$ . If  $k \geq \delta n$  and  $n - k \geq \delta n$ , then, uniformly on  $|z| \leq R$ ,*

$$H_{n,k}(z) = e^{Lz/2} \left[ 1 + \frac{L^2 z^2 / 12 - Lz/2}{n} + O_{R,\delta}(n^{-2}) \right]. \quad (7.8)$$

*The same expansion may be differentiated any fixed number of times, with the corresponding differentiated error bound.*

*Proof.* The unsigned Stirling coefficient is the elementary symmetric function  $[^n_{n-d}] = e_d(1, \dots, n-1)$ . Therefore

$$e_d(1, \dots, n-1) \leq \frac{(n(n-1)/2)^d}{d!} \leq \frac{n^{2d}}{2^d d!}.$$

Because  $n - d \geq k \geq \delta n$ , the absolute value of the  $d$ th term in (7.6) is at most  $(LR/(2\delta^2))^d / d!$ . Its tail for  $d > n^{1/10}$  is smaller than every fixed inverse power of  $n$ .

For  $d \leq n^{1/10}$  we claim the uniform expansions

$$\begin{bmatrix} n \\ n-d \end{bmatrix} = \frac{n^{2d}}{2^d d!} \left[ 1 - \frac{d(2d+1)}{3n} + O\left(\frac{(1+d)^4}{n^2}\right) \right], \quad (7.9)$$

$$\left(\frac{(n-d)!}{n!}\right)^2 = n^{-2d} \left[1 + \frac{d(d-1)}{n} + O\left(\frac{(1+d)^4}{n^2}\right)\right]. \quad (7.10)$$

For the first, set  $S_1 = n(n-1)/2$  and sample  $d$  independent indices in  $\{1, \dots, n-1\}$  with probabilities  $p_j = j/S_1$ . The probability of no equal pair is  $d!e_d/S_1^d$ . The first two inclusion–exclusion bounds for pair coincidences give

$$\mathbb{P}(\text{no coincidences}) = 1 - \binom{d}{2} \sum_j p_j^2 + O\left(d^4 \left(\sum_j p_j^2\right)^2 + d^3 \sum_j p_j^3\right).$$

Here  $\sum p_j^2 = 4/(3n) + O(n^{-2})$  and  $\sum p_j^3 = O(n^{-2})$ . Expanding  $S_1^d$  proves (7.9). For the second, expand  $-2 \sum_{j=0}^{d-1} \log(1 - j/n)$  with its uniformly bounded Taylor remainder.

Multiplication gives

$$L^d \binom{n}{n-d} \left(\frac{(n-d)!}{n!}\right)^2 = \frac{(L/2)^d}{d!} \left[1 + \frac{d(d-4)}{3n} + O\left(\frac{(1+d)^8}{n^2}\right)\right].$$

The factorial majorant permits summation of the errors and extension of the main terms to all  $d \geq 0$ . Since  $\sum d(d-4)x^d/d! = (x^2 - 3x)e^x$ , this proves (7.8). Establishing the same uniform estimate on a slightly larger disk and using Cauchy’s derivative estimate proves the differentiated versions.  $\square$

*Remark 7.3* (The hypothesis  $\delta$  is not needed). [Added 3 October 2026, batch 85B.] The coefficient of  $z^d$  in (7.6) is  $\chi_{n,d} L^d$ , where  $\chi_{n,d} = \lfloor \frac{n}{n-d} \rfloor / (n)_d^2$  does not depend on  $k$ ; so  $H_{n,k}(z)$  is the truncation at degree  $n-k$  of the entire polynomial  $W_n(Lz)$  of Part II,  $W_n(w) = \sum_{d=0}^{n-1} \chi_{n,d} w^d$ . Since  $f(z)^k$  begins in degree  $k$ , the truncation does not change the coefficient in (7.7), which therefore holds with  $W_n(Lz)$  in place of  $H_{n,k}(z)$ ; this is manuscript 01’s form of the reduction. Manuscript 01 proves the expansion of  $W_n$  to every order, uniformly on every compact subset of  $\mathbb{C}$  and *with no hypothesis on  $k$* , together with an exact differential recurrence in  $n$  (Theorem 14.3 and Lemma 14.2). That is the general form of Lemma 7.2; the conditions  $k \geq \delta n$ ,  $n-k \geq \delta n$  above serve only the truncated polynomial, and on compact rays, where both apply, the two statements coincide. The polynomials  $P_1, P_2$  of (7.15) are manuscript 01’s  $p_1, p_2$ ; Part II adds  $P_3$ .

### 7.3 The saddle calculation and its error

The function  $r/(1 - e^{-r})$  increases from 1 to  $\infty$ , because its derivative is

$$\frac{1 - (1+r)e^{-r}}{(1 - e^{-r})^2} > 0.$$

Hence the positive saddle is unique. For  $\lambda \in K$  it lies in a compact positive interval and  $b = r \, d\lambda/dr$  is bounded away from zero.

On the circle  $z = re^{i\theta}$ , the coefficient integral has phase

$$\log \frac{f(re^{i\theta})}{f(r)} - i\lambda\theta = -\frac{b}{2}\theta^2 - \frac{i\kappa_3}{6}\theta^3 + \frac{\kappa_4}{24}\theta^4 + O_K(\theta^5). \quad (7.11)$$

Every coefficient of  $f$  in positive degree is strictly positive. Equality in  $|f(re^{i\theta})| \leq f(r)$  can therefore occur only at  $\theta = 0$  modulo  $2\pi$ . Compactness gives a uniform strict gap on every fixed complementary arc. Near zero, the modulus is bounded by  $f(r)e^{-c\theta^2}$  for some uniform  $c > 0$ .

For  $g(z) = e^{Lz/2}$ ,

$$\frac{g(re^{i\theta})}{g(r)} = 1 + iv\theta - \frac{v + v^2}{2}\theta^2 + O_K(\theta^3).$$

Expanding (7.11), multiplying by this amplitude, and setting  $\theta = x/\sqrt{k}$  gives the order-1/ $k$  contribution

$$\frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b}.$$

These constants follow from the Gaussian moments  $\mathbb{E}X^2 = b^{-1}$ ,  $\mathbb{E}X^4 = 3b^{-2}$ , and  $\mathbb{E}X^6 = 15b^{-3}$  for  $X \sim N(0, b^{-1})$ . The order-1/ $n$  term of Lemma 7.2, evaluated at the saddle, is  $(v^2/3 - v)/n$ . Since  $n = \lambda k$ , this proves the coefficient in (7.4).

For completeness, the error is relative  $O_K(k^{-2})$ . Expand the analytic phase and amplitude through sufficiently many terms, for example through degree eight, on a shrinking central arc. Odd terms integrate to zero by symmetry, and the Gaussian modulus bound dominates the integrated remainders. The fixed complementary arc is exponentially small; the intervening arc is controlled by the same Gaussian bound. Finally, an analytic amplitude error uniformly  $O(n^{-2})$  contributes at most

$$O(n^{-2})f(r)^k r^{-n} \int_{-\pi}^{\pi} \left| \frac{f(re^{i\theta})}{f(r)} \right|^k d\theta = O(n^{-2}k^{-1/2})f(r)^k r^{-n}.$$

Retaining this Gaussian factor proves the required relative error; a bare maximum-modulus estimate would lose an unnecessary factor  $\sqrt{k}$ . Together with (7.7), this proves the first-order assertion of Theorem 7.1.

[Added 3 October 2026, batch 85B: Theorem 7.1, including its all-orders form (7.17), is also manuscript 01's main theorem on compact rays, with the leading approximation  $\mathcal{B}(n, k)$  of (15.1) and the same first coefficient: its  $A_1(\rho)$  is  $E_1(\lambda)$  of (7.4), term for term, after the renames  $\rho \rightarrow \lambda$ ,  $\tau \rightarrow r$ ,  $\mu \rightarrow v$ . Its route differs in two places. It integrates the untruncated kernel  $W_n(Lz)$  on  $|z| = r$  and controls the tails with  $|W_n(w)| \leq e^{|w|}$  (Lemma 14.2); on the central arc  $|\theta| \leq k^{-2/5}$  it expands two orders beyond the target and bounds the intermediate region by  $\exp(-ck^{1/5})$ . It also proves the uniqueness of the maximum on the circle directly: equality in the triangle inequality for  $\sum_{j \geq 1} r^j e^{ij\theta}/j!$  forces all  $e^{ij\theta}$  to agree, and consecutive powers force  $e^{i\theta} = 1$ .]

## 7.4 A constructive expansion to every fixed order

Here is a precise algorithm for the higher coefficients, together with the uniform remainder argument. Put  $w = Lz$  and  $\epsilon = 1/n$ . The exact polynomial

$$E_n(w) = \prod_{j=1}^{n-1} \left( 1 + \frac{jw}{n^2} \right)$$

has, on every fixed disk, an expansion  $E_n(w) \sim e^{w/2} \sum_{p \geq 0} R_p(w) n^{-p}$ , with polynomial  $R_p$ . Compute them from

$$\log E_n(w) = \sum_{m \geq 1} \frac{(-1)^{m-1} w^m}{m n^{2m}} \sum_{j=1}^{n-1} j^m. \quad (7.12)$$

The inner power sum is a finite polynomial in  $n$ ; at order  $n^{-p}$  only  $m \leq p + 1$  is needed. For example,

$$R_0 = 1, \quad R_1 = -\frac{w}{2} - \frac{w^2}{6}, \quad R_2 = \frac{w^2}{4} + \frac{w^3}{12} + \frac{1}{2} \left( \frac{w}{2} + \frac{w^2}{6} \right)^2.$$

Define polynomials  $Q_q(d)$  by the formal identity

$$\exp \left( 2 \sum_{m \geq 1} \frac{\epsilon^m}{m} \sum_{j=0}^{d-1} j^m \right) = \sum_{q \geq 0} Q_q(d) \epsilon^q. \quad (7.13)$$

Thus  $Q_0 = 1$ ,  $Q_1 = d(d-1)$ , and

$$Q_2 = \frac{d(d-1)(2d-1)}{6} + \frac{d^2(d-1)^2}{2}.$$

With  $D_w = w \, d/dw$ , put

$$P_j(w) = e^{-w/2} \sum_{p+q=j} Q_q(D_w) \left[ e^{w/2} R_p(w) \right]. \quad (7.14)$$

These are polynomials, and

$$H_{n,k}(z) \sim e^{Lz/2} \sum_{j \geq 0} \frac{P_j(Lz)}{n^j}, \quad P_0 = 1, \quad P_1(w) = \frac{w^2}{12} - \frac{w}{2}, \quad P_2(w) = \frac{w^4}{288} - \frac{w^2}{12}. \quad (7.15)$$

To justify this algorithm to any fixed order  $p$ , set  $F_d(x) = \prod_{j=0}^{d-1} (1-jx)^{-2}$ . For  $d \leq n-k \leq (1-\delta)n$  and  $0 \leq x \leq 1/n$ , every  $1-jx \geq \delta$ . The  $m$ th derivative of its logarithm is

$$2(m-1)! \sum_{j < d} \frac{j^m}{(1-jx)^m} = O_{m,\delta}((1+d)^{m+1}).$$

The finite product rule for derivatives of an exponential then gives

$$|F_d^{(p+1)}(x)| \leq C_{p,\delta} \delta^{-2d} (1+d)^{2p+2}.$$

Taylor's theorem bounds the remainder after the  $p$ th term by  $C_{p,\delta} n^{-p-1} \delta^{-2d} (1+d)^{2p+2}$ . Multiplication by the exact coefficient  $e_d(1, \dots, n-1)/n^{2d} \leq 1/(2^d d!)$  makes this error summable on every fixed  $w$ -disk. Multiplying its  $d$ th coefficient by  $Q_q(d)$  is exactly the operation  $Q_q(D_w)$ . Extending the sum beyond  $n-k \geq \delta n$  adds only a factorially small tail. Independently, expanding (7.12) and its derivatives on a slightly larger disk has a uniform Taylor remainder. Combining the two steps proves (7.15), with error  $O(n^{-p-1})$  after any fixed order  $p$ .

The remaining Gaussian integrations are also explicit. Define the linear functional on polynomials in  $y$  by

$$\mathcal{G}_b(y^{2m}) = (-1)^m (2m-1)!! \, b^{-m}, \quad \mathcal{G}_b(y^{2m+1}) = 0,$$

with  $(-1)!! = 1$ . It is integration after the substitution  $y = iX$ ,  $X \sim N(0, b^{-1})$ . Introduce a formal variable  $\tau$  and set

$$\begin{aligned} E_j(\lambda) = \mathcal{G}_b \left( [\tau^{2j}] \exp \left\{ \sum_{m \geq 3} \frac{\kappa_m y^m \tau^{m-2}}{m!} + v(e^{\tau y} - 1) \right\} \right. \\ \left. \times \sum_{p \geq 0} \frac{\tau^{2p}}{\lambda^p} P_p(Lre^{\tau y}) \right). \end{aligned} \quad (7.16)$$

At each order this is a finite polynomial calculation: it uses  $p \leq j$  and  $m \leq 2j + 2$ . Odd powers vanish after applying  $\mathcal{G}_b$ . It gives  $E_0 = 1$  and the  $E_1$  in (7.4). Therefore for every fixed  $J \geq 0$ ,

$$T(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}} \left[ \sum_{j=0}^J \frac{E_j(\lambda)}{k^j} + O_{K,J}(k^{-J-1}) \right]. \quad (7.17)$$

The same arc estimates as above justify termwise finite integration. This is an asymptotic expansion to each finite order; convergence of the infinite displayed series is not asserted.

[Added 3 October 2026, batch 85B: manuscript 01 gives the same finite prescription with its expansion parameter called  $\varepsilon$  instead of  $\tau$ : its  $H_M(\varepsilon, y)$  is the expression inside  $\mathcal{G}_b$  in (7.16), truncated at  $m \leq 2J + 2$  and  $p \leq J$ , because  $P_p(Lre^{\tau y})$  is its  $p_p(2\mu e^{\varepsilon y})$  with  $2\mu = Lr$ . It stresses the order of operations: first expand and truncate in  $\tau$ , then apply  $\mathcal{G}_b$  to each polynomial coefficient; no Gaussian expectation of an untruncated exponential of a high-degree polynomial is intended, and such an integral need not exist. It computes the kernel polynomials by a different finite route, (14.11)–(14.12), and adds a polynomial recursion for the  $\kappa_j$  ((15.2)); both are in Part II.]

## 7.5 The A261784 diagonal

Put  $n = 2N$ ,  $k = N$ , and

$$r = 2 + W_0(-2e^{-2}) = 1.5936242600400400923230 \dots \quad (7.18)$$

The principal real Lambert branch is necessary; the other real branch gives the excluded boundary value  $r = 0$ . Equation (7.1) becomes  $r/(1 - e^{-r}) = 2$ . Define

$$d_* = \frac{4(e^r - 1)}{L^2 r^2} = \frac{4}{L^2 r(2 - r)} = 12.8556891736661213193266 \dots, \quad (7.19)$$

$$c_* = \frac{2^{r/2}}{4\pi L \sqrt{r - 1}} = 0.2588649231114655502517752 \dots \quad (7.20)$$

**Corollary 7.4** (Exact amplitude and correction for A261784). *The sequence  $a_N = T(2N, N)$  satisfies*

$$a_N = c_* d_*^N \frac{(N!)^2}{N} \left( 1 + \frac{\beta}{N} + O(N^{-2}) \right), \quad (7.21)$$

where, with  $v = Lr/2$ ,

$$\begin{aligned} b &= 2(r - 1), \quad \kappa_3 = 2(r^2 - 3r + 3), \quad \kappa_4 = 2(r^3 - 8r^2 + 19r - 13), \\ \beta &= \frac{\kappa_4}{8b^2} - \frac{5\kappa_3^2}{24b^3} + \frac{v\kappa_3}{2b^2} - \frac{v + v^2}{2b} + \frac{v^2/3 - v}{2} - \frac{1}{8} \\ &= -0.5218067015564826638891441 \dots \end{aligned}$$

*Proof.* Specialize Theorem 7.1 at  $\lambda = 2$  and use

$$\frac{(2N)!}{(N!)^2} = \frac{4^N}{\sqrt{\pi N}} \left( 1 - \frac{1}{8N} + O(N^{-2}) \right).$$

The saddle prefactor has  $b = 2(r - 1)$ . Collecting the two square roots gives (7.20), and the central-binomial correction accounts for the final  $-1/8$  in  $\beta$ .  $\square$

The OEIS entry already gives the leading equivalent in (7.21), with  $d_*$  in Lambert- $W$  form and  $c_*$  numerically [3]. The contribution here is its derivation from the positive transform, the exact expression (7.20), the explicit correction, and its place in the uniform all-orders theorem.

[Added 3 October 2026, batch 85B: manuscript 01 proves the same corollary to every fixed order, with the same  $r$ ,  $d_*$ ,  $c_*$  and first correction; written out with  $\kappa_3 = 2q$ ,  $\kappa_4 = 2s$ ,  $q = r^2 - 3r + 3$ ,  $s = r^3 - 8r^2 + 19r - 13$ , its closed form is  $\beta_1 = \frac{s}{16(r-1)^2} - \frac{5q^2}{48(r-1)^3} - \frac{v^2+v}{4(r-1)} + \frac{qv}{4(r-1)^2} + \frac{v^2}{6} - \frac{v}{2} - \frac{1}{8}$ , which is the display above with  $b = 2(r-1)$  substituted. Both manuscripts note that the expression for  $d_*$  agrees algebraically with the one posted in A261784 and that  $c_*$  reproduces its decimal prefactor. The two computations of  $r$ ,  $d_*$ ,  $c_*$ ,  $\beta_1$  and  $\beta_2$  were made by independent programs and agree in every printed digit. Manuscript 01 adds the third correction  $\beta_3$ , the recipe for every further one, and fifty-digit values (Section 17).]

[Added 6 October 2026, batch 108: Report 169 (Part IV) proves the corollary to every fixed order by a third route, a residue in the variable  $Q$  of Section 37 with Euler–Maclaurin summation (Theorem 35.1); its closed form (35.4) is the expression for  $\beta$  above, expanded, and its coefficients are the  $\beta_j$  of Corollary 17.1 (Remark 35.2).]

## 7.6 Column counts under the dense row condition

There is a further consequence: enforcing every row to be nonzero in this regime still leaves a universal leading Gaussian column law.

**Corollary 7.5.** *For a uniformly chosen matrix counted by  $T(n, k)$ , with  $n/k \in K \Subset (1, \infty)$ , its number of columns satisfies*

$$\frac{C - n/(2L)}{\sqrt{n(1-L)/(4L^2)}} \xrightarrow{d} N(0, 1). \quad (7.22)$$

*Uniformly in this regime,*

$$\mathbb{E}C = \frac{n+1}{2L} - \frac{1}{2} - \frac{r}{4} + O_K(k^{-1}), \quad (7.23)$$

$$\text{Var}(C) = \frac{(n+1)(1-L)}{4L^2} - \frac{1}{4} + \frac{r}{8} + O_K(k^{-1}). \quad (7.24)$$

*Proof.* The weighted version of (2.4) replaces  $F_q$  by  $F_q(u) = \sum_c c! \{c\}^q u^c$ , whose exponential generating function is  $1/(1+u-ue^z)$ . Its dominant pole is  $L_u = \log(1+1/u)$  with residue  $-1/(1+u)$ ; the other poles are  $L_u + 2\pi i j$ . The same positive-sum and saddle proof works uniformly for real  $u$  in a compact interval around 1. It gives

$$T(n, k; u) = \frac{n! f(r)^k e^{L_u r/2}}{(1+u) L_u^{n+1} r^n \sqrt{2\pi k b}} (1 + O_K(k^{-1})). \quad (7.25)$$

In the ratio of (7.25) to its value at  $u = 1$ , put  $u = e^{t/\sqrt{n}}$ . Direct Taylor expansion of the elementary function  $\log(L/L_{e^v})$  has first derivative  $1/(2L)$  and second derivative  $(1-L)/(4L^2)$  at zero. Hence

$$\log \mathbb{E} \exp \left( \frac{t(C - n/(2L))}{\sqrt{n}} \right) \longrightarrow \frac{t^2(1-L)}{8L^2}.$$

This proves the Gaussian limit by real moment generating functions; no differentiation of an unspecified asymptotic error is used.

For the sharper moment statements, use an exact auxiliary mixture. Let  $Q$  have probabilities proportional to  $\begin{bmatrix} n \\ q \end{bmatrix} \{q\} F_q$ ,  $k \leq q \leq n$ , and put  $D = n - Q$ . Conditional on  $Q = q$ , the distribution of

$C$  is the block-count distribution  $J_q$  in a uniform ordered set partition of  $q$  elements, because its probability generating polynomial is  $F_q(u)/F_q$ . This specifies an exact distributional mixture; no canonical matrix statistic is being claimed for  $D$ .

Marking  $D$  with a fresh variable  $\xi$  replaces  $H_{n,k}(z)$  by  $H_{n,k}(\xi z)$ . Applying the coefficient proof separately to  $zH'_{n,k}(z)$  and  $z^2H''_{n,k}(z)$  gives

$$\mathbb{E}D = \frac{Lr}{2} + O_K(k^{-1}), \quad \text{Var}(D) = \frac{Lr}{2} + O_K(k^{-1}).$$

Here the differentiated analytic kernel estimates follow from Lemma 7.2; the exponentially small Fubini replacement error is also valid for each of these nonnegative weighted sums. The exact identities

$$F'_q(1) = \frac{F_{q+1} - F_q}{2}, \quad F''_q(1) = \frac{F_{q+2} - 5F_{q+1} + 2F_q}{4}$$

follow from the Stirling recurrence or direct differentiation of the generating function. Using (7.5), they yield

$$\mathbb{E}J_q = \frac{q+1}{2L} - \frac{1}{2} + O(q\vartheta^q), \quad \text{Var}(J_q) = \frac{(q+1)(1-L)}{4L^2} - \frac{1}{4} + O(q^2\vartheta^q).$$

Since  $Q \geq k$ , the exponential errors stay negligible after mixing. The laws of total expectation and total variance now give (7.23)–(7.24).  $\square$

[Added 3 October 2026, batch 85B: the auxiliary variable  $D = n - Q$  of this proof is manuscript 01's *Stirling defect*  $\Delta_{n,k}$  (with  $Q = Q_{n,k}$ ). Manuscript 01 proves more than its two moments:  $\Delta_{n,k}$  converges in law to  $\text{Poisson}(Lr/2)$ , and its probability generating function is  $e^{v(\xi-1)}(1 + O(k^{-1}))$  uniformly for complex  $\xi$  in compact sets, with an expansion to every fixed order (Theorem 16.2). The two displayed moments follow from it. Both manuscripts state that no canonical matrix statistic is claimed for this variable.]

[Added 6 October 2026, batch 108: at  $n/k = 2$  Report 169 (Part IV) proves (7.23) and (7.24) again, term for term, from an expansion uniform for complex column weights  $|u - 1| \leq 0.02$  (Theorem 41.1, Remark 41.2), and adds the joint limit of the column count with the number of cells with entry at least 2, which is  $\text{Poisson}$  of mean  $Lr = 2v$  and independent of the column count. That number is a statistic of the matrices themselves; it is not  $D$ , and no relation between the two is claimed (Remark 41.3).]

The compact restriction  $K \subseteq (1, \infty)$  is essential in all these statements. At  $k = n$  the Stirling kernel collapses and  $T(n, n) = F_n$  exactly. At the coverage window  $n/k \sim \log k$  the saddle leaves compact sets. Neither boundary is covered by the uniformity asserted in Theorem 7.1.

## 8 Prescribing the number of columns changes the threshold

The preceding models allow the column count to fluctuate. A different coverage scale appears when it is prescribed to be much smaller than the number of rows. The exact stars-and-bars identities underlying this model are classical [4, Proposition 28]; the following limit follows from direct uniform estimates on them.

**Theorem 8.1** (Fixed-column coverage). *Let  $k \rightarrow \infty$ , let  $m = m(k) \geq 1$  be integers with  $m = o(k)$ , and let  $n = n(k) \geq 1$ . Choose uniformly a  $k \times m$  matrix of nonnegative integers of total sum  $n$ , conditioned on no zero column. Let  $M$  count its zero rows and put*

$$h = \frac{n}{k}, \quad \theta = m \log \left( 1 + \frac{n}{km} \right). \quad (8.1)$$

If  $\theta - \log k \rightarrow s \in \mathbb{R}$ , then

$$M \xrightarrow{d} \text{Poisson}(e^{-s}), \quad \mathbb{P}(M = 0) \rightarrow \exp(-e^{-s}).$$

Thus an exact parametrization of the critical mass, up to integer rounding, is

$$n = km \left( \exp \left( \frac{\log k + s}{m} \right) - 1 \right). \quad (8.2)$$

*Proof.* First allow zero columns, and call the zero-row count  $M_0$ . There are  $\binom{n+km-1}{n}$  matrices. If  $j$  designated rows are zero, deleting them and comparing these binomial coefficients gives, with  $N = km$ ,

$$\mathbb{E}[(M_0)_j] = (k)_j \prod_{a=1}^{jm} \frac{N - a}{N + n - a}. \quad (8.3)$$

Fix  $j$  and take  $k \geq 2j$ . For  $f(a) = \log(N - a) - \log(N + n - a)$  one has  $f(0) = -\log(1 + n/N)$  and

$$f'(a) = -\frac{n}{(N - a)(N + n - a)}.$$

The mean value theorem and summation imply

$$\sum_{a=1}^{jm} (f(a) - f(0)) = O_j \left( \frac{nm^2}{N(N + n)} \right) = O_j \left( \frac{mh}{k(m + h)} \right).$$

Also  $\log(k)_j = j \log k + O_j(k^{-1})$ . Therefore

$$\log \mathbb{E}[(M_0)_j] = j(\log k - \theta) + O_j \left( \frac{1 + \min(m, h)}{k} \right).$$

For  $x \geq 0$ ,  $\log(1 + x) \geq (\log 2) \min(1, x)$ , so  $\theta \geq (\log 2) \min(m, h)$ . At the assumed critical scale, the error is  $O_j((\log k)/k)$ . The factorial moments tend to  $e^{-js}$ ; the finite Taylor/Bonferroni argument in Lemma 6.1 proves the Poisson limit.

It remains to justify conditioning. If  $m = 1$ , the sole column cannot be zero because  $n > 0$ . If  $m \geq 2$ , the probability that one designated column is zero is at most

$$\prod_{a=1}^k \frac{N - a}{N + n - a} \leq \left( \frac{N}{N + n} \right)^k = \exp(-k\theta/m).$$

The union bound gives probability at most  $m \exp(-k\theta/m)$  for any zero column. Its logarithm tends to  $-\infty$ , since  $m = o(k)$  and  $\theta = \log k + O(1)$ . Conditioning on an event whose probability tends to one changes the law of any statistic in total variation by a quantity tending to zero. Thus  $M$  and  $M_0$  have the same limit. Rounding  $n$  in (8.2) changes  $\theta$  by at most  $1/k$ .  $\square$

**Corollary 8.2** (A shifted window). *Suppose  $m/(\log k)^2 \rightarrow \gamma \in (0, \infty)$  and  $n/k - \log k \rightarrow s$ . In the preceding fixed-column model,*

$$M \xrightarrow{d} \text{Poisson} \left( e^{-s+1/(2\gamma)} \right), \quad \mathbb{P}(M = 0) \rightarrow \exp \left( -e^{-s+1/(2\gamma)} \right). \quad (8.4)$$

*Proof.* Expanding (8.1) gives

$$\theta = h - \frac{h^2}{2m} + O(h^3/m^2) = \log k + s - \frac{1}{2\gamma} + o(1).$$

Apply Theorem 8.1 with this shifted constant.  $\square$

Other regimes follow from the exact parametrization. If  $(\log k)^2 = o(m)$  and  $m = o(k)$ , the usual critical mass  $k(\log k + s) + o(k)$  returns. If  $m \sim \gamma \log k$ , the leading critical mass is  $\gamma(e^{1/\gamma} - 1)k \log k$ ; the smaller order- $k$  centering depends on the actual integer sequence  $m(k)$ , so the exact formula should be retained for that precision. For a fixed positive integer  $m$ , the critical mass is

$$n = me^{s/m}k^{1+1/m} - mk + O(1).$$

In particular, one column requires mass of order  $k^2$  for coverage.

This is a theorem about a prescribed column count. It is not a theorem about a column weight  $u = u_k \rightarrow 0$ ; relating those two ensembles is a separate research problem.

## 9 Exact checks and numerical diagnostics

The accompanying programs reproduce the integer checks and numerical tables independently of the typesetting. Integer recurrences, polynomial identities, and determinants are evaluated exactly. The subsequent asymptotic comparisons use 120 decimal digits of working precision. Their role is to detect implementation errors and illustrate the proved estimates; the numerical comparisons do not constitute proofs.

[Added 3 October 2026, batch 85B: this section uses manuscript 06's delivery names. In the report, `code/verify.py` is `code/06-hankel-verify.py`, `code/make_figures.py` is `code/06-hankel-make_figures.py`, each `data/X` is `data/06-hankel-X`, `code/requirements.txt` is `data/06-hankel-requirements.txt`, the figure is `figures/06-hankel-coupon_window.pdf`, and `README.txt` is replaced by the report's `README.md`, which gives rerun instructions that work with these names. Manuscript 01's checks are described in Section 19.]

### 9.1 Independent exact calculations

The program `code/verify.py` constructs  $A(n, k)$  from the nonzero-column composition recurrence (2.2) and obtains  $T(n, k)$  by binomial inversion (2.3). Separately, it constructs both Stirling triangles and the Fubini numbers by their integer recurrences, and evaluates the positive transform (2.4). These two routes agree for all  $0 \leq n \leq 30$ ,  $0 \leq k \leq 10$ : 341 comparisons for each of  $A$  and  $T$ . All nine supplied rows  $n = 0, \dots, 8$  of A261781 also agree exactly [2].

The recurrence checks cover  $k = 1, \dots, 10$ , with coefficient arrays computed through  $n = 85$ . They verify coprimality of the exact numerator and denominator, the nonzero impulse at degree  $D = k(k+1)/2$ , and the homogeneous recurrence beginning at  $D+1$ , as in Theorem 2.2. The determinant checks verify Theorem 3.1 for  $k = 1, \dots, 5$  and shifts 1, 2, 3; the next-size tail determinants vanish for  $k \leq 4$ . They also check the initial size- $(D+1)$  identity (3.10) at  $u = 1$  for  $k \leq 4$ . Five further cases are exact polynomial identities in the column weight  $u$ , including

$$H_{3,1}(u) = 27u^{14}(1+u)^8(1+2u)^2.$$

These finite checks agree with the general algebraic proofs; they do not replace the uniform arguments in those proofs.

### 9.2 The coverage window

Write  $p_{n,k} = T(n, k)/A(n, k)$  and, using the actual integer value of  $n$ , set

$$h = \frac{n}{k}, \quad \eta = ke^{-h}, \quad p_0 = e^{-\eta}, \quad p_1 = e^{-\eta} \left( 1 + \frac{(1-L)h\eta - (1+h)\eta^2}{2k} \right).$$

Thus  $p_0$  is the moving-mean Poisson approximation and  $p_1$  includes the first correction in Theorem 5.1. Table 3 gives the signed relative errors  $\delta_0 = p_{n,k}/p_0 - 1$  and  $\delta_1 = p_{n,k}/p_1 - 1$ . Here  $n$  is the nearest integer to  $k \log k$ , so the target window coordinate is zero but its actual value  $s = n/k - \log k$  is retained. This avoids assigning integer-rounding effects to the asymptotic error.

| $k$  | $n$  | Actual $s$ | $p_{n,k}$      | $\delta_0$                | $\delta_1$                |
|------|------|------------|----------------|---------------------------|---------------------------|
| 20   | 60   | +0.004268  | 0.339714114626 | $-8.04855 \times 10^{-2}$ | $-4.60223 \times 10^{-3}$ |
| 50   | 196  | +0.007977  | 0.356935471459 | $-3.74272 \times 10^{-2}$ | $-9.74476 \times 10^{-4}$ |
| 100  | 461  | +0.004830  | 0.361879493451 | $-2.10377 \times 10^{-2}$ | $-3.02538 \times 10^{-4}$ |
| 500  | 3107 | -0.000608  | 0.365694896898 | $-5.33335 \times 10^{-3}$ | $-1.86151 \times 10^{-5}$ |
| 1000 | 6908 | +0.000245  | 0.366903156844 | $-2.89783 \times 10^{-3}$ | $-5.38847 \times 10^{-6}$ |

Table 3: Coverage diagnostics at the rounded central window. The displayed probabilities and errors are rounded from 120-digit evaluations; both approximations use the actual  $n$ .

The probability calculation uses explicit error accounting. Put  $E_n = B\rho^{n-1}$ , with the constants of Theorem 4.1. For  $1 \leq j < k$  define

$$\tilde{m}_j = \binom{k}{j} \frac{c_{k-j} d_{k-j}^n}{c_k d_k^n}, \quad m_j = \mathbb{E} \binom{M}{j}.$$

Whenever  $E_n < 1$ , the dominant-pole estimate gives

$$|m_j - \tilde{m}_j| \leq \frac{2E_n}{1 - E_n} \tilde{m}_j.$$

Odd and even Bonferroni sums, enlarged by the summed error on their terms, therefore give analytic lower and upper bounds. The computation stops when the last retained moment is below  $10^{-70}$ , or at  $j = k$ , where the finite inclusion-exclusion sum is complete. All 15 tabulated cases at target  $s = -1, 0, 1$  are supplied in the data files. Five cases with  $n \leq 200$  were also compared against independently computed exact integer ratios. Thirty additional evaluations check the uniform pole bound against exact unrestricted coefficients.

The truncation and pole-replacement errors are thus bounded analytically. The decimal endpoints themselves use high-precision floating arithmetic and have *not* been certified by directed rounding. The stated numerical precision is a diagnostic setting, not an interval-arithmetic certificate.

### 9.3 The dense diagonal and two correction coefficients

For A261784 define the normalized exact count

$$\mathcal{R}_N = \frac{T(2N, N)}{c_* d_*^N (N!)^2 / N},$$

with  $c_*, d_*$  from (7.20) and (7.19). The all-orders expansion (7.17), specialized at  $\lambda = 2$ , gives

$$\mathcal{R}_N = 1 + \frac{\beta_1}{N} + \frac{\beta_2}{N^2} + O(N^{-3}), \quad \beta_1 = \beta.$$

There is an exact coefficient rule for the second constant:

$$\beta_1 = E_1(2) - \frac{1}{8}, \quad \beta_2 = E_2(2) - \frac{E_1(2)}{8} + \frac{1}{128}. \quad (9.1)$$

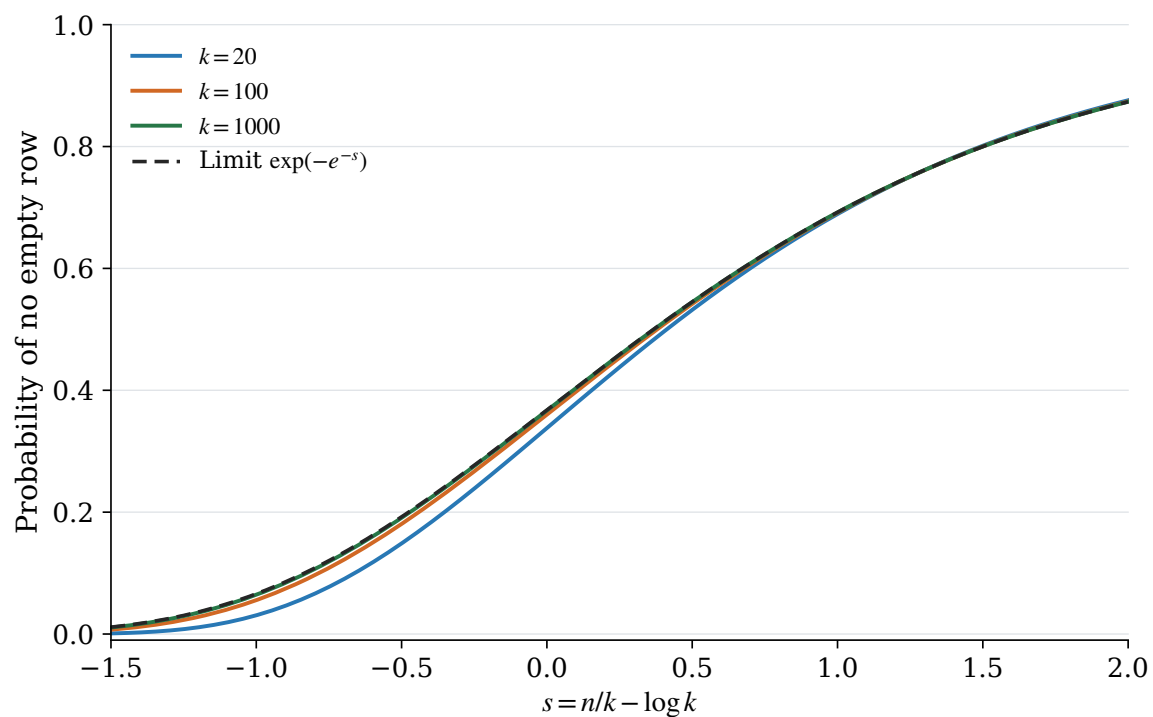


Figure 1: The empty-row transition. Finite-height curves connect evaluations at integer masses for  $k = 20, 100, 1000$ ; the dashed curve is  $\exp(-e^{-s})$ . The horizontal coordinate always uses the actual  $s = n/k - \log k$ . The probabilities are evaluated by the bounded Bonferroni calculation described above.

Here  $E_1, E_2$  are obtained from the finite Gaussian-moment rule (7.16); the last two terms arise from the central-binomial expansion. Evaluating that rule, independently of the diagonal integer data, gives

$$\begin{aligned}\beta_1 &= -0.521806701556482663889144103069\dots, \\ \beta_2 &= 0.128288828088705778459796704234\dots\end{aligned}$$

Thus  $\beta_2$  is a derived coefficient rather than a fitted constant. The finite arithmetic expression used for its evaluation is included in `code/verify.py`.

To display the convergence, define

$$\begin{aligned}U_N &= N(\mathcal{R}_N - 1), & V_N &= N^2 \left( \mathcal{R}_N - 1 - \frac{\beta_1}{N} \right), \\ W_N &= N^3 \left( \mathcal{R}_N - 1 - \frac{\beta_1}{N} - \frac{\beta_2}{N^2} \right).\end{aligned}$$

Theorem 7.1 and its all-orders refinement imply  $U_N \rightarrow \beta_1$ ,  $V_N \rightarrow \beta_2$ , and boundedness of  $W_N$ . The numerator  $T(2N, N)$  in every row of Table 4 is computed as an exact integer from (2.4); only the normalization uses floating arithmetic.

| $N$ | $U_N$       | $V_N$      | $W_N$       |
|-----|-------------|------------|-------------|
| 5   | -0.49620219 | 0.12802256 | -0.00133133 |
| 10  | -0.50899740 | 0.12809304 | -0.00195793 |
| 20  | -0.51539780 | 0.12817802 | -0.00221621 |
| 30  | -0.51753296 | 0.12821232 | -0.00229519 |
| 50  | -0.51924187 | 0.12824171 | -0.00235602 |
| 75  | -0.52009661 | 0.12825702 | -0.00238568 |
| 100 | -0.52052405 | 0.12826482 | -0.00240032 |

Table 4: Dense-diagonal diagnostics from exact integer counts. The first two columns approach the independently derived correction coefficients; the final column records the next scaled remainder.

At  $N = 100$ , the signed relative error after including  $\beta_1/N$  is  $1.28938 \times 10^{-5}$ ; after also including  $\beta_2/N^2$  it is  $-2.41288 \times 10^{-9}$ . These are finite numerical comparisons, not uniform error constants for the asymptotic theorem.

[Added 3 October 2026, batch 85B: manuscript 01's Table 7 reports the same comparisons with the opposite sign convention, approximation divided by exact count minus one; at  $N = 100$  it lists  $-1.289 \times 10^{-5}$  and  $+2.413 \times 10^{-9}$ , the same two numbers to first order. Its exact counts  $a_N$ ,  $N \leq 200$ , come from its own program.]

## 9.4 Reproduction and data

The package includes the complete L<sup>A</sup>T<sub>E</sub>X source, compiled article, both Python programs, their pinned dependencies, the figure in PDF and PNG form, and CSV/JSON outputs. From the package root, `python3 code/verify.py` regenerates the checks and tables; `python3 code/make_figures.py` regenerates the plot. The `README.txt` gives the dependency and L<sup>A</sup>T<sub>E</sub>X build commands. The programs use no network access after their dependencies are installed. The supplied run records report Python 3.12.14, SymPy 1.14.0, mpmath 1.3.0, and matplotlib 3.10.8.

[Added 3 October 2026, batch 85B: in this report the compiled article is a build of the merged text, not manuscript 06's delivered PDF, and the names are those listed at the start of this section.]

Both programs write unprefixed files under their output root, which defaults to the directory above `code/`, and `06-hankel-make_figures.py` imports its sibling as `verify`; they must therefore be rerun on a copy with the delivered names restored, as the report README describes, never in the report directory. At the intake's rerun on a copy (Python 3.13.5) `verify.py` passed in about 5 s of Python time and the figure program took about 74 s, against the 0.3 s and 2.3 s of the delivered README; the six CSV files were byte-identical, five JSON files equal apart from line endings, and the PNG figure pixel-identical.]

## 10 Further research: precise questions and useful next steps

The following are proposed research problems motivated by the results above. They are not assertions that every formulation is open throughout the literature. The source audit was narrower than that. Each question identifies a statement not proved in this article and a concrete way in which its resolution would extend the present results.

**Research question 1** (A uniform approximation across the triangle). Can one obtain relative asymptotics for  $T(n, k)$  that remain valid from the near-diagonal boundary  $n - k = o(k)$  through the dense rays and into the coverage window  $n/k = \log k + O(1)$ ?

There are two distinct boundaries to resolve. As  $n/k \downarrow 1$ , the saddle  $r$  tends to zero, the variance parameter  $b$  tends to zero, and the compact-set saddle proof no longer applies. At the endpoint,  $T(n, n) = F_n$  exactly. As  $n/k$  grows, the saddle escapes to infinity and the kernel estimates must hold on expanding disks. A useful first deliverable is a uniform formula for  $n = k + \ell$  with  $\ell$  growing through the range where a fixed-defect expansion ceases to be accurate. A second is a proved overlap region in which the positive-transform saddle estimate and the empty-row probability estimate agree, including their relative errors. Such a result would turn the separate formulas in this article into a controlled asymptotic description of a much larger part of A261781.

[Added 6 October 2026, batch 108: Report 169 (Part IV) asks for a variable row-density regime,  $k/n$  in a compact subset of  $(0, 1)$ , and for the limits toward sparse and nearly unit row occupancy (its question 2). For the unmarked count the compact part is Theorem 7.1, and the two limits are this question; Research question 18 keeps only its marked form.]

**Research question 2** (Wider windows and higher coverage corrections). For which explicit ranges of  $s_k = n/k - \log k$  can the Poisson–Charlier expansion of Section 5 be extended, and can all its correction polynomials be generated with a uniform summable remainder?

The theorem here keeps  $\eta = ke^{-n/k}$  in a fixed compact subset of  $(0, \infty)$ . If  $\eta$  grows, the factorial-moment sums are concentrated at growing indices, so the same fixed-disk argument does not directly give a useful relative estimate for the small probability  $M = 0$ . If  $\eta$  tends to zero, a different error scale is appropriate. Expanding the exact ratios  $c_{k-j}d_{k-j}^n/(c_k d_k^n)$  suggests a constructive series in Poisson–Charlier polynomials, but the admissible joint growth of  $j$ ,  $\eta$ , and  $h$  must be proved. A satisfactory result would state both an explicit range and a remainder that remains smaller than the probabilities or distances being approximated.

[Added 3 October 2026, batch 85B: for the single event  $M = 0$  and bounded  $s$ , the second half of this question is answered by manuscript 01: Theorem 13.1 gives every correction polynomial of  $\mathbb{P}(M = 0)$  by a finite algorithm, with a uniform remainder  $O((1+h)^{J+1}/k^{J+1})$  after order  $J$ . The question stays open for the whole law of  $M$  (the generating function (5.3) and the probability-mass expansion beyond the first correction) and for every window wider than bounded  $s$ .]

**Research question 3** (Joint local limits and residual dependence). Can the Gaussian–Poisson convergence in Section 6 be strengthened to a joint local limit theorem with an explicit first correction?

For example, one can ask for a uniform approximation to  $\mathbb{P}(C = c, M = m)$  for integer  $c$  in a growing central range and  $m$  in a specified range, and for the corresponding law conditioned on  $M = 0$ . The present moment-generating-function proof establishes distributional independence after normalization, but it does not control the Fourier integral away from the origin. A local theorem requires both this off-axis control and a lattice analysis. The first mixed correction would quantify the small dependence that survives before taking the limit, and would refine the order-one column-moment formulas on dense rays. This is a natural setting in which to compare joint Edgeworth and Poisson–Charlier expansions.

[Added 6 October 2026, batch 108: the analogue on the ray  $n/k = 2$ , with the number of cells with entry at least 2 in place of the zero-row count, is Research question 19 (Report 169); there too only the distributional limit is proved (Theorem 41.1).]

**Research question 4** (Small column weights and equivalence of ensembles). How does the coverage transition change when the column weight  $u = u_k$  tends to zero, and when does conditioning its weighted model on  $C = m$  preserve the corresponding asymptotic laws?

For any fixed  $m$ , conditioning on  $C = m$  removes the factor  $u^m$ , so the resulting distribution is exactly the prescribed-column model. The analytic difficulty is making that conditioning asymptotically uniform when its probability may be small. The weighted pole estimate in Section 4 is uniform on compact positive weight intervals; it does not supply a uniform spectral gap as  $u \downarrow 0$ . The explicit critical parameter  $m \log(1 + n/(km))$  and its shift at  $m \asymp (\log k)^2$  provide testable targets for a joint saddle analysis in  $z$  and  $u$ . Extending the prescribed-column theorem from  $m = o(k)$  to comparable or larger column counts is another concrete part of this problem.

**Research question 5** (A combinatorial explanation of determinant positivity). Is there a direct combinatorial model for the coefficients of  $H_{k,s}(u)$ , or a sign-reversing involution proving the product in Theorem 3.1?

The entries of the determinant count matrices, but the determinant itself is a signed sum of products of such counts. Its coefficientwise positivity is presently explained by spectral factorization. A direct model would explain the factors  $\binom{k}{j}^j$  and the powers of  $(1 + u)^m - u^m$ , and might expose refinements that the one-variable spectrum does not reveal. The negative smaller minor exhibited in Section 3 rules out a blanket appeal to total positivity. One should seek a model specific to these maximal minors, or a precise classification of which other minors have a definite sign.

[Added 5 October 2026, batch 98: manuscript 08, the source of Part III, asks the same question (its question 8, printed only here) and adds which arithmetic factors a successful model would have to explain: the gcd multiplicities, the degree lost at the trivial root of unity, and the positive factors  $(1 + u)^q - u^q$  (Section 25); it stresses that the target is the maximal determinant itself, since positivity of all smaller Hankel minors is not available as a general shortcut. Its zero analysis does not answer the question, which stays open. The classification of other minors is continued by Research question 15.]

**Research question 6** (Recurrences in finite characteristic). What is the minimal eventual recurrence of the column  $T(n, k)$  over  $\mathbb{F}_p$ , and how does it depend on  $k$  and the prime  $p$ ?

Over characteristic zero every pole survives and all root sets are disjoint. Reduction modulo a prime can merge roots, make multiplicities inseparable, or cancel numerator and denominator factors.

The determinant product isolates the possible exceptional primes through binomial coefficients, powers of 2, and factors  $2^m - 1$ . It therefore supplies an arithmetic guide, but does not by itself classify the remaining rank or the reduced rational function at each exceptional prime. An exact classification could yield new divisibility and periodicity statements for the triangle, with fast recurrence-based proofs suitable for individual OEIS entries.

**Research question 7** (The limiting distribution of determinant zeros). After removing the factors at  $u = 0$  and  $u = -1$ , what is the limiting zero-counting measure of  $H_{k,s}(u)$  as  $k \rightarrow \infty$ ?

Every remaining zero has real part  $-1/2$  and is the image of a root of unity under  $\zeta \mapsto 1/(\zeta - 1)$ . Its multiplicity is controlled by the weighted gcd sum over pairs  $1 \leq i < j \leq k$ . Determining the normalized measure therefore reduces to a specific arithmetic counting problem, rather than an unknown root-location problem. Useful targets include an explicit density along the vertical line, tail estimates for zeros with large imaginary part, and the contribution of roots of small order. The answer should specify whether the zero measure is normalized by the full degree or by the degree remaining after the two exceptional factors are removed.

[Added 5 October 2026, batch 98: answered in Part III. Normalized by the degree remaining after the two exceptional factors are removed, the zero measure converges to the image on  $\operatorname{Re} u = -\frac{1}{2}$  of the Cauchy law of scale  $\frac{1}{2}$ , with density  $2/(\pi(1 + 4y^2))$  in the imaginary coordinate, and the Kolmogorov distance is  $\sim (9/\pi^2)(\log k)/k$  (Theorem 24.1, Corollary 28.3). Roots of order  $m$  carry the normalized mass  $3 \log k / (\zeta(2)\psi(m)k) + O_m(1/k)$  each (Corollary 25.4); the tails at height  $\asymp k$  follow a divisor-sum law, matched to the Cauchy tail at every intermediate height (Theorems 29.2 and 30.2); and under the full degree the limit is a three-component mixture determined by  $\lim s/k^2$  (Theorem 24.3). Finer questions, among them the second term of the distance and the last few zeros, are Research questions 9–15.]

**Research question 8** (Which column models have the same coverage law?). Which replacements for the unrestricted column generating function  $(1 - z)^{-k} - 1$  retain the window  $n = k(\log k + s)$ , and which alter its first correction or its scale?

Natural examples impose bounded entries, bounded column sums, or a weight depending on column size. The row symmetry needed for exact factorial moments may survive even when the root-of-unity spectrum does not. One can aim for sufficient analytic conditions on the dominant pole and its row-deletion ratios that imply a Poisson law, then identify which derivatives determine the first correction. A theorem formulated this way would separate properties special to matrix compositions from properties shared by a wider class of symmetric occupancy models.

## 10.1 A practical order of attack

For a continuation in ProveIt, the most direct algebraic project is to formalize the reduced denominator and its no-cancellation proof, followed by the resultant and maximal Hankel determinant identities. These statements are finite algebra and have explicit base conventions. They provide a manageable route to a proof-assistant-checked contribution without first formalizing saddle-point analysis.

[Added 3 October 2026, batch 85B: manuscript 01 names the same first target (Section 20). Nothing of either manuscript is formalized; the only formal neighbour is the Lean Fubini generating function cited in Section 1, and this paragraph is a proposal, not a plan of the repository.] [Added 5 October 2026, batch 98: manuscript 08 extends the same first stage by the cyclotomic multiplicities, the exact degree identities and the full-support CDF comparisons, then the primitive-lattice error and the elementary CDF bound (Research question 16); also a proposal only.]

For analytic work, the highest-value next targets are the near-diagonal boundary and a proved overlap with the coverage regime. The positive Stirling transform avoids cancellation and gives an explicit starting point in both cases. The small-weight problem is a second substantial direction, because the prescribed-column result already identifies a change of scale that a uniform two-variable theory must explain. The computation scripts in the accompanying package can supply regression examples for these projects, but every extension needs a new proof of its claimed uniformity and error bounds.

## Part II

# Finite spectra, uniform asymptotics, and two Poisson limits

## 11 Provenance, scope and notation of Part II

[Added 3 October 2026, batch 85B.] Part II was added in the write of batch 85B. It is built from manuscript 01 of batch 85 and prints everything that manuscript adds to Part I. Where the two manuscripts prove the same statement, the statement is printed once, in Part I, with a bracketed note naming manuscript 01's version; Table 5 lists those second routes. Everything in Part II is written in Part I's letters (Table 2).

**Standing assumption:**  $u = 1$ . Manuscript 01 never weights columns. *Every statement of Part II is at column weight  $u = 1$ :* its  $A(n, k)$ ,  $T(n, k)$ ,  $c_k$ ,  $d_k$ ,  $F_k(z; 1)$  and  $M$  are the unweighted objects of Part I, and its generating variable  $\xi$  in Section 16 marks the Stirling defect, not columns. No weighted analogue of a Part II result is claimed. Several of its proofs use  $u = 1$  explicitly: the residue  $-a_k/(2k)$ , the bound  $a_k \geq \frac{1}{2}$ , the Fubini pole at  $L = \log 2$  and the constants of Section 13 are the  $u = 1$  values of Part I's  $-a_{k,u}/((1+u)k)$ ,  $a_{k,u} \geq b$ ,  $L_u$  and  $\mu_k(u)$ ,  $d_k(u)$ .

### 11.1 The manuscript

Manuscript 01 of batch 85 is the archive `OEIS_Matrix_Compositions_Research.zip` (wrapper directory `matrix_compositions_research/`, 26 files), which arrived in commit `317c1ce2e` and was placed in this report by commit `fa2f3e419`. Its article is titled *Matrix compositions: finite spectra, uniform asymptotics, and two Poisson limits*, subtitled *Proofs for OEIS A261781 and all-orders refinements for A261784* (22 pages, dated 3 October 2026). Its title page and PDF metadata read “Research report prepared for Vladimir Reshetnikov”; it names no author and no tool. Its pin is the repository tree revision `d68b65ea064084fb121df9bb431b21d086f7be81` (2026-10-03, 17:08), at which it read the series-and-transseries overview `Analysis/Transseries/docs/series-and-transseries/README.md` (blob `7bdc83aeb`) as motivation for its inversion section; its indexed searches for A261781 and A261784 returned nothing and exposed a different revision, `ce37e13f4` [9].

It was written without sight of manuscript 06, the source of Part I, which arrived thirteen minutes later in commit `345284ac8`. The two texts share 0.7% (01 in 06) and 0.5% (06 in 01) of their word 8-grams, almost all of them bibliography entries. Every number they share agrees: the constants  $r$ ,  $b$ ,  $v$ ,  $d_*$ ,  $c_*$ ,  $\beta_1$ ,  $\beta_2$  in about seventy printed digits, computed by two independent programs, and the coverage probabilities they both tabulate.

**Status of manuscript 01, as delivered.** It is a proof-bearing research report, not a Lean-certified or independently refereed publication. The two OEIS conjecture labels were present in the entry it inspected. The A261784 leading asymptotic was already posted; it is not claimed as a new prediction, and the entry is not said to label it a conjecture. Priority for the additional uniform results is not established by its bounded literature search. It separates the *proof* of a statement from *priority* for that statement, and says that no uninspected repository theorem is assumed and no Lean certification is inherited merely by association.

## 11.2 What is new, and a correction

Manuscript 01's abstract says that it proves the two conjectures posted in A261781, the recurrence order  $k(k+1)/2$  of column  $k$  and the limit  $d_{k+1} - d_k \rightarrow 1/L$ ; its delivered README and its OEIS draft (Section 22) say the same. That claim is narrowed here.

- A homogeneous constant-coefficient recurrence of order  $\binom{k+1}{2}$  for column  $k$ , with the product denominator  $Q_{k,1}(z) = \prod_{j=1}^k (2(1-z)^j - 1)$ , is Proposition 25 of Munarini, Poneti and Rinaldi [4] (printed page 16), published in 2009, eight years before the conjecture; their Remark 26 prints the cases of two and three rows, the first of which is manuscript 01's example (2.8). Manuscript 01 cites Propositions 1, 12, 13 and 29 of that paper, but not Proposition 25 or Remark 26. What it proves that is new is the *minimality*: the product is the reduced denominator, so no eventual constant-coefficient recurrence of smaller order exists, even over  $\mathbb{C}$ . Manuscript 06 states this correctly (Section 1 and Remark 2.3).
- The limit  $d_{k+1} - d_k \rightarrow 1/L$  is immediate from the closed form of [4, Proposition 13], as both manuscripts say. Manuscript 01's genuine strengthening is that the increments *increase strictly* and stay below  $1/L$  (Theorem 12.1).
- Manuscript 01 says that Louchard's paper [6] was identified through A261780 but not retrieved. That remains a true statement about its own preparation and its non-claim stands; but the paper is retrievable, and manuscript 06 inspected the author's long version and found no matching coverage-window statement (Section 1). The intake confirmed that the author's publication list carries it as item 99 but did not read it.

The generating functions and the positive Stirling–Fubini transform are Munarini, Poneti and Rinaldi's, and both manuscripts credit them; the leading A261784 law,  $d_*$  and the decimal value of  $c_*$  are Kotěšovec's [3].

New in manuscript 01 relative to manuscript 06, and printed below: strict monotonicity of the increments (Section 12); every fixed order of the coverage probability (Section 13); the entire Stirling kernel with its differential recurrence, its third polynomial and a second exact algorithm (Section 14); a recursion for the cumulants (Section 15); the Poisson law of the Stirling defect (Section 16); the third diagonal correction and fifty-digit constants (Section 17); an asymptotic inversion with integer threshold brackets, which is an instance of the repository's inversion apparatus (Section 18); exact-ratio coverage and defect tables (Sections 16 and 19); and its research questions and proof audit (Sections 20–22).

## 11.3 Manuscript 01's second routes to Part I's results

Results of Part I that manuscript 01 does not have: the column weight  $u$  throughout; the initial impulse (Remark 2.4); the whole Hankel section (Theorem 3.1); the uniform constant of Theorem 4.1;

| Part I                       | Manuscript 01's version and route                                                                                                                                                                                               |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (2.1)–(2.3), (2.4)           | the same classical identities at $u = 1$ , credited to [4]; the Stirling transform proved by the same geometric expansion and finite difference                                                                                 |
| Theorem 2.2                  | the same theorem at $u = 1$ , same proof (Remark 2.3); the recurrence written from $n > D$                                                                                                                                      |
| (2.9), Corollary 2.5         | the same constants and expansions of $d_k, c_k$ ; adds strict monotonicity (Theorem 12.1); lacks (2.11)                                                                                                                         |
| Theorem 4.1                  | the bound $(k - 1)\rho^{n+1}$ , $u = 1$ only (Remark 4.2)                                                                                                                                                                       |
| (5.4), Corollary 5.3         | the same expansion of $\mathbb{P}(M = 0)$ and Poisson limit, by Bonferroni truncation of inclusion–exclusion (note after Theorem 5.1); no generating function in complex $w$ , no probability-mass or total-variation statement |
| (7.5)                        | the same lemma, same proof                                                                                                                                                                                                      |
| (7.7), Lemma 7.2             | the untruncated kernel $W_n$ , with no hypothesis on $k$ (Remark 7.3, Section 14)                                                                                                                                               |
| Theorem 7.1, (7.16), (7.17)  | the same theorem and prescription; tails by $ W_n(w)  \leq e^{ w }$ (notes after the proof of Theorem 7.1 and after (7.17))                                                                                                     |
| Corollary 7.4, (9.1)         | the same constants and $\beta_1, \beta_2$ ; adds $\beta_3$ and all orders (Section 17)                                                                                                                                          |
| ED, Var $D$ in Corollary 7.5 | implied by the Poisson law of $\Delta_{n,k} = D$ (Theorem 16.2)                                                                                                                                                                 |
| Table 3, Table 4             | Tables 8 and 7, from its own exact counts; equal where they overlap                                                                                                                                                             |

Table 5: Results of Part I that manuscript 01 proves as well, and its version of each. None is reprinted in Part II.

the coverage generating function, the probability-mass and total-variation expansions and the moving-mean remark of Section 5; the joint Gaussian–Poisson law of Section 6; the column central limit theorem on dense rays (Corollary 7.5); and the prescribed-column threshold of Section 8.

## 12 Strict monotonicity of the growth-constant increments

Extend  $d_k$  to real  $x > 0$  by  $d(x) = (1 - 2^{-1/x})^{-1}$ , so that  $d_k = d(k)$ .

**Theorem 12.1** (Strengthened increment statement). *The sequence  $d_{k+1} - d_k$ ,  $k \geq 1$ , is strictly increasing, is always below  $1/L$ , and tends to  $1/L$ .*

*Proof.* Put  $t = L/(2x)$ . Since  $d(x) = \frac{1}{2} + \frac{1}{2} \coth t$ ,

$$d'(x) = \frac{1}{L} \left( \frac{t}{\sinh t} \right)^2. \quad (12.1)$$

This lies strictly between 0 and  $1/L$ . The function  $t/\sinh t$  is strictly decreasing for  $t > 0$ , because  $\sinh t - t \cosh t < 0$ ; the latter expression vanishes at 0 and has derivative  $-t \sinh t < 0$ . As  $x$  increases,  $t$  decreases; thus  $d'$  increases strictly to  $1/L$ . Integrating  $d'$  over  $[k, k + 1]$  proves the claim.  $\square$

*Remark 12.2* (Further details from manuscript 01). (i) The inverse of (2.3) is  $A(n, k) = \sum_{j=0}^k \binom{k}{j} T(n, j)$ : choose the nonzero rows. Also  $T(n, k) = 0$  for  $k > n$ , and the rational expression (2.1) at  $u = 1$  equals 1 for  $k = 0$ , as it must. (ii) The bijection with the sorted-word compositions of A261780 and A261781 lists the row label  $i$  as many times as the entry in row  $i$  of

a column, in increasing order; column boundaries become part boundaries, and the construction is reversible. (iii) At the unique smallest-modulus pole  $1 - a_k$  only the  $j = k$  summand of (2.3) contributes, with residue  $-a_k/(2k)$ , so its coefficient contribution is  $\frac{a_k}{2k}(1 - a_k)^{-n-1} = c_k d_k^n$ . (iv) The alternating sum (2.3) explains why cancellation is a numerical concern when all rows must be occupied; the positive transform (2.4) is an exact computational alternative to it.

### 13 Every fixed order in the coverage window

For real  $x \rightarrow \infty$  the exact constants have the expansions

$$\log d(x) = \log \frac{x}{L} + \frac{L}{2x} - \frac{L^2}{24x^2} + O(x^{-4}), \quad (13.1)$$

$$\log c(x) = -\log(2L) - \frac{L}{2x} - \frac{L^2}{24x^2} + O(x^{-4}), \quad (13.2)$$

where  $c(x) = (d(x) - 1)/(2x)$ ; these are the identities for  $\log d_m$  and  $\log c_m$  in the proof of Theorem 5.1, with  $g_0(x) = -L^2 x^2/24 + O(x^4)$ , and both differences from the leading logarithms are analytic in  $1/x$  at 0. Let  $\mathcal{T}_a(x)$  denote the Touchard polynomial, defined by  $\sum_{j \geq 0} j^a x^j / j! = e^x \mathcal{T}_a(x)$ .

**Theorem 13.1** (All fixed orders for coverage). *Let  $k \rightarrow \infty$  through integers, with  $h = n/k = \log k + s$  and  $\eta = ke^{-h} = e^{-s}$ , where  $s$  stays in a fixed compact real interval. For every fixed  $J \geq 0$  there are polynomials  $\mathcal{Q}_1, \dots, \mathcal{Q}_J$  in  $h$  and  $\eta$ , with coefficients polynomial in  $L$  over  $\mathbb{Q}$ , such that*

$$\frac{T(n, k)}{A(n, k)} = \mathbb{P}(M = 0) = e^{-\eta} \left[ 1 + \sum_{j=1}^J \frac{\mathcal{Q}_j(h, \eta; L)}{k^j} + O\left(\frac{(1+h)^{J+1}}{k^{J+1}}\right) \right], \quad (13.3)$$

uniformly in such  $s$ . Each  $\mathcal{Q}_j$  has degree at most  $j$  in  $h$ , and

$$\mathcal{Q}_1(h, \eta; L) = \frac{1}{2}((1-L)h\eta - (h+1)\eta^2).$$

For  $J = 1$  this is (5.4).

*Proof.* Put  $\epsilon = 1/k$  (a local letter; Part I's  $\epsilon$  in Section 7.4 is  $1/n$ ). Using the analytic expansions behind (13.1)–(13.2), form the formal power series in  $\epsilon$

$$\Psi(j, h, \epsilon) = \sum_{i=0}^{j-1} \log(1 - i\epsilon) + \log \frac{c(\epsilon^{-1} - j)}{c(\epsilon^{-1})} + \frac{h}{\epsilon} \left( \log \frac{d(\epsilon^{-1} - j)}{d(\epsilon^{-1})} + j\epsilon \right). \quad (13.4)$$

By (5.2) and Theorem 4.1,  $\mathbb{E}(M)_j = \eta^j \exp \Psi(j, h, 1/k)$  up to the exponentially small spectral errors. The cancellation in the last parentheses removes its linear leading term. Every coefficient of  $\Psi$  is a polynomial in  $j$ ,  $h$ ,  $L$ , and its dependence on  $h$  is at most linear. Hence  $[\epsilon^i]e^\Psi$  is a polynomial in  $j$ ,  $h$ ,  $L$  of degree at most  $i$  in  $h$ . Replace each monomial  $j^a$  in that coefficient by  $\mathcal{T}_a(-\eta)$  to obtain  $\mathcal{Q}_i$ ; this is the summation  $\sum_j (-\eta)^j j^a / j! = e^{-\eta} \mathcal{T}_a(-\eta)$  of the inclusion–exclusion series  $\mathbb{P}(M = 0) = \sum_j (-1)^j \mathbb{E}(M)_j / j!$ .

To justify the operation, truncate at  $j \leq C \log k$ . Taylor's formula applied to the finitely many logarithms and to (13.1)–(13.2), continued to the required order, bounds the moment remainder by

$$\frac{\eta^j}{k^{J+1}} C_J (1+h)^{J+1} (1+j)^{C'_J}$$

for constants  $C_J, C'_J$  on the chosen  $s$ -interval. The exponential of the small discarded logarithm is uniformly bounded in this range. Summing against  $1/j!$  preserves the stated error, and the Bonferroni inequalities bound the tail superalgebraically, as in the second route noted after Theorem 5.1. This proves both the algorithm and the uniform remainder.  $\square$

The appearance of  $\log k$  inside the coefficients, through  $h$ , is intrinsic to the transition scale. Theorem 13.1 is an all-orders Poincaré expansion of the single probability  $\mathbb{P}(M = 0)$ ; it does not assert convergence as  $J \rightarrow \infty$ , nor a classification of exponentially small terms, and it does not extend Part I's expansion of the whole law of  $M$  beyond the first correction (see the note to Research question 2 in Section 10).

## 14 The entire Stirling kernel

For  $n \geq 1$  define, with the falling factorial  $(n)_\ell$  of Part I and  $(n)_0 = 1$ ,

$$\chi_{n,\ell} = \frac{\left[ \begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right]}{(n)_\ell^2}, \quad W_n(w) = \sum_{\ell=0}^{n-1} \chi_{n,\ell} w^\ell, \quad (14.1)$$

and set  $\chi_{n,\ell} = 0$  outside  $0 \leq \ell < n$ . By Remark 7.3, Part I's kernel is the truncation  $H_{n,k}(z) = \sum_{\ell \leq n-k} \chi_{n,\ell} (Lz)^\ell$ .

**Proposition 14.1** (Positive coefficient representation). *Let  $f(z) = e^z - 1$  and*

$$\tilde{T}(n, k) = \frac{n!}{2L^{n+1}} [z^n] f(z)^k W_n(Lz). \quad (14.2)$$

For  $n \geq k \geq 1$ ,  $T(n, k) = \tilde{T}(n, k)(1 + O_\vartheta(\vartheta^k))$ .

*Proof.* In (2.4) replace  $F_q$  using (7.5), use  $\{q\}_k = q! [z^q] f(z)^k / k!$  and put  $q = n - \ell$ . The resulting exact main sum is

$$\frac{1}{2n! L^{n+1}} \sum_{\ell=0}^{n-1} \left[ \begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right] ((n-\ell)!)^2 L^\ell [z^{n-\ell}] f(z)^k,$$

which is (14.2). Positivity of every summand transports the relative error.  $\square$

**Lemma 14.2** (Bounds and an exact differential recurrence). *For  $0 \leq \ell < n$ ,  $0 \leq \chi_{n,\ell} \leq 1/\ell!$ ; consequently  $|W_n(w)| \leq e^{|w|}$  for every complex  $w$ . With  $D_w = w \, d/dw$ ,*

$$(n+1)^2 W_{n+1}(w) = ((n+1 - D_w)^2 + nw) W_n(w), \quad W_1(w) = 1. \quad (14.3)$$

*Proof.* The first-kind coefficient  $\left[ \begin{smallmatrix} n \\ n-\ell \end{smallmatrix} \right]$  is the  $\ell$ th elementary symmetric polynomial of  $1, 2, \dots, n-1$ . Each of its  $\binom{n-1}{\ell}$  summands is at most  $(n-1)_\ell$ . Hence

$$\chi_{n,\ell} \leq \frac{(n-1)_\ell^2}{\ell! (n)_\ell^2} \leq \frac{1}{\ell!}.$$

The first-kind recurrence gives

$$\chi_{n+1,\ell} = \frac{(n+1-\ell)^2}{(n+1)^2} \chi_{n,\ell} + \frac{n}{(n+1)^2} \chi_{n,\ell-1}.$$

Multiplying by  $w^\ell$  and summing proves (14.3).  $\square$

**Theorem 14.3** (Uniform entire kernel expansion). *For each fixed  $J \geq 0$  there are rational polynomials  $P_0, \dots, P_J$ , with  $P_0 = 1$  and  $\deg P_r \leq 2r$ , such that*

$$W_n(w) = e^{w/2} \left( \sum_{r=0}^J \frac{P_r(w)}{n^r} + O_{J,K}(n^{-J-1}) \right) \quad (14.4)$$

*uniformly on every compact  $K \subset \mathbb{C}$ , with no condition relating  $n$  to any  $k$ . The expansion may be differentiated any fixed number of times, uniformly on compact sets. The polynomials are those of (7.15):*

$$\begin{aligned} P_1(w) &= \frac{w^2}{12} - \frac{w}{2}, & P_2(w) &= \frac{w^4}{288} - \frac{w^2}{12}, \\ P_3(w) &= \frac{w^6}{10368} + \frac{w^5}{576} + \frac{w^4}{480} - \frac{w^3}{24} - \frac{w^2}{12}. \end{aligned} \quad (14.5)$$

*Proof.* This is a coefficient-level proof with a uniform remainder. Let  $\Sigma_j(n) = \sum_{a=1}^{n-1} a^j$ , regarded as its Faulhaber polynomial in  $n$ . With  $\epsilon = 1/n$  put  $\mathbf{a}_j(\epsilon) = \epsilon^{2j} \Sigma_j(1/\epsilon)$ . This is a polynomial beginning in degree  $j-1$ , and  $\mathbf{a}_1(\epsilon) = (1-\epsilon)/2$ . For a fixed integer  $\ell \geq 0$  define

$$\mathbf{e}_\ell(\epsilon) = [t^\ell] \exp \left( \sum_{j=1}^{\ell} \frac{(-1)^{j-1}}{j} \mathbf{a}_j(\epsilon) t^j \right), \quad \mathbf{r}_\ell(\epsilon) = \mathbf{e}_\ell(\epsilon) \prod_{a=0}^{\ell-1} (1 - a\epsilon)^{-2}. \quad (14.6)$$

Newton's identity for elementary symmetric functions shows that  $\mathbf{r}_\ell(1/n) = \chi_{n,\ell}$  whenever  $n > \ell$ . In particular  $\mathbf{r}_\ell(0) = 1/(2^\ell \ell!)$ .

The exponent in (14.6) is  $t/2 + \sum_{r \geq 1} \epsilon^r \mathbf{b}_r(t)$  with  $\deg \mathbf{b}_r \leq r+1$ . Thus at total order  $\epsilon^r$  the part multiplying  $e^{t/2}$  has degree at most  $2r$  in  $t$ , and extracting  $t^\ell$  converts  $t^q e^{t/2}$  into  $2^q(\ell)_q/(2^\ell \ell!)$ . The other factor of  $\mathbf{r}_\ell$  has logarithm  $2 \sum_{r \geq 1} \frac{\epsilon^r}{r} \sum_{a=0}^{\ell-1} a^r$ , whose coefficient at order  $r$  is a polynomial in  $\ell$  of degree  $r+1 \leq 2r$ . Therefore

$$[\epsilon^r] \mathbf{r}_\ell(\epsilon) = \frac{\Pi_r(\ell)}{2^\ell \ell!}, \quad \Pi_r \in \mathbb{Q}[\ell], \quad \deg \Pi_r \leq 2r; \quad (14.7)$$

for example  $\Pi_0 = 1$  and  $\Pi_1(\ell) = \ell(\ell-4)/3$ , the polynomial  $d(d-4)/3$  of the proof of Lemma 7.2.

Next, the Taylor remainders are bounded uniformly in  $\ell$ . There are absolute  $c, C > 0$  such that, for  $|\epsilon| \leq c/(\ell+1)^2$  and  $2 \leq j \leq \ell$ ,

$$|\mathbf{a}_j(\epsilon)| \leq C|\epsilon|^{j-1}. \quad (14.8)$$

For completeness: Faulhaber's formula writes  $\mathbf{a}_j$  as  $\frac{1}{j+1} \sum_{a=0}^j \binom{j+1}{a} \mathbf{B}_a \epsilon^{j-1+a}$ , with Bernoulli numbers  $\mathbf{B}_a$ . The coefficients of  $t/(e^t - 1)$  obey  $|\mathbf{B}_a|/a! \leq C_0$  by a Cauchy bound on a fixed circle of radius 1. Since  $j!/(j+1-a)! \leq j^{a-1}$  for  $a \geq 1$ , the sum of the terms after the first is bounded by a geometric series in  $j|\epsilon|$ .

For  $\ell \geq 1$ , apply Cauchy's coefficient bound in (14.6) on  $|t| = 2\ell$ . The  $j=1$  term contributes at most  $\ell + O(1)$  to the real part of the exponent, while (14.8) bounds the sum of the terms  $j \geq 2$  by a constant, because  $|\epsilon|\ell^2$  is bounded; the product  $\prod(1 - a\epsilon)^{-2}$  is bounded for the same reason. Stirling's factorial estimate then gives

$$\sup_{|\epsilon| \leq c/(\ell+1)^2} |\mathbf{r}_\ell(\epsilon)| \leq \frac{C\sqrt{\ell+1}}{2^\ell \ell!}; \quad (14.9)$$

for  $\ell = 0$  this is immediate. Cauchy's Taylor-remainder estimate at  $\epsilon = 1/n$  consequently yields

$$\chi_{n,\ell} = \frac{1}{2^\ell \ell!} \sum_{r=0}^J \frac{\Pi_r(\ell)}{n^r} + O_J \left( \frac{(\ell+1)^{2J+5/2}}{2^\ell \ell! n^{J+1}} \right) \quad (14.10)$$

whenever  $\ell \leq \delta_0 \sqrt{n}$ , for a sufficiently small absolute  $\delta_0 > 0$  and all sufficiently large  $n$ .

Multiply (14.10) by  $w^\ell$  and sum. On  $|w| \leq V$  the remainder is summable uniformly. Beyond  $\delta_0 \sqrt{n}$  the bound  $\chi_{n,\ell} \leq 1/\ell!$  makes the original tail smaller than every power of  $n^{-1}$ , and the same holds for the polynomially weighted Taylor coefficients in (14.7). Hence

$$\mathbf{W}_n(w) = \sum_{r=0}^J \frac{1}{n^r} \sum_{\ell \geq 0} \frac{\Pi_r(\ell)}{\ell!} \left( \frac{w}{2} \right)^\ell + O_{J,V}(n^{-J-1}).$$

The inner sum is  $e^{w/2}$  times a polynomial of degree at most  $2r$ , by the Touchard identity. This proves (14.4); expanding (14.6) gives  $P_1, P_2, P_3$ . Cauchy's derivative estimate on a slightly larger compact disk gives the differentiated expansions.  $\square$

**A finite exact coefficient algorithm.** The polynomials can be computed without a symbolic parameter  $\ell$ . For a requested order  $J$ , compute  $e_\ell(n) = \lfloor \frac{n}{n-\ell} \rfloor$  for  $0 \leq \ell \leq 2J$  by

$$e_0(n) = 1, \quad \ell e_\ell(n) = \sum_{j=1}^{\ell} (-1)^{j-1} e_{\ell-j}(n) \Sigma_j(n), \quad (14.11)$$

expand  $e_\ell(n)/(n)_\ell^2$  in powers of  $1/n$  to order  $J$ , and use  $\deg P_r \leq 2r$ :

$$P_r(w) = \sum_{m=0}^{2r} w^m \sum_{\ell=0}^m \frac{(-1/2)^{m-\ell}}{(m-\ell)!} [n^{-r}] \frac{e_\ell(n)}{(n)_\ell^2}. \quad (14.12)$$

Here  $[n^{-r}]$  is the coefficient in the expansion at infinity, not a coefficient of the polynomial numerator alone. Manuscript 01's program (`code/01-two-poisson-verify.py`) runs (14.11)–(14.12) exactly over  $\mathbb{Q}$  and checks (14.3) coefficientwise through  $n = 80$ . This is a second finite route to Part I's algorithm (7.12)–(7.14). [Added 3 October 2026, batch 85B: the intake checked  $P_3$  numerically: at  $w = 1.3$ ,  $n^3(e^{-w/2}\mathbf{W}_n(w) - 1 - P_1/n - P_2/n^2)$  is  $-0.22070$ ,  $-0.22010$ ,  $-0.21981$  for  $n = 400, 800, 1600$ , against  $P_3(1.3) = -0.21951$ , with the expected  $O(1/n)$  approach.]

## 15 Complements on compact rays

Manuscript 01 writes the leading factor of Theorem 7.1 as

$$\mathcal{B}(n, k) = \frac{n! f(r)^k e^v}{2L^{n+1} r^n \sqrt{2\pi k b}}, \quad T(n, k) = \mathcal{B}(n, k) \left[ 1 + \sum_{j=1}^J \frac{E_j(\lambda)}{k^j} + O_{J,K}(k^{-J-1}) \right]. \quad (15.1)$$

It adds two observations and a recursion.

**The variance  $b$ .**  $b = \kappa_2 = \lambda(1 + r - \lambda)$  is the variance of the positive-integer random variable with probabilities proportional to  $r^j/j!$ ,  $j \geq 1$ , which explains its positivity; the proof needs only the monotonicity of  $r/(1 - e^{-r})$ .

**A recursion for the cumulants.** With independent variables  $z, a$ , define

$$\mathcal{K}_1(z, a) = a, \quad \mathcal{K}_{j+1} = z \partial_z \mathcal{K}_j + a(1 + z - a) \partial_a \mathcal{K}_j. \quad (15.2)$$

Then  $\kappa_j = \mathcal{K}_j(r, \lambda)$ . This follows from  $\mathcal{D}\lambda = \lambda(1 + z - \lambda)$  for  $\lambda(z) = \mathcal{D} \log f(z) = z/(1 - e^{-z})$ , before evaluation at the saddle. One must not differentiate a specialized formula while holding  $\lambda$  fixed. (For example  $\mathcal{K}_3 = a(2a^2 - 3az - 3a + z^2 + 3z + 1)$ , which is the expanded  $\kappa_3$  printed after Theorem 7.1.) Together with (14.12) and (7.16), this specifies every  $E_j$  by finite rational operations and the evaluation of  $r$ ; in fact  $E_j$  is a rational function of  $r, \lambda, L$  whose denominators are powers of  $b$  and  $\lambda$ .

*Remark 15.1* (Floors and endpoints). The ratio in Theorem 7.1 is the *actual*  $n/k$ . For  $n = \lfloor \lambda_0 k \rfloor$ , replacing it prematurely by  $\lambda_0$  inside exponential factors can lose a nonvanishing factor. The theorem is not uniform as  $\lambda \downarrow 1$  or  $\lambda \rightarrow \infty$ ; at the boundary  $n = k$  the exact identity is  $T(n, n) = F_n$ , and the coverage window, where  $\lambda \sim \log k$ , needs the separate proof of Section 5. Manuscript 01's optional program `code/01-two-poisson-ray_spotchecks.py` tests the expansion at  $\lambda = 3/2, 3$  and  $5$  for  $40 \leq k \leq 200$  with 75-digit arithmetic (`data/01-two-poisson-ray_spot_checks.csv`); the relative error after the third correction is between  $2.38 \times 10^{-12}$  and  $1.63 \times 10^{-9}$  in absolute value at the seven tested points.

## 16 The Stirling defect: a second Poisson limit

The positive identity (2.4) supplies an auxiliary probability space. It should not be confused with the zero-row statistic  $M$ , which vanishes identically on the objects counted by  $T$ .

**Definition 16.1.** For  $n \geq k \geq 1$ , define  $Q_{n,k}$  on  $\{k, \dots, n\}$  by

$$\mathbb{P}(Q_{n,k} = q) = \frac{\begin{bmatrix} n \\ q \end{bmatrix} \{q\} F_q}{\sum_{p=k}^n \begin{bmatrix} n \\ p \end{bmatrix} \{p\} F_p}, \quad \Delta_{n,k} = n - Q_{n,k}. \quad (16.1)$$

$\Delta_{n,k}$  is the *Stirling defect* of the transform.  $Q_{n,k}$  and  $\Delta_{n,k}$  are Part I's  $Q$  and  $D$  of Section 7.6.

**Theorem 16.2** (Poisson defect and analytic generating functions). *Suppose  $n, k \rightarrow \infty$  and  $n/k \rightarrow \lambda_0 \in (1, \infty)$ . Then*

$$\Delta_{n,k} \xrightarrow{d} \text{Poisson}\left(\frac{Lr(\lambda_0)}{2}\right). \quad (16.2)$$

*More precisely, uniformly for complex  $\xi$  in compact sets and for  $n/k$  in compact subsets of  $(1, \infty)$ ,*

$$\mathbb{E} \xi^{\Delta_{n,k}} = e^{v(\xi-1)} (1 + O(k^{-1})), \quad v = \frac{Lr(n/k)}{2}. \quad (16.3)$$

*There is an expansion through every fixed order, locally uniformly in  $\xi$ , and the same holds for each fixed derivative with respect to  $\xi$ .*

*Proof.* Marking  $n - q = \ell$  in the main transform replaces  $W_n(Lz)$  by  $W_n(\xi Lz)$ . Lemma 14.2 and positivity bound the Fubini replacement error, also for bounded complex  $\xi$  by using absolute values. The proof of Theorem 7.1, with  $\xi$  in a compact set, now replaces  $e^{Lr/2}$  by  $e^{\xi Lr/2}$ ; this uses Theorem 14.3 on compact sets of  $\mathbb{C}$ . Division by the unmarked coefficient gives (16.3); the denominator's leading factor is positive and bounded away from zero after normalization, so the division is uniform. Replacing  $P_p(Lre^{\tau y})$  by  $P_p(\xi Lre^{\tau y})$  in (7.16) gives every higher order. Uniform holomorphy on a larger compact set permits differentiation. Coefficient extraction at  $\xi = 0$  yields convergence of each probability to the corresponding Poisson mass, and these limiting masses sum to one.  $\square$

Differentiating (16.3) once and twice at  $\xi = 1$  gives  $\mathbb{E}\Delta_{n,k} = v + O(k^{-1})$  and  $\text{Var } \Delta_{n,k} = v + O(k^{-1})$ , the two moments of Corollary 7.5. The theorem explains why only a bounded distance below the top Stirling index contributes at leading order, although the positive sum ranges from  $k$  to  $n$ : the factor  $e^v = e^{Lr/2}$  in (15.1) is exactly the aggregate contribution of the limiting Poisson cloud. The report does not assert that  $\Delta_{n,k}$  is a canonical statistic on unadorned matrices; finding such a statistic is a separate combinatorial problem (Section 20).

[Added 5 October 2026, batch 98: the collection's report **a122399-surjection-diagonal**, Part II (its Theorems 13.1 and 15.1, labels **a122:pd:thm:trichotomy** and **a122:pd:thm:allorders**), proves the analogous law for a different array: for a uniformly chosen pair of an ordered partition of an  $m$ -set and a map of an  $n$ -set to its blocks, the defect ( $m$  minus the number of blocks) tends to a Poisson law of parameter  $e^{-c}/2$  at  $n = m(\log m + c)$ , with an expansion through every fixed order summed by Touchard polynomials. There the factor  $1/2$  comes from defects that are pairs (its Proposition 16.1); a dated note of its batch-98 write, on its source audit, names the same mechanism, pairs weighted by  $1/2!$ , for the parameter  $Lr/2$  of Theorem 16.2. No theorem is shared. (Wording corrected after the independent check of 7 October 2026; the note first read “the defect (the number of missing blocks,  $m$  minus the block count)” and “its source note names the same mechanism”. The defect  $m - K$  is not a number of missing blocks, and the delivered source audit of that report does not mention this report: the sentence is in the write's note about that audit.)]

For  $n = 2N$ ,  $k = N$  the limiting mean is  $v = 0.55230808135934145323\dots$ . Direct evaluation of the positive weights in (16.1) gives Table 6.

| $\ell$ | $N = 10$    | $N = 50$    | $N = 200$   | $e^{-v}v^\ell/\ell!$ |
|--------|-------------|-------------|-------------|----------------------|
| 0      | 0.592485680 | 0.578956893 | 0.576452261 | 0.575619699          |
| 1      | 0.314681461 | 0.317313214 | 0.317769736 | 0.317919411          |
| 2      | 0.078997588 | 0.086058333 | 0.087361728 | 0.087794730          |
| 3      | 0.012377926 | 0.015394446 | 0.015970554 | 0.016163246          |
| 4      | 0.001344681 | 0.002042736 | 0.002184003 | 0.002231773          |
| 5      | 0.000106192 | 0.000214396 | 0.000238309 | 0.000246525          |

Table 6: Convergence of  $\mathbb{P}(\Delta_{2N,N} = \ell)$  to its Poisson limit (manuscript 01).

## 17 A261784: the third correction and every further one

**Corollary 17.1** (Diagonal expansion to every order). *For every fixed  $J \geq 0$ ,*

$$a_N = c_* d_*^N \frac{(N!)^2}{N} \left( 1 + \sum_{j=1}^J \frac{\beta_j}{N^j} + O_J(N^{-J-1}) \right). \quad (17.1)$$

*Every  $\beta_j$  is a rational function of  $L$  and  $r$ , specified exactly by the prescription (7.16) (with (15.2) and (14.12)) and the central binomial expansion. Besides  $\beta_1 = \beta$  and  $\beta_2$  of Part I,*

$$\beta_3 = -0.00244353842284859500374038095590013033570627782694\dots \quad (17.2)$$

*Proof.* At  $\lambda = 2$  one has  $f(r) = r/(2-r)$  and  $b = 2(r-1)$ . Substitute these in (15.1) and use

$$\frac{(2N)!}{(N!)^2} = \frac{4^N}{\sqrt{\pi N}} \left( 1 - \frac{1}{8N} + \frac{1}{128N^2} + \frac{5}{1024N^3} + O(N^{-4}) \right). \quad (17.3)$$

With  $E_j = E_j(2)$  this gives

$$\beta_1 = E_1 - \frac{1}{8}, \quad \beta_2 = E_2 - \frac{E_1}{8} + \frac{1}{128}, \quad \beta_3 = E_3 - \frac{E_2}{8} + \frac{E_1}{128} + \frac{5}{1024}, \quad (17.4)$$

the first two being (9.1). The same multiplication determines every further coefficient. All computations are finite; the displayed decimals are evaluations of these exact prescriptions, not fitted constants.  $\square$

Manuscript 01's ninety-digit evaluation gives, truncated to fifty decimals,

$$\begin{aligned} r &= 1.59362426004004009232304187587516024178900242481885\dots, \\ d_* &= 12.85568917366612131932660300054233483234414963116428\dots, \\ c_* &= 0.25886492311146555025177523170232718427705044811049\dots, \\ E_1(2) &= -0.39680670155648266388914410306946945484974842970590\dots, \\ E_2(2) &= 0.07087549039414544547365369135067624979402488769624\dots, \\ E_3(2) &= 0.00463313773232960649210026876816463100456049274216\dots, \\ \beta_1 &= -0.52180670155648266388914410306946945484974842970590\dots, \\ \beta_2 &= 0.12828882808870577845979670423435993165024344140947\dots \end{aligned}$$

[Added 3 October 2026, batch 85B: the intake compared these with manuscript 06's independent seventy-digit values in `data/06-hankel-diagonal_constants.json`;  $r, b, v, d_*, c_*, \beta_1$  and  $\beta_2$  agree in every printed digit.] [Added 6 October 2026, batch 108: Report 169's exact expressions for  $\beta_1, \beta_2, \beta_3$  (Section 38) agree with  $\beta_1, \beta_2$  and (17.2) to within  $10^{-50}$ , as they must (Remark 35.2).]

Let  $a_N^{[i]}$  denote (17.1) truncated after  $\beta_i/N^i$ , with  $i = 0$  the leading approximation. Table 7 reports  $a_N^{[i]}/a_N - 1$ , using exact integer  $a_N$  and 90-digit working precision, and Figure 2 plots the same quantities for more  $N$ . (Part I's Table 4 uses the reciprocal convention.) The very small

| $N$ | $i = 0$                 | $i = 1$                 | $i = 2$                  | $i = 3$                  |
|-----|-------------------------|-------------------------|--------------------------|--------------------------|
| 10  | $+5.363 \times 10^{-2}$ | $-1.350 \times 10^{-3}$ | $+2.063 \times 10^{-6}$  | $-5.117 \times 10^{-7}$  |
| 20  | $+2.645 \times 10^{-2}$ | $-3.289 \times 10^{-4}$ | $+2.844 \times 10^{-7}$  | $-2.917 \times 10^{-8}$  |
| 50  | $+1.049 \times 10^{-2}$ | $-5.183 \times 10^{-5}$ | $+1.905 \times 10^{-8}$  | $-7.075 \times 10^{-10}$ |
| 100 | $+5.232 \times 10^{-3}$ | $-1.289 \times 10^{-5}$ | $+2.413 \times 10^{-9}$  | $-4.344 \times 10^{-11}$ |
| 200 | $+2.613 \times 10^{-3}$ | $-3.215 \times 10^{-6}$ | $+3.035 \times 10^{-10}$ | $-2.691 \times 10^{-12}$ |

Table 7: Relative errors of successive asymptotic approximations to A261784 (manuscript 01).

third-order residual at  $N = 200$  is consistent with the proved  $O(N^{-4})$  relative remainder; between  $N = 100$  and  $N = 200$  it falls by a factor of about 16. It does not by itself establish that order, which comes from the uniform analytic proof.

## 18 Asymptotic inversion and integer thresholds

[Added 3 October 2026, batch 85B: this section is an instance of the inversion apparatus of the repository's transseries volume [11]: Proposition 18.1 is a first step of its perturbed inversion, Theorem `p0:thm:perturbed-inversion`, for the core  $2x \log(\alpha x)$ , and Proposition 18.2 is a case of the separation condition of its staircase theorem, `p0:thm:staircase`. Manuscript 01 does not cite

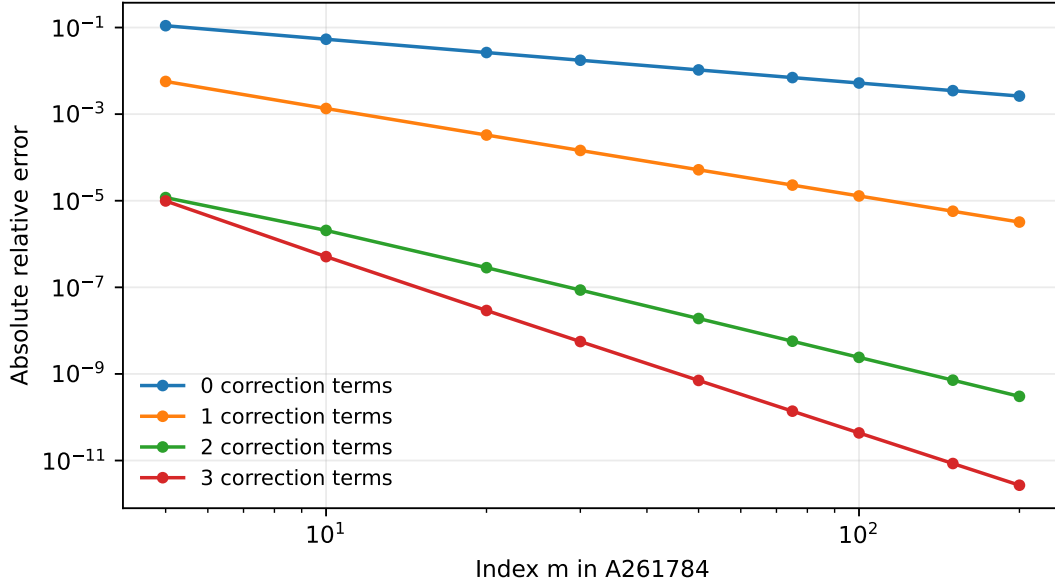


Figure 2: Absolute relative errors from the same exact data as Table 7, with additional intermediate indices (manuscript 01). These are numerical validations, not certified interval error bounds.

either; it was motivated by the volume’s overview README, read at its pin. No novelty is claimed for the method; the constants are specific to A261784.]

A rapidly growing sequence has a natural discrete inverse, but no unique smooth interpolation; the two objects are kept separate.

**A selected smooth comparator.** For fixed  $J \geq 1$  and sufficiently large real  $x$  define

$$\mathcal{A}^{[J]}(x) = c_* d_*^x \frac{\Gamma(x+1)^2}{x} \left( 1 + \sum_{j=1}^J \beta_j x^{-j} \right). \quad (18.1)$$

It is positive and eventually strictly increasing, because the derivative of its logarithm is  $2 \log x + \log d_* + O(x^{-1}) > 0$ . Set

$$\alpha = \frac{\sqrt{d_*}}{e}, \quad \tilde{\beta}_1 = \beta_1 + \frac{1}{6} = -0.35514003488981599722247743640280278818 \dots \quad (18.2)$$

( $\tilde{\beta}_1$  is manuscript 01’s “ $\beta_1$ ”; it is not Part I’s  $\beta_1$ .) Stirling’s expansion and (17.1) give, at integer  $N$ ,

$$\log a_N = 2N \log(\alpha N) + \log(2\pi c_*) + \frac{\tilde{\beta}_1}{N} + O(N^{-2}). \quad (18.3)$$

The cancellation of the  $\log N$  term is specific to the  $(N!)^2/N$  normalization.

**Proposition 18.1** (Lambert- $W$  inverse and first residual correction). *Let  $Y \rightarrow \infty$  and put*

$$\Lambda = \frac{\log Y - \log(2\pi c_*)}{2}, \quad \varpi = W_0(\alpha \Lambda), \quad x_0 = \frac{\Lambda}{\varpi}. \quad (18.4)$$

If  $x_J(Y)$  solves  $\mathcal{A}^{[J]}(x_J) = Y$ , then

$$x_J(Y) = x_0 - \frac{\tilde{\beta}_1}{2x_0(1+\varpi)} + O\left(\frac{x_0^{-2}}{\log x_0}\right). \quad (18.5)$$

Further terms are obtained by substituting a formal correction into the logarithm of (18.1) and cancelling successive residuals.

*Proof.* The definition of  $\varpi$  gives  $\alpha x_0 = e^\varpi$ , so  $x_0 \log(\alpha x_0) = \Lambda$  and  $x_0$  inverts the dominant core exactly. Substitution in the logarithm of (18.1) leaves the residual  $\tilde{\beta}_1/x_0 + O(x_0^{-2})$ . The derivative of the core is  $2(1+\log(\alpha x_0)) = 2(1+\varpi)$ , and the displayed correction cancels the residual to  $O(x_0^{-2})$ ; quadratic errors in the correction are smaller. The derivative stays comparable to  $\log x_0$ , and the mean-value theorem transports the remaining logarithmic residual to the claimed root error.  $\square$

**The discrete threshold.** Define

$$\mathcal{N}(Y) = \min\{N \geq 0 : a_N \geq Y\}. \quad (18.6)$$

The sequence  $a_N$  is strictly increasing. Indeed, pad every matrix counted by  $a_N$  with a new zero row and append two columns, each having a single 1 in that new row. This is an injection into the matrices counted by  $a_{N+1}$ . It is not surjective: a positive one-column matrix with  $N+1$  rows and total  $2N+2$  is not in its image.

**Proposition 18.2** (Residual brackets for integer inversion). *For each fixed  $J \geq 1$  there is a constant  $C_J$  such that, for all sufficiently large  $Y$ , if  $\epsilon_J = C_J x_J(Y)^{-J-1} / \log x_J(Y)$ , then*

$$\lceil x_J(Y) - \epsilon_J \rceil \leq \mathcal{N}(Y) \leq \lceil x_J(Y) + \epsilon_J \rceil. \quad (18.7)$$

In particular  $\mathcal{N}(Y) = \Lambda/W_0(\alpha\Lambda) + O(1)$ .

*Proof.* At integer  $N$  the logarithmic difference between  $a_N$  and  $\mathcal{A}^{[J]}(N)$  is  $O(N^{-J-1})$  by Corollary 17.1. The derivative of  $\log \mathcal{A}^{[J]}$  is bounded below by a positive constant times  $\log x_J$  in a fixed-width interval around  $x_J$ . Choosing  $C_J$  large enough, the model's change across  $\epsilon_J$  dominates that logarithmic error at every integer in the interval. Integers strictly below  $x_J - \epsilon_J$  cannot attain  $Y$ ; every integer at least  $x_J + \epsilon_J$  does. Monotonicity handles integers farther from the interval. Taking ceilings gives (18.7).  $\square$

No unconditional formula  $\mathcal{N}(Y) = \lceil x_J(Y) \rceil$  follows. When the smooth root lies extremely close to an integer, even a tiny asymptotic error can decide which of two consecutive integer thresholds is correct; the bracket is a valid statement that does not hide that ambiguity. Its constants are existential consequences of the asymptotic proof; no numerically certified value of  $C_J$  is claimed.

[Added 6 October 2026, batch 108: Report 169 (Part IV) proves Propositions 18.1 and 18.2 again and carries the inversion two terms further (Proposition 39.1, Theorem 39.2): with  $\tilde{\beta}_2 = \beta_2 - \frac{1}{2}\beta_1^2$  and  $\tilde{\beta}_3 = \beta_3 - \beta_1\beta_2 + \frac{1}{3}\beta_1^3 - \frac{1}{180}$ , the root has the further terms  $-\tilde{\beta}_2/(2x_0^2(1+\varpi))$  and  $-x_0^{-3}(\tilde{\beta}_3/(2(1+\varpi)) + \tilde{\beta}_1^2/(4(1+\varpi)^2) + \tilde{\beta}_1^3/(8(1+\varpi)^3))$ , with error  $O(x_0^{-4}/(1+\varpi))$ ; Remark 39.3 shows that this holds for  $x_J(Y)$ ,  $J \geq 3$ .]

## 19 Manuscript 01's checks

Manuscript 01's program `code/01-two-poisson-verify.py` (delivered as `scripts/verify.py`) uses Python integers for exact combinatorics, SymPy for rational polynomial operations, and mpmath at 90 decimal digits for evaluation. It (1) matches eight displayed rows of A261781 and all fifteen displayed terms of A261784, finite transcriptions marked as such in the source; (2) compares the positive Stirling transform with a separately implemented generating-function recurrence followed by inclusion–exclusion for every  $0 \leq k \leq n \leq 40$ ; (3) computes the reduced rational functions for  $1 \leq k \leq 8$ , checks the denominator degrees and coprimality, and verifies later terms against the extracted recurrence; (4) checks every coefficient of (14.3) through  $n = 80$ , constructs  $P_1, P_2, P_3$  over  $\mathbb{Q}$ , and compares the finite Gaussian algorithm's  $E_1$  with the closed formula (7.4); and (5) computes  $a_0, \dots, a_{200}$  exactly (`data/01-two-poisson-a261784_exact.txt`) and produces the diagonal, coverage and defect tables, which are floating-point tables, not interval-arithmetic certificates. Its record (`data/01-two-poisson-verification.json`) says “ALL CHECKS PASSED”. Finite tests do not prove the infinite statements; they check algebra, indexing, normalization and implementation.

The coverage table takes  $n$  to be the nearest integer to  $k \log k$  and uses the actual  $h = n/k$  and  $\eta = ke^{-h}$  in both approximations; the probabilities in its third column are ratios of exact integers. The correction is not advertised as monotone in accuracy for every small input: the theorem is a

| $k$ | $n$ | Exact coverage | $e^{-\eta}$ | First correction |
|-----|-----|----------------|-------------|------------------|
| 10  | 23  | 0.314344674    | 0.366928440 | 0.319053131      |
| 20  | 60  | 0.339714115    | 0.369449445 | 0.341284787      |
| 40  | 148 | 0.355320143    | 0.371970377 | 0.355818395      |
| 80  | 351 | 0.360509443    | 0.369892970 | 0.360669046      |
| 120 | 574 | 0.359614880    | 0.366349652 | 0.359697754      |

Table 8: The first correction of (5.4) near the centre of the coverage transition (manuscript 01). Rounding  $n$  explains why the leading values are not all exactly  $e^{-1}$ .

uniform asymptotic statement for bounded  $s$ , not an approximation inequality for all  $k$ .

[Added 3 October 2026, batch 85B: at the intake's rerun on a copy, `verify.py -max-m 200` passed all checks in 162s (its README says about 16s), with every output byte-identical to the shipped one or equal apart from line endings and elapsed time; `ray_spotchecks.py`, which imports `verify` and so needs SymPy as well, reproduced its CSV byte for byte in 207s; and the figure program reproduced the PNG pixel for pixel. The 201 computed terms of A261784 were not compared with the OEIS b-file [8], and manuscript 01 does not claim that they were.]

## 20 Further research questions of manuscript 01

Manuscript 01 orders these by proximity to its methods, not by a claim that any is easy or guaranteed to be new. The bracketed notes re-scope them against Part I.

**Uniformity between compact rays and the coverage window.** Prove a single relative asymptotic valid on an expanding interval such as  $1 + \varepsilon \leq n/k \leq \log k + C$ , with an error that remains informative when coverage is exponentially rare. The kernel suggests extending (14.4) from fixed  $w$  to growing  $w = Lr$ , while the saddle must be controlled as  $r \rightarrow \infty$ ; a successful result should reproduce the polynomial corrections of Theorem 13.1, not only the limiting curve. A concrete first

step is a bound for the kernel remainder on  $|w| \leq C \log n$  that keeps its exponential dependence on  $|w|$ . [Added: this is the second deliverable of Research question 1 of Part I; open.]

**The near-diagonal boundary.** At  $n = k$  the answer is  $F_n$ , whereas the saddle variance  $b$  vanishes as  $n/k \downarrow 1$ . Determine expansions uniform when  $\ell = n - k$  is fixed, when  $\ell \rightarrow \infty$  with  $\ell = o(k)$ , and across their transition. The positive Stirling sum has only  $\ell + 1$  summands in the fixed- $\ell$  regime, so exact polynomial descriptions may be feasible before any saddle approximation; matching them to a large- $\ell$  saddle expansion would complete the present analysis at the endpoint. [Added: this is the first deliverable of Research question 1 of Part I; open.]

**A combinatorial realization of the latent defect.** Find a direct decorated-matrix construction whose forgetful map induces exactly the weights (16.1). Ideally the defect would count a familiar local feature and make the factor  $e^{Lr/2}$  visible; such a construction could also explain the polynomials  $P_r$  and the negative first-order term in  $P_1$  without Faulhaber expansion. [Added: open; both manuscripts decline to claim a canonical statistic.] [Added 6 October 2026, batch 108: Report 169 (Part IV) bears on this question without answering it. On the ray  $n/k = 2$  the number  $\mathcal{J}_N$  of cells with entry at least 2, a familiar local feature of the matrices, has a generating function carrying the factor  $e^{Lr(\omega-1/2)}$ , so that weighting each such cell by  $\frac{1}{2}$  removes the factor  $e^{Lr/2}$  from the amplitude;  $\mathcal{J}_N$  is asymptotically Poisson of mean  $Lr$ , twice the mean of the defect (Remark 41.3). It is not the defect, no decorated construction inducing the weights (16.1) is given, and nothing is proved on other rays; the question stays open.]

**Joint laws with the number of columns.** Mark the columns by  $u$ ; before requiring nonzero rows the bivariate generating function is  $F_k(z; u)$  of (2.1). Study the joint distribution of the column count and the zero-row count in the coverage window, and of the column count together with a combinatorial defect on compact rays; the principal pole is governed by  $\log(1 + 1/u)$ , and uniform control near  $u = 1$  is needed before claiming a central limit theorem or asymptotic independence. [Added 3 October 2026, batch 85B: the first half is answered by Part I. Theorem 4.1 gives the uniform control in  $u$  on compact intervals, Lemma 6.1 the Poisson law under column weights, and Theorem 6.2 the joint limit of the normalized column count and the zero-row count, with independence and the conditional Gaussian law given  $M = m$ . On compact rays Corollary 7.5 gives the column central limit theorem and its proof the exact mixture over  $Q_{n,k}$ ; the joint law of the column count and the defect  $\Delta_{n,k}$ , or a combinatorial defect, remains open, as do the local refinements of Research question 3.] [Added 6 October 2026, batch 108: Report 169 (Part IV) answers an analogue of the second half on the single ray  $n/k = 2$ : the column count and the number of cells with entry at least 2 are asymptotically independent, Gaussian and Poisson( $Lr$ ) (Theorem 41.1, Remark 41.2). That statistic is not a realization of the defect (its limiting mean is  $Lr = 2v$ , the defect's is  $v$ ); the joint law with  $\Delta_{n,k}$  or with a combinatorial realization of it, every other compact ray, and the local refinements stay open. (Wording corrected after the independent check of 6 October 2026; the note first said that Part IV “settles one instance of the second half”, with the repeated-cell count “a combinatorial statistic in place of the defect”).]

**Weighted rows, columns, and restricted alphabets.** Replace the identical row weights or the unrestricted column entries by a specified weight system, and determine which features of the recurrence-order proof survive when distinct factors can share poles. The proof suggests a precise obstruction: cancellations and coincident spectra, rather than the mere presence of a rational generating function. A classification for finite sets of permitted column weights would be a useful

structural extension. [Added: related to Research question 8 of Part I, which asks the same about the coverage law; open.]

**Beyond all algebraic orders.** The Fubini generating function has further poles at  $L + 2\pi ij$ . The present proofs bound their total effect exponentially but do not extract their separate saddle contributions. Derive those sectors, with their oscillatory phases, and compare their scale with the late terms of the algebraic expansion; this is where a genuinely multi-sector transseries, rather than a Poincaré series, should enter, and optimal truncation and rigorous remainder bounds would be part of that problem. [Added 3 October 2026, batch 85B: for the Fubini numbers themselves the sectors are exact and convergent, Theorem `q2:thm:fubini` of [11]; the open problem is to carry them through the Stirling transform and the saddle. The report on A260700 [10] lists the same open problem for its sequence.] [Added 6 October 2026, batch 108: Report 169 (Part IV) asks the same question for A261784 (its question 6, printed only here), adding the growth of the coefficients  $\beta_j$  and possible Gevrey bounds, and the role of complex saddles other than the Fubini sectors; it notes that its explicit pole-tail bound (Lemma 36.1) identifies one exponentially small scale but supplies no exponentially improved expansion. Open.]

**Effective certified computation and formalization.** Make the constants in the compact-ray error bound and in (18.7) explicit; interval estimates on the central arc and a certified minor-arc gap could turn the inversion into a proof-producing threshold algorithm. The exact recurrence theorem is a suitable first formalization target: its core is pairwise disjoint simple poles and the equivalence between rational generating functions and eventual linear recurrences. The analytic kernel and saddle theorems are substantially larger tasks and should be scoped separately. [Added: the same target as Part I’s “practical order of attack”; nothing is formalized yet.]

## 21 Hypotheses that cannot be dropped

Manuscript 01 ends with a proof audit, kept here in full.

**Constant versus polynomial coefficients.** The minimality assertion is for homogeneous recurrences with constant coefficients. It is not a lower bound for recurrences with polynomial coefficients in  $n$ , nonlinear recurrences, or recurrences involving several columns  $k$  at once.

**Nonnegative versus strictly positive entries.** Entries are nonnegative; the restrictions are on entire rows and columns, not on each cell. Replacing cells by positive integers changes the generating functions and the sequence.

**The polynomial part of a rational function.** The fixed-spectrum formula for  $A(n, k)$  applies to  $n \geq 1$ , and the recurrence for  $T(n, k)$  starts after the numerator has ceased contributing. Ignoring either point creates incorrect initial conditions.

**Positivity before relative error transport.** The approximation of  $F_q$  is transported through the positive identity (2.4), not through the alternating inclusion–exclusion sum: an exponentially accurate approximation term by term in an alternating sum need not be relatively accurate after cancellation. The coverage calculation instead controls factorial moments and then uses Bonferroni truncation.

**Compactness in the ray theorem.** The uniform constants require  $n/k$  to stay away from 1 and infinity. Neither endpoint is covered by substituting it in (15.1), and the actual rational  $n/k$  must be used when  $n$  is rounded.

**Two different Poisson variables.**  $M$  counts zero rows in uniformly sampled matrices that may have zero rows;  $\Delta_{n,k}$  is an auxiliary variable defined by positive transform weights. Their means, parameter regimes and sample spaces are different.

**Formal Gaussian algebra.** The prescription (7.16) means coefficient extraction first, Gaussian evaluation second; it does not require the integral of the untruncated exponential expression to converge.

**Asymptotic expansion versus convergent series.** Every order  $J$  is fixed before the size parameter tends to infinity. Neither growth estimates for the coefficients as  $J \rightarrow \infty$  nor optimal truncation are asserted.

**Numerics and proof status.** Exact integer checks certify their finite inputs. The 90-digit numerical calculations are independent consistency checks, not rigorous floating-point enclosures or a machine-checked proof of the theorems.

## 22 Manuscript 01's draft OEIS additions (not submitted)

Manuscript 01 includes draft additions to two OEIS entries, shipped as `data/01-two-poisson-OEIS_updates_draft.txt`. *They were not submitted*, by manuscript 01 or by this report; manuscript 01 itself asks for independent review and a stable public manuscript link first, and for the earlier source and conjecture attributions to be preserved. Their substance, in this report's letters:

**A261781.** For  $k > 0$  the reduced denominator of the ordinary generating function of column  $k$  is  $\prod_{j=1}^k (2(1-x)^j - 1)$ , so its minimal eventual homogeneous constant-coefficient recurrence has order  $k(k+1)/2$ : inclusion–exclusion expresses the column generating function as the alternating binomial sum of  $(1-x)^j/(2(1-x)^j - 1)$ , whose  $j$ th summand has simple poles at  $1 - 2^{-1/j}e^{2\pi i\ell/j}$ , and different summands have disjoint pole sets. The fixed-column constants are  $c_k = 1/(2k(2^{1/k} - 1))$  and  $d_k = 1/(1 - 2^{-1/k})$ , and  $d_{k+1} - d_k$  increases strictly to  $1/\log 2$ , with  $d'(x)$  as in (12.1). Attribution should retain the existing conjecture attribution and cite the generating-function source.

[Added 3 October 2026, batch 85B: *before any use* this draft must be corrected as in Section 11.2. Its sentence saying that these statements prove the two claims labelled conjectures overstates them: a recurrence of order  $k(k+1)/2$  with this denominator is [4, Proposition 25], with the case  $k = 2$  in Remark 26, and the limit follows from Proposition 13 there; what is new is the minimality and the strict monotonicity. The shipped file is kept byte-identical and still contains the overstated sentence.]

**A261784.** With  $L = \log 2$  and  $r = 2 + W_0(-2e^{-2})$ , the posted prefactor is  $c_* = 2^{r/2}/(4\pi L\sqrt{r-1})$  and the growth rate  $d_* = 4/(L^2r(2-r))$ ; the relative correction begins  $1 + \beta_1/N + \beta_2/N^2 + \beta_3/N^3$  with the values of Section 17 (the draft's index  $n$  is  $N$  here). The decimals evaluate finite exact prescriptions and are not fitted. The leading asymptotic already in the entry, Kotěšovec's

(18 February 2017, updated 20 April 2024), is not to be presented as new, nor described as labelled a conjecture.

## Part III

# Cauchy limits, arithmetic atoms, and edge laws for the Hankel zeros

## 23 Provenance, scope and notation of Part III

[Added 5 October 2026, batch 98.] Part III was added in the write of batch 98. It is built from manuscript 08 of that batch, which answers Research question 7 of Part I, “The limiting distribution of determinant zeros” (Section 10). After the factors at  $u = 0$  and  $u = -1$  are removed, the zeros of the maximal Hankel determinant  $H_{k,s}(u)$  of Theorem 3.1 converge to the image on the line  $\operatorname{Re} u = -\frac{1}{2}$  of a Cauchy law of scale  $\frac{1}{2}$ . Manuscript 08 also determines the multiplicity of every root of small order, the sharp Kolmogorov rate, the tails at every height, the transition of the absolute moments and the limit under the full-degree normalization. Its Section 2 re-proves Theorem 3.1 by the argument of Section 3; that proof is not reprinted (Remark 24.4). Everything else of manuscript 08 is printed below, in the letters of Table 10. Bracketed notes marked “[Note of the batch-98 write]” and remarks marked “[write]” are the write’s; all other text of Part III is manuscript 08’s.

### 23.1 The manuscript

Manuscript 08 of batch 98 is the archive `a261781_hankel_zero_laws.zip` (30 files in a wrapper directory of the same name), which arrived in commit `2172df76a` and was placed in this report by commit `0fba5167f`. Its article is titled *Cauchy Limits, Arithmetic Atoms, and Edge Laws for Matrix-Composition Hankel Zeros*, subtitled “An explicit continuation of the ProveIt study of OEIS A261781” (29 pages; main file `a261781_hankel_zero_laws.tex`, 2,041 lines, 110 labels; dated October 5, 2026). Its author line and PDF metadata read “OpenAI ChatGPT”, with “An OEIS-related research article”. Its pin is the repository commit `d18416ec7e187a0948248cb3077e37d7b089a8f9` (2026-10-04, 19:16 –0700) [15], at which this report consisted of Parts I and II exactly as they stand: the text of this report did not change between the pin and the batch-98 write. Its source note, shipped as `07-zero-laws-PROVENANCE.md`, names the labels `mxc:thm:hankel` and `mxc:eq:weighted-hankel-product` and the question it answers.

**Status and non-claims of manuscript 08, kept.** Its proofs are mathematical proofs in an unrefereed research article; its computations are independent checks, not proofs of limits and not proof-assistant certificates. No exhaustive search for historical priority was made: targeted repository and public-source searches checked overlap and attribution only. Its proof tools are standard (finite spectral algebra, Möbius inversion, elementary Fourier approximation and real-variable estimates); its contribution is the family-specific theorem set, not those tools. The exact determinant product is prior repository mathematics, which it re-derives for self-containment and credits to this report; the enumeration and the existence of an order- $\binom{k+1}{2}$  recurrence are published ([4, Proposition 25]), and it does not present recurrence existence as an OEIS conjecture newly solved. It claims no Lean or Rocq formalization and no OEIS submission. Its further non-claims are kept where they stand: the

error in (24.6) is absolute, not a uniform relative equivalent (Section 24); total variation is not the right distance (Remark 26.1); normalizing the reduced zeros by the full degree gives a subprobability (Section 26); the smooth correction does not extend to indicators (Section 27); the sharp constant comes without a quantitative error and the finite data lie far below it (Sections 28 and 32); the edge law does not describe the finitely many largest zeros (Section 29); and its questions are “not claims that every formulation has been checked against all literature” (Section 33). Its source note adds that the statements were cross-checked by independent reviews in its own session, whose integration review corrected the range of a primitive angle, the absolute-value convention of the Kolmogorov distance, the nearest-integer notation and the domain of the primitive triangle count.

### 23.2 What Part III adds

Table 9 sets Part III against what Parts I and II already contain.

Table 9: What Part III adds. Research question 7 is answered; its finer continuations are Research questions 9 to 16.

| Topic                       | Parts I and II                                                                                                                                                                            | Part III (manuscript 08 of batch 98)                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location and multiplicities | zeros other than $0, -1$ on $\operatorname{Re} u = -\frac{1}{2}$ ; exact multiplicities at $0$ and $-1$ ; the others “follow by collecting the cyclotomic difference factors” (Section 3) | the cyclotomic factorization (25.2), every order $2 \leq m \leq k-1$ present, the exact count of distinct zeros, and $E_m(k) = k^2 \log k / (\zeta(2)\psi(m)) + O(k^2)$ uniformly in $m$ (Section 25)                                |
| Limit law                   | asked in Research question 7                                                                                                                                                              | the Cauchy law of scale $\frac{1}{2}$ , with the elementary bound $2G_k/\bar{D}_k$ and the sharp Kolmogorov constant $9/\pi^2$ (Theorem 24.1, Corollary 28.3)                                                                        |
| Corrections                 | —                                                                                                                                                                                         | the first correction of smooth statistics (Theorem 27.1); arithmetic CDF corrections at fixed and moving endpoints (Theorems 28.1 and 28.2)                                                                                          |
| Tails                       | asked in Research question 7                                                                                                                                                              | the divisor-sum edge law at height $\asymp k$ (Theorem 29.2), the exact extreme zeros (Proposition 29.3), the expansion of the edge profile at the origin and a uniform transition formula for every height (Theorems 30.1 and 30.2) |
| Moments                     | —                                                                                                                                                                                         | the transition at $p = 1$ , with the constant term at $p = 1$ and the constants above it (Theorem 24.2)                                                                                                                              |
| Normalization               | asked in Research question 7                                                                                                                                                              | the three-component limit of the full zero measure as $s/k^2 \rightarrow \varkappa$ (Theorem 24.3)                                                                                                                                   |

### 23.3 Notation of Part III

Manuscript 08 uses many letters that Parts I and II use for other objects. Part III renames every such letter; the renames are changes of letter only, and no normalization is changed. Objects of manuscript 08 that collide with Latin letters of Parts I and II are written in sans-serif type, as Part II did for  $F_q$  and  $W_n$ ; the three zero measures carry a superscript  $z$ . Table 10 lists every rename

with the false reading the delivered letter would have in this report. The figures of Section 32 are delivered files and keep manuscript 08's letters; their captions translate them.

Table 10: Manuscript 08's symbols and the letters used for them in Part III. "False reading" gives what the manuscript-08 letter means elsewhere in this report.

| Notion                                                           | Part III                                                                   | Ms. 08                                                         | False reading if copied                                                                         |
|------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Hankel size $\binom{k+1}{2}$                                     | written out ( $d$ of Section 3)                                            | $d_k$                                                          | $d_k$ is the growth constant (2.9); in Part III a bare $d$ is only a summation or divisor index |
| $\binom{k+1}{3}$                                                 | $c$ , as in Section 3                                                      | $c_k$                                                          | $c_k$ is the amplitude (2.9)                                                                    |
| Reduced factor; its degree                                       | $R_k(u); D_k$                                                              | $R_k(u); D_k$                                                  | $R_p(w)$ are kernel remainders; $D, D_u, D(\eta)$ are Part I objects                            |
| Exceptional exponents; constant                                  | $A_k, B_{k,s}; C_k$                                                        | $A_k, B_{k,s}; C_k^{(0)}$                                      | $A(n, k)$ is a count, $B, B(u)$ are constants, $C$ counts columns                               |
| Full degree $\deg H_{k,s}$                                       | $\deg H_{k,s}$                                                             | $L_{k,s}$                                                      | $L = \log 2$                                                                                    |
| Difference factor $(1+u)^q - u^q$                                | $U_q(u)$                                                                   | $F_q(u)$                                                       | $F_k(z; u)$ is a generating function                                                            |
| Primitive multiplicity                                           | $E_m(k)$                                                                   | $E_m(k)$                                                       | $E_n(w), E_j(\lambda), E_n$                                                                     |
| Grid weight                                                      | $W_k(q)$                                                                   | $W_k(q)$                                                       | $W_n(w)$ is the Stirling kernel, $W_N$ a diagonal diagnostic                                    |
| Zero measures on angles, on $Y$ , on the line; full zero measure | $\rho_k^z, \nu_k^z, \widehat{\nu}_k^z; \mu_{k,s}^z$                        | $\rho_k, \nu_k, \widehat{\nu}_k; \mu_{k,s}$                    | $\rho$ is the pole ratio, $\nu_k = \mathbb{E}M$ in Section 5, $\mu_k(u)$ an amplitude           |
| Dirac masses                                                     | $\delta_0, \delta_{-1}$                                                    | $\delta_0, \delta_{-1}$                                        | $\delta_k$ below is a displacement                                                              |
| Angular distribution function                                    | $\mathcal{F}_k(t)$                                                         | $F_k(t)$                                                       | $F_k(z; u); F_q$ are Fubini numbers                                                             |
| Kolmogorov distance $d_K(\nu_k^z, C)$                            | $\Delta_k^K$                                                               | $\Delta_k$                                                     | $\Delta_{n,k}$ is the Stirling defect                                                           |
| Absolute moments; their constant                                 | $M_p(k); \mathcal{M}_p$                                                    | $M_p(k); C_p$                                                  | $M$ is the zero-row count, $C_J$ a bracket constant                                             |
| Euler's constant                                                 | $\gamma_E$                                                                 | $\gamma$                                                       | $\gamma = \lim m/(\log k)^2$ in Corollary 8.2                                                   |
| von Mangoldt function                                            | $\Lambda_{\text{vM}}$                                                      | $\Lambda$                                                      | $\Lambda$ is Part II's inverse core                                                             |
| $\lim s_k/k^2; \lim s/k^{p+1}$                                   | $\varkappa; \varkappa_p$                                                   | $\tau; a$                                                      | $\tau$ is a formal variable in (7.16); $a = 1 + u$ in Section 3                                 |
| Sign of a displacement                                           | $\varsigma$                                                                | $\sigma$                                                       | $\sigma_{-1}(n), \sigma_1(n)$ are divisor functions                                             |
| $\pi/\arctan(1/(2y))$ ; its reciprocal                           | $\Omega(y); 1/\Omega(y)$                                                   | $\beta(y); \delta(y)$                                          | $\beta_1, \beta_2, \beta_3$ are diagonal corrections                                            |
| $k/\Omega(y)$                                                    | $\mathbf{m}$                                                               | $M$                                                            | $M$ is the zero-row count                                                                       |
| Tail count                                                       | $Z_k(y)$                                                                   | $\mathcal{T}_k(y)$                                             | $\mathcal{T}_a(x)$ are Part II polynomials                                                      |
| Counts without coprimality                                       | $P_\ell^*(Y), \mathcal{P}^*(K, Q)$                                         | $B_\ell(Y), \mathcal{N}(K, Q)$                                 | $\mathcal{N}(Y)$ is Part II's integer threshold                                                 |
| Primitive congruence count                                       | $P_m(X)$                                                                   | $P_m(X)$                                                       | $P_j(t), P_j(w), P_r(w)$ are polynomials                                                        |
| Fractional-part sum; logarithmic sums                            | $\mathcal{S}_k(t); \mathcal{S}_k^{\text{gap}}, \mathcal{S}_k^{\text{gcd}}$ | $\mathcal{A}_k(t); \mathcal{A}_k^{\log}, \mathcal{B}_k^{\log}$ | $\mathcal{A}^{[J]}(x)$ is a comparator, $\mathcal{B}(n, k)$ a saddle approximation              |
| Harmonic sawtooth sum and auxiliaries; harmonic numbers          | $Q_N(\theta), K_N(\theta), J_b(t); H_N$                                    | $Q_N, K_N, J_b; H_N$                                           | $Q_{n,k}; K$ a compact set; $J, J_q$ ; $H_{k,s}$ the determinant                                |
| Sawtooth; trigonometric bounds; Bernoulli polynomials            | $\overline{B}_1(x); T^\pm; B_3(v)$                                         | $s(x); P^\pm; B_3(v)$                                          | $s$ is the Hankel shift                                                                         |

| Notion                                                         | Part III                                                 | Ms. 08                               | False reading if copied                                                                          |
|----------------------------------------------------------------|----------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------|
| Sawtooth lemma:<br>displacement, exponent,<br>constant, cutoff | $\tilde{\delta}_k, \tilde{\alpha}, \kappa, \mathbf{N}_k$ | $\eta_k, \beta, c, N_k$              | $\eta$ is the Poisson mean, $\beta = \beta_1$ ,<br>$c = \binom{k+1}{3}$ , $N$ the diagonal index |
| Edge density; scaled measures                                  | $-\Xi'(x); \nu_k^e$                                      | $\omega_{\text{edge}}(x); \lambda_k$ | $\omega_j$ are roots of unity, $\lambda = n/k$                                                   |
| Grid error; cotangent sum; its<br>regular part                 | $\text{err}_q; \mathbf{S}_1(q); \mathbf{h}(v)$           | $\epsilon_q; S_1(q); h(v)$           | $\epsilon = 1/k; h = n/k$                                                                        |
| Bernoulli-proof auxiliaries                                    | $\Upsilon(v), \mathcal{I}_3; n = d\ell$                  | $F(v), I;$<br>$n = dm$               | $m$ is the cyclotomic order                                                                      |
| Homogeneous cyclotomic<br>polynomial                           | $\Phi_m(X_1, X_2)$                                       | $\Phi_m(X, Y)$                       | $Y$ is the imaginary coordinate                                                                  |

Unchanged, with their meaning in Part III:  $g_{ij} = \gcd(i, j)$ , as in Section 3;  $q_{ij} = (j - i)/g_{ij}$  and a grid denominator  $q$ , not the denominator coefficients  $q_i$  of Section 2;  $H_{k,s}$  and  $T_u(n, k)$ , the objects of Section 3; the zero  $u(t)$  and its angle  $t$ ; the imaginary coordinate  $Y$ , not Part II's threshold argument;  $\mathcal{C}, \hat{\mathcal{C}}, f_{\mathcal{C}}, F_{\mathcal{C}}$  for the Cauchy law;  $d_K$  for the Kolmogorov distance (Part I writes  $d_{TV}$  for total variation);  $\psi$  (Dedekind),  $\varphi$  (Euler) and  $\mu(\cdot)$  (Möbius, only inside arithmetic sums, not the amplitude  $\mu_k(u)$ );  $\sigma_{-1}(n)$  and  $\sigma_1(n)$ , divisor functions with their argument  $(n)$ , not the variance  $\sigma_k^2$  of Section 6;  $\Xi, \mathcal{H}$  and  $\mathcal{P}(K, Q)$ ; and the local letters  $a, b, h, r, \ell, p, x, X, \alpha, \theta$ , whose meaning is fixed where they occur ( $a, b$  are coprime integers, not  $1 + u$  and  $u$  of Section 3).

## 24 The zero measures and the main conclusions

### 24.1 An OEIS family with an arithmetic zero problem

Let  $T_u(n, k)$  be the polynomial enumerating matrices with  $k$  ordered rows, an arbitrary number of ordered columns, nonnegative integer entries, total sum  $n$ , and no zero row or column, each column receiving weight  $u$  (Section 3). At  $u = 1$  this is the triangle A261781 [2], and the diagonal  $T_1(2n, n)$  is A261784 [3]. Matrix compositions were studied by Munarini, Poneti and Rinaldi [4], whose generating functions and recurrences are the enumerative foundation of the subject. Part I supplies the weighted maximal Hankel factorization and asks what the limiting zero-counting measure is after the factors at  $u = 0$  and  $u = -1$  have been removed, with tail information and the contribution of roots of small order. Part III answers those parts of the question. It does not relabel the known enumeration or the determinant product as a new result, and it is an explicit extension of this report, not a claim that an exhaustive search has established historical priority.

### 24.2 Notation and the three zero measures

Throughout,  $k, s$  are positive integers,  $c = \binom{k+1}{3}$ ,  $g_{ij} = \gcd(i, j)$  and  $q_{ij} = (j - i)/g_{ij}$ . With  $H_{k,s}(u)$  the determinant of Theorem 3.1, define

$$\mathbf{R}_k(u) = \prod_{1 \leq i < j \leq k} ((1 + u)^{q_{ij}} - u^{q_{ij}})^{2g_{ij}}, \quad (24.1)$$

$$G_k = \sum_{i < j \leq k} g_{ij}, \quad \mathbf{D}_k = \deg \mathbf{R}_k = 2(c - G_k). \quad (24.2)$$

The polynomials  $\mathbf{R}_1$  and  $\mathbf{R}_2$  are constant. Hence all probability measures normalized by  $\mathbf{D}_k$  are defined only for  $k \geq 3$ .

Every zero of  $R_k$  has the form

$$u(t) = \frac{1}{e^{2\pi it} - 1} = -\frac{1}{2} - \frac{i}{2} \cot(\pi t), \quad 0 < t < 1. \quad (24.3)$$

We use three measures, always counting multiplicity:  $\rho_k^z$  on angles  $t \in (0, 1)$ ;  $\nu_k^z$  on the imaginary coordinate  $Y = -\frac{1}{2} \cot(\pi t)$ ; and  $\widehat{\nu}_k^z$  on the complex line  $\operatorname{Re} u = -\frac{1}{2}$ . They are pushforwards of one another. The full zero measure of  $H_{k,s}$ , normalized by its full degree, is denoted by  $\mu_{k,s}^z$ .

Write  $\mathcal{C}$  for the probability distribution on  $\mathbb{R}$  with density and cumulative distribution function

$$f_{\mathcal{C}}(y) = \frac{2}{\pi(1 + 4y^2)}, \quad F_{\mathcal{C}}(y) = \frac{1}{2} + \frac{1}{\pi} \arctan(2y). \quad (24.4)$$

Its image on the vertical line is  $\widehat{\mathcal{C}}$ . For real probability laws,  $d_K$  is the supremum of the absolute difference of their cumulative distribution functions. We reserve  $\mu(n)$  inside arithmetic sums for the Möbius function; this is distinct from the measure  $\mu_{k,s}^z$ .

### 24.3 The main conclusions

**Theorem 24.1** (Bulk limit and arithmetic multiplicities). *As  $k \rightarrow \infty$ ,*

$$D_k = \frac{k^3}{3} - \frac{k^2 \log k}{\zeta(2)} + O(k^2), \quad \nu_k^z \implies \mathcal{C}. \quad (24.5)$$

*If  $E_m(k)$  is the multiplicity of any zero associated with a primitive  $m$ -th root of unity,  $m \geq 2$ , then*

$$E_m(k) = \frac{k^2 \log k}{\zeta(2) \psi(m)} + O(k^2), \quad \psi(m) = m \prod_{p|m} \left(1 + \frac{1}{p}\right), \quad (24.6)$$

*with an absolute implied constant, uniform in  $m$ . Moreover,*

$$d_K(\nu_k^z, \mathcal{C}) \sim \frac{3}{2\zeta(2)} \frac{\log k}{k} = \frac{9}{\pi^2} \frac{\log k}{k}. \quad (24.7)$$

The proofs are in Sections 25–28. The uniform error in (24.6) is an absolute error, not a uniform relative equivalent for all growing  $m$ . It implies a relative equivalent when  $\psi(m) = o(\log k)$ ; for larger  $m$  it may be too large to reveal the leading term. The function  $\psi$  is itself OEIS A001615 [13].

**Theorem 24.2** (Extreme tails and moments). *For each fixed  $x > 0$ ,*

$$k \nu_k^z((kx, \infty)) \longrightarrow \Xi(x) := \frac{3}{\zeta(2)} \sum_{n \geq 1} \sigma_{-1}(n) (1 - 2\pi n x)_+^2, \quad \sigma_{-1}(n) = \sum_{d|n} \frac{1}{d}. \quad (24.8)$$

*The sum is finite. For every sequence  $y_k \rightarrow \infty$  with  $y_k = o(k)$ ,*

$$\nu_k^z((y_k, \infty)) \sim \frac{1}{2\pi y_k}. \quad (24.9)$$

*For the absolute moments  $M_p(k) = \int |Y|^p d\nu_k^z(Y)$ ,*

$$M_p(k) \longrightarrow \frac{2^{-p}}{\cos(\pi p/2)}, \quad 0 < p < 1, \quad (24.10)$$

$$M_1(k) = \frac{1}{\pi} \left( \log k + \gamma_E - \log \pi - \frac{5}{6} + \frac{\zeta'(2)}{\zeta(2)} \right) + O\left(\frac{\log^2 k}{k}\right), \quad (24.11)$$

$$M_p(k) \sim \mathcal{M}_p k^{p-1}, \quad p > 1, \quad (24.12)$$

$$\mathcal{M}_p = \frac{12\zeta(p)\zeta(p+1)}{(2\pi)^p(p+1)(p+2)\zeta(2)}. \quad (24.13)$$

Here  $(z)_+ = \max(z, 0)$ , and  $\gamma_E$  is Euler's constant. Sections 29–31 give proofs, including a quantitative edge-to-bulk matching formula. The finite sum in (24.8) retains arithmetic structure that weak convergence cannot see.

**Theorem 24.3** (Retaining the two exceptional factors). *If  $s = s_k \geq 1$  and  $s_k/k^2 \rightarrow \varkappa \in [0, \infty)$ , then*

$$\mu_{k,s_k}^z \Rightarrow \frac{\delta_0}{3(1+\varkappa)} + \frac{1+3\varkappa}{3(1+\varkappa)} \delta_{-1} + \frac{\hat{C}}{3(1+\varkappa)}. \quad (24.14)$$

*If  $s_k/k^2 \rightarrow \infty$ , the limit is  $\delta_{-1}$ . For any fixed shift  $s$ , the limiting masses of the two point components and the Cauchy component are all  $\frac{1}{3}$ .*

| Scale or statistic              | Result                                   | Information retained                              |
|---------------------------------|------------------------------------------|---------------------------------------------------|
| Fixed imaginary coordinate      | Cauchy law                               | Overall distribution                              |
| Fixed cyclotomic order          | Multiplicity proportional to $1/\psi(m)$ | Arithmetic concentration                          |
| CDF error                       | Exact constant $9/\pi^2$                 | Sharp global convergence rate                     |
| Smooth statistic                | Explicit $(\log k)/k$ correction         | A universal first bias                            |
| Height proportional to $k$      | Quadratic divisor-sum tail               | Extreme zeros and cutoff                          |
| Moment order $p$                | Transition at $p = 1$                    | Failure of moment convergence above the threshold |
| Shift $s$ proportional to $k^2$ | Three-component mixture                  | Dependence on full-degree normalization           |

Table 11: The results of Part III concern different asymptotic questions about the same exact polynomial (manuscript 08).

## 24.4 The determinant foundation

Manuscript 08's Section 2 proves the determinant identity with explicit conventions, to connect the arithmetic to the OEIS objects without depending on an unexamined factorization. In Part III's letters it reads

$$H_{k,s}(u) = C_k u^{A_k} (1+u)^{B_{k,s}} R_k(u), \quad A_k = \binom{k+1}{2} + 2c, \quad B_{k,s} = k(s-1) + 2c, \quad C_k = \prod_{j=1}^k \binom{k}{j}^j, \quad (24.15)$$

which is (3.1) of Theorem 3.1, with  $A_k = \sum_{j \leq k} j^2$ .

*Remark 24.4* (Manuscript 08's re-proof of Theorem 3.1; a pointer). [Note of the batch-98 write.] Manuscript 08's Proposition 2.1 ("Prior maximal Hankel identity") and its proof are a second derivation of Theorem 3.1 by the same argument as Section 3, and it credits the identity to this report: the generating function and inclusion–exclusion (2.1)–(2.3); the spectrum and residue weights (3.2); the Vandermonde factorization (3.7); the root products, discriminant and resultant (3.4)–(3.6); and the sign count  $\sum_j (j(k-j) + \binom{j}{2}) = 2c$ . It is not reprinted. Also shared with Section 3: the remark that the size  $\binom{k+1}{2}$  is maximal for the positive-index Hankel matrix at  $u > 0$ , that this does not assert positive definiteness of smaller minors, the reference to the different Cassini-type determinant of [4, Proposition 21], and  $H_{3,s}(u) = 27u^{14}(1+u)^{3s+5}(1+2u)^2$ . Manuscript 08 adds the small cases

$$H_{1,s}(u) = u(1+u)^{s-1}, \quad H_{2,s}(u) = 2u^5(1+u)^{2s},$$

which explain why reduced zeros first occur at  $k = 3$ , the example  $T_u(2, 2) = u + 2u^2$ , and the observation that the convention  $s \geq 1$  is essential because the constant polynomial part of the generating function contributes at  $n = 0$  (Part I's (3.9)). The write checked  $H_{1,s}$  and  $H_{2,s}$  against (24.15).

## 25 Cyclotomic multiplicities and the degree loss

### 25.1 An exact arithmetic description

Put  $U_q(u) = (1 + u)^q - u^q$ . For  $q \geq 2$ , it has exactly  $q - 1$  simple roots  $u(\ell/q)$ ,  $1 \leq \ell < q$ ;  $U_1 = 1$ . Indeed  $((1 + u)/u)^q = 1$ , and the root of unity 1 has no finite preimage. Its absence is the degree cancellation in  $U_q$ .

For every integer  $m \geq 1$ , define

$$E_m(k) = 2 \sum_{\substack{1 \leq i < j \leq k \\ m | q_{ij}}} g_{ij}. \quad (25.1)$$

For  $m \geq 2$ , this is the multiplicity of each primitive order- $m$  zero. The auxiliary value  $E_1(k) = 2G_k$  counts the weighted degree lost at the excluded root of unity 1; it does not describe a zero at a finite point.

Let  $\Phi_m(X_1, X_2)$  be the homogeneous cyclotomic polynomial

$$\Phi_m(X_1, X_2) = \prod_{\substack{1 \leq a \leq m \\ (a, m) = 1}} (X_1 - e^{2\pi i a/m} X_2).$$

Factoring  $X_1^q - X_2^q$ , and using  $X_1 - X_2 = 1$  at  $(X_1, X_2) = (1 + u, u)$ , proves the exact identities

$$R_k(u) = \prod_{m=2}^{k-1} \Phi_m(1 + u, u)^{E_m(k)}, \quad (25.2)$$

$$D_k = \sum_{m=2}^{k-1} \varphi(m) E_m(k) = 2c - E_1(k). \quad (25.3)$$

The factors in (25.2) need not be monic: their leading coefficients are  $\Phi_m(1, 1)$ , which are nonzero. The degree is nevertheless  $\varphi(m)$ .

Every order  $2 \leq m \leq k - 1$  occurs, since the pair  $(i, j) = (1, m + 1)$  has  $q_{ij} = m$ . Thus the number of distinct reduced zeros, ignoring multiplicity, is exactly  $\sum_{m=2}^{k-1} \varphi(m)$ . This is much smaller than  $D_k$ , of cubic order, and explains why ignoring multiplicity changes the finite distribution substantially.

For computation, it is useful to group first by the unreduced grid denominator  $q$ . Write

$$\mathcal{W}_k(q) = \sum_{\substack{q < b \leq k \\ (b, q) = 1}} \left\lfloor \frac{k}{b} \right\rfloor \left( \left\lfloor \frac{k}{b} \right\rfloor + 1 \right). \quad (25.4)$$

Then

$$R_k = \prod_{q=1}^{k-1} U_q^{\mathcal{W}_k(q)}, \quad E_m(k) = \sum_{\substack{q \leq k-1 \\ m | q}} \mathcal{W}_k(q).$$

To see this, set  $i = ga$ ,  $j = g(a + q)$ , where  $(a, q) = 1$ , and sum the multiplicity  $2g$  over  $1 \leq g \leq \lfloor k/(a + q) \rfloor$ .

## 25.2 A primitive-lattice estimate with uniform error

**Lemma 25.1.** *For  $X \geq 1$  and any integer  $m \geq 1$ , let*

$$P_m(X) = \#\{1 \leq a < b \leq X : (a, b) = 1, m \mid b - a\}.$$

*Uniformly in  $m$ ,*

$$P_m(X) = \frac{X^2}{2\zeta(2)\psi(m)} + O(X \log(2X)). \quad (25.5)$$

*Proof.* Without the coprimality condition, let

$$P_\ell^*(Y) = \#\{1 \leq a < b \leq Y : \ell \mid b - a\}.$$

Writing the difference as  $\ell h$  gives

$$P_\ell^*(Y) = \sum_{\ell h < \lfloor Y \rfloor} (\lfloor Y \rfloor - \ell h) = \frac{Y^2}{2\ell} + O(Y),$$

with an absolute error constant for all  $\ell$ , including  $\ell > Y$ . Möbius inversion now gives

$$\begin{aligned} P_m(X) &= \sum_{d \leq X} \mu(d) P_{m/(m,d)}^*(X/d) \\ &= \frac{X^2}{2m} \sum_{d \leq X} \frac{\mu(d)(m, d)}{d^2} + O\left(X \sum_{d \leq X} \frac{1}{d}\right). \end{aligned}$$

The infinite-sum tail costs  $O(X)$ , uniformly in  $m$ , because  $(m, d)/m \leq 1$ . Its absolutely convergent Euler product is

$$\frac{1}{m} \sum_{d \geq 1} \frac{\mu(d)(m, d)}{d^2} = \frac{1}{m\zeta(2)} \prod_{p \mid m} \frac{1 - 1/p}{1 - 1/p^2} = \frac{1}{\zeta(2)\psi(m)}.$$

This proves the estimate. □

**Proposition 25.2** (Uniform multiplicity asymptotic). *Equation (24.6) holds, including  $m = 1$  under the auxiliary convention  $E_1 = 2G_k$ .*

*Proof.* Separate each pair by its gcd:

$$E_m(k) = 2 \sum_{g \leq k} g P_m(k/g).$$

Insert Lemma 25.1. The main term is

$$\frac{k^2}{\zeta(2)\psi(m)} \sum_{g \leq k} \frac{1}{g}.$$

The total error is bounded by a constant times

$$k \sum_{g \leq k} \log(2k/g) = k\{k \log(2k) - \log(k!)\} = O(k^2).$$

Replacing the harmonic sum by  $\log k + O(1)$  preserves that error, uniformly in  $m$ . □

*Remark 25.3* (write). [Note of the batch-98 write.] The uniformity in  $m$  includes the orders  $m \geq k$ , at which no pair has  $m \mid q_{ij}$  and  $E_m(k) = 0$ : there  $\psi(m) \geq m \geq k$ , so the main term is at most  $k \log k / \zeta(2)$  and is itself  $O(k^2)$ . The statement is true there, and empty.

The  $m = 1$  case and (25.3) give (24.5). The underlying average-gcd phenomenon is classical; see Tóth [14] for context. The calculation above supplies the particular congruence refinement and all uniformity needed here.

**Corollary 25.4** (Macroscopic multiplicity of a fixed arithmetic zero). *For each fixed  $m \geq 2$  and integer  $a$  with  $1 \leq a < m$  and  $(a, m) = 1$ ,*

$$\rho_k^z(\{a/m\}) = \frac{3}{\zeta(2)} \frac{\log k}{\psi(m)} \frac{1}{k} + O_m(1/k). \quad (25.6)$$

*In particular,*

$$\nu_k^z(\{0\}) = \frac{1}{\zeta(2)} \frac{\log k}{k} + O(1/k). \quad (25.7)$$

*Proof.* Divide (24.6) by  $D_k = k^3/3 + O(k^2 \log k)$ . The extra  $O(\log^2 k/k^2)$  term is  $O(1/k)$ . Use  $\psi(2) = 3$  for the second assertion.  $\square$

| $m$ | $\psi(m)$ | Coefficient of $(\log k)/k$ per zero | Imaginary coordinates for small orders     |
|-----|-----------|--------------------------------------|--------------------------------------------|
| 2   | 3         | $1/\zeta(2)$                         | 0                                          |
| 3   | 4         | $3/(4\zeta(2))$                      | $\pm 1/(2\sqrt{3})$                        |
| 4   | 6         | $1/(2\zeta(2))$                      | $\pm 1/2$                                  |
| 5   | 6         | $1/(2\zeta(2))$                      | $-\frac{1}{2} \cot(\pi a/5), 1 \leq a < 5$ |
| 6   | 12        | $1/(4\zeta(2))$                      | $\pm \sqrt{3}/2$                           |

Table 12: Small-order arithmetic concentrations (manuscript 08). Multiplicity is stated per primitive zero, not as the sum over all  $\varphi(m)$  such zeros.

## 26 The Cauchy limit and full normalization

### 26.1 A direct quantitative proof

Let  $\mathcal{F}_k(t) = \rho_k^z((0, t])$ ,  $0 \leq t \leq 1$ . For  $0 \leq t < 1$ , the  $q$ -grid contains  $\lfloor qt \rfloor$  points among  $1/q, \dots, (q-1)/q$  at or below  $t$ . Thus

$$D_k(\mathcal{F}_k(t) - t) = \sum_{i < j} 2g_{ij}(t - \{q_{ij}t\}). \quad (26.1)$$

Since the absolute value of each bracket is at most one,

$$\sup_{0 \leq t \leq 1} |\mathcal{F}_k(t) - t| \leq \frac{2G_k}{D_k} = \frac{3}{\zeta(2)} \frac{\log k}{k} + O(1/k). \quad (26.2)$$

The identity at  $t = 1$  is interpreted separately: both cumulative distribution functions equal one.

The function  $Y(t) = -\frac{1}{2} \cot(\pi t)$  is a strictly increasing bijection from  $(0, 1)$  to  $\mathbb{R}$ . Its inverse is  $t = \frac{1}{2} + \pi^{-1} \arctan(2y)$ , so it sends the uniform distribution exactly to  $\mathcal{C}$ . It also preserves the Kolmogorov discrepancy. Equation (26.2) therefore proves the weak limit and an effective, finite- $k$  bound.

The logarithm cannot be removed from the rate. Symmetry about  $t = \frac{1}{2}$ , with the central atom retained, gives

$$\mathcal{F}_k(1/2) - 1/2 = \frac{E_2(k)}{2D_k} = \frac{1}{2\zeta(2)} \frac{\log k}{k} + O(1/k).$$

The exact optimal leading constant requires a more delicate argument, given in Section 28. In particular the central atom supplies a useful lower bound, but it does not locate the largest discrepancy.

*Remark 26.1* (The mode of convergence). Every  $\nu_k^z$  is discrete, whereas  $\mathcal{C}$  has a density. Their total-variation distance, defined as the supremum over measurable events, is therefore exactly one. Weak convergence and uniform cumulative-distribution convergence are the appropriate statements here. Similarly, weak convergence alone gives no convergence of the first or higher absolute moments.

## 26.2 Exceptional roots and growing Hankel shifts

By (24.15), the full degree is

$$\deg H_{k,s} = A_k + B_{k,s} + D_k = \binom{k+1}{2} + 6c + k(s-1) - 2G_k. \quad (26.3)$$

The full zero measure satisfies the exact decomposition

$$\mu_{k,s}^z = \frac{A_k}{\deg H_{k,s}} \delta_0 + \frac{B_{k,s}}{\deg H_{k,s}} \delta_{-1} + \frac{D_k}{\deg H_{k,s}} \hat{\nu}_k^z. \quad (26.4)$$

For  $k \geq 3$ , the three components have disjoint supports. For  $k = 1, 2$ , the last component is absent; a zero exponent in an exceptional factor simply contributes no root.

*Proof of Theorem 24.3.* We have

$$A_k/k^3 \rightarrow 1/3, \quad D_k/k^3 \rightarrow 1/3, \quad B_{k,s}/k^3 = 1/3 + s/k^2 + o(1).$$

If  $s/k^2 \rightarrow \varkappa < \infty$ , divide (26.4) by its total degree and use  $\hat{\nu}_k^z \implies \hat{\mathcal{C}}$ . This gives (24.14). If  $s/k^2 \rightarrow \infty$ , the coefficient of  $\delta_{-1}$  tends to one, which proves the remaining assertion.  $\square$

Thus the shift has no effect whatever on the reduced zero measure, but has a genuine effect on full-degree normalization. In particular, normalizing only the retained zeros by the original full degree would produce a subprobability measure, with mass  $1/[3(1 + \varkappa)]$ , rather than a Cauchy probability measure.

## 27 The first correction for smooth statistics

The arithmetic jumps in Table 12 coexist with a particularly simple first correction after smooth averaging.

**Theorem 27.1.** *If  $f$  is a twice continuously differentiable periodic function on  $\mathbb{R}/\mathbb{Z}$ , and  $I_f = \int_0^1 f(t) dt$ , then*

$$\int f d\rho_k^z = I_f + \frac{3}{\zeta(2)} \frac{\log k}{k} (I_f - f(0)) + O_f(1/k). \quad (27.1)$$

*Consequently, if  $\phi \in C_c^2(\mathbb{R})$ , then*

$$\int \phi(Y) d\nu_k^z(Y) = \left(1 + \frac{3}{\zeta(2)} \frac{\log k}{k}\right) \int_{\mathbb{R}} \phi(y) \frac{2 dy}{\pi(1 + 4y^2)} + O_\phi(1/k). \quad (27.2)$$

*Proof.* The Fourier series of  $f$  is absolutely convergent, and

$$|\widehat{f}(h)| \leq \frac{\|f''\|_{L^1}}{4\pi^2 h^2}, \quad h \neq 0.$$

Summing the Fourier series over a regular  $q$ -grid therefore gives

$$\sum_{\ell=0}^{q-1} f(\ell/q) = qI_f + \text{err}_q, \quad |\text{err}_q| \leq \frac{\|f''\|_{L^1}}{12q}.$$

After removing the term at zero and summing with the weights  $\mathcal{W}_k(q)$ , we obtain

$$\mathsf{D}_k \int f \, d\rho_k^z = 2cI_f - 2G_k f(0) + \sum_q \mathcal{W}_k(q) \text{err}_q. \quad (27.3)$$

From (25.4),

$$\mathcal{W}_k(q) \leq \sum_{b=q+1}^k \left( \frac{k^2}{b^2} + \frac{k}{b} \right) \leq \frac{2k^2}{q}.$$

Thus the last sum in (27.3) is  $O_f(k^2)$ . Divide by  $\mathsf{D}_k$ , and use

$$\frac{2G_k}{\mathsf{D}_k} = \frac{3}{\zeta(2)} \frac{\log k}{k} + O(1/k).$$

This proves (27.1). For  $\phi \in C_c^2(\mathbb{R})$ , the pullback  $f(t) = \phi(-\frac{1}{2} \cot(\pi t))$  is periodic  $C^2$  and is zero near  $t = 0, 1$ . Hence  $f(0) = 0$ , and the change of variables in (24.4) gives (27.2).  $\square$

The coefficient in (27.1) has a direct interpretation: the regular grids would include the angle zero, but that angle corresponds to the point at infinity under (24.3). Removing it and renormalizing creates the positive first bias in compactly supported smooth statistics. A discontinuous indicator cannot be substituted into this theorem. Its endpoint may hit a root of small order, and then the correction changes.

## 28 Arithmetic CDF corrections and the sharp constant

We now refine the elementary bound (26.2). For  $0 \leq t < 1$ , write

$$\mathsf{D}_k(\mathcal{F}_k(t) - t) = 2G_k t - \mathcal{S}_k(t), \quad \mathcal{S}_k(t) = \sum_{i < j} 2g_{ij} \{q_{ij}t\}. \quad (28.1)$$

The endpoint  $t = 1$  is treated separately. All limits below use right-continuous CDFs unless a left limit is displayed.

**Theorem 28.1** (Pointwise arithmetic correction). *For every irrational  $t \in (0, 1)$ ,*

$$\lim_{k \rightarrow \infty} \frac{k}{\log k} (\mathcal{F}_k(t) - t) = \frac{3}{\zeta(2)} \left( t - \frac{1}{2} \right).$$

*For a reduced rational  $t = r/m \in (0, 1)$ ,*

$$\lim_{k \rightarrow \infty} \frac{k}{\log k} (\mathcal{F}_k(t) - t) = \frac{3}{\zeta(2)} \left( t - \frac{1}{2} + \frac{1}{2\psi(m)} \right).$$

*The corresponding left-continuous value  $\mathcal{F}_k(t-)$  has the same limit with a minus sign before  $1/(2\psi(m))$ .*

The right/left distinction records the atom at the primitive root: subtracting the formulas yields asymptotic normalized atom mass  $3 \log k / (\zeta(2)\psi(m)k)$ .

**Theorem 28.2** (Moving angles near rational points). *Suppose  $t_k \in [0, 1)$  and  $t_k \rightarrow t_0$ . If  $t_0$  is irrational, then*

$$\frac{k}{\log k} (\mathcal{F}_k(t_k) - t_k) \longrightarrow \frac{3}{\zeta(2)} (t_0 - 1/2).$$

*If  $t_0 = r/m \in [0, 1]$  is rational in lowest terms, assume  $\delta_k = t_k - t_0 \neq 0$  has an eventually constant sign  $\varsigma$ , and*

$$\alpha = \lim_{k \rightarrow \infty} \min \left\{ 1, \frac{\log(1/|\delta_k|)}{\log k} \right\} \quad \text{exists.}$$

*Then  $0 \leq \alpha \leq 1$  and*

$$\frac{k}{\log k} (\mathcal{F}_k(t_k) - t_k) \longrightarrow \frac{3}{\zeta(2)} \left( t_0 - \frac{1}{2} + \frac{\varsigma \alpha}{2\psi(m)} \right). \quad (28.2)$$

*If  $t_k = t_0 < 1$  identically, the formula holds with  $\varsigma = +1$ ,  $\alpha = 1$ . Statements with eventual equality have the same meaning. For the endpoints  $0 = 0/1$  and  $1 = 1/1$ , the permitted signs are, respectively,  $+1$  and  $-1$ .*

**Corollary 28.3** (Sharp Kolmogorov constant).

$$\sup_{0 \leq t \leq 1} |\mathcal{F}_k(t) - t| \sim \frac{3}{2\zeta(2)} \frac{\log k}{k} = \frac{9}{\pi^2} \frac{\log k}{k}.$$

*The same constant holds after the monotone change of coordinate  $y = -\frac{1}{2} \cot(\pi t)$ , relative to the Cauchy distribution of scale  $\frac{1}{2}$ .*

## 28.1 An exact uniform reduction

Write  $H_N = \sum_{q \leq N} q^{-1}$  and  $Q_N(\theta) = \sum_{q \leq N} \{q\theta\}/q$ . All  $O(1)$  constants in the following lemma are absolute and uniform in the real argument  $t$ .

**Lemma 28.4.**

$$2G_k = \frac{k^2}{\zeta(2)} \log k + O(k^2), \quad (28.3)$$

$$D_k = \frac{k^3}{3} + O(k^2 \log k), \quad (28.4)$$

$$\frac{\mathcal{S}_k(t)}{k^2} = \sum_{d=1}^{\infty} \frac{\mu(d)}{d^2} Q_k(dt) + O(1). \quad (28.5)$$

*Proof.* Write  $i = ga$ ,  $j = gb$  with  $(a, b) = 1$ . With  $q = b - a$ , coprimality is equivalent to  $(q, b) = 1$ . Consequently

$$\mathcal{S}_k(t) = \sum_{b=2}^k h_b(h_b + 1) J_b(t), \quad h_b = \lfloor k/b \rfloor, \quad J_b(t) = \sum_{\substack{1 \leq q < b \\ (q, b) = 1}} \{qt\}.$$

Since  $h_b(h_b + 1) = (k/b)^2 + O(k/b)$  and  $0 \leq J_b \leq b$ , this gives

$$\frac{\mathcal{S}_k(t)}{k^2} = \sum_{b \leq k} \frac{J_b(t)}{b^2} + O(1).$$

Möbius inversion, including the harmless  $b = 1$  term, yields

$$\sum_{b \leq k} \frac{J_b(t)}{b^2} = \sum_{d \leq k} \frac{\mu(d)}{d^2} K_{\lfloor k/d \rfloor}(dt), \quad K_N(\theta) = \sum_{n \leq N} \frac{1}{n^2} \sum_{q < n} \{q\theta\}.$$

Interchanging the finite sums and using

$$\sum_{n=q+1}^N \frac{1}{n^2} = \frac{1}{q} + O(q^{-2}) + O(N^{-1}) \quad (1 \leq q < N)$$

shows  $K_N(\theta) = Q_N(\theta) + O(1)$  uniformly in  $\theta$  and  $N$ . Moreover,

$$|Q_k(\theta) - Q_{\lfloor k/d \rfloor}(\theta)| \leq O(1) + \log(2d).$$

The convergent series  $\sum_d (1 + \log d)/d^2$  therefore allows us to replace all cutoffs by  $k$ . Extending the outer sum to infinity changes the result by at most  $H_k \sum_{d > k} d^{-2} = O(1)$ , proving (28.5).

The gcd estimate is Proposition 25.2 at  $m = 1$ , and the degree estimate follows from (24.2).  $\square$

## 28.2 The harmonic sawtooth lemma

This is the key step responsible for the sharp uniform constant. It uses only approximation of a Riemann-integrable periodic function by trigonometric upper and lower bounds, and a geometric-sum estimate. It does not use an unproved cancellation statement for Möbius sums.

**Lemma 28.5** (Harmonic sawtooth near a limiting angle). *Let  $\theta_k \rightarrow \theta_0$  be a real sequence. If  $\theta_0$  is irrational, then*

$$\frac{Q_k(\theta_k)}{\log k} \rightarrow \frac{1}{2}.$$

*Suppose instead that  $\theta_0 = a/b$  in lowest terms,  $b \geq 1$ ,  $\tilde{\delta}_k = \theta_k - \theta_0 \neq 0$  has eventually constant sign  $\varsigma$ , and*

$$\tilde{\alpha} = \lim_{k \rightarrow \infty} \min\{1, \log(1/|\tilde{\delta}_k|)/\log k\}$$

*exists. Then*

$$\frac{Q_k(\theta_k)}{\log k} \rightarrow \frac{1}{2} - \frac{\varsigma \tilde{\alpha}}{2b}. \quad (28.6)$$

*For  $\theta_k = a/b$  exactly, the limit is  $1/2 - 1/(2b)$ .*

*Proof.* Let  $\bar{B}_1(x) = \{x\} - 1/2$ . For every  $\varepsilon > 0$  there exist real trigonometric polynomials  $T^-$  and  $T^+$  with

$$T^-(x) \leq \bar{B}_1(x) \leq T^+(x) \quad (x \in \mathbb{R}), \quad \left| \int_0^1 T^\pm(x) dx \right| \leq \varepsilon.$$

For completeness, first choose continuous periodic lower and upper bounds that agree with the sawtooth away from an arbitrarily small interval around its jump; then approximate these continuous functions uniformly by trigonometric polynomials and move the approximants down or up by their uniform approximation error. Their integrals can be made arbitrarily close to the sawtooth's zero integral.

Here  $\|x\| = \min_{n \in \mathbb{Z}} |x - n|$  denotes distance to the nearest integer. For  $h \neq 0$  and  $\|h\theta\| > 0$ , a geometric sum followed by partial summation gives

$$\left| \sum_{q=M'+1}^N \frac{\exp(2\pi i h q \theta)}{q} \right| \leq \frac{C}{(M'+1)\|h\theta\|}, \quad 0 \leq M' < N, \quad (28.7)$$

with an absolute constant  $C$ . In the irrational case, for the finitely many nonzero frequencies in  $T^\pm$ , each  $\|h\theta_k\|$  is bounded away from zero for all sufficiently large  $k$ . The polynomial inequalities and (28.7) with  $M' = 0$  give

$$\left| \sum_{q \leq k} \frac{\overline{B}_1(q\theta_k)}{q} \right| \leq \varepsilon H_k + O_{\varepsilon, \theta_0}(1).$$

Divide by  $\log k$  and let  $\varepsilon \downarrow 0$ .

Now suppose  $\theta_0 = a/b$  and  $\tilde{\delta}_k \neq 0$ . Fix  $0 < \kappa < 1/(4b)$ , and write

$$N_k = \min\{k, \lfloor \kappa/|\tilde{\delta}_k| \rfloor\}.$$

For all large  $k$ ,  $N_k \rightarrow \infty$ . For  $q \leq N_k$ , the displacement  $|q\tilde{\delta}_k|$  is too small to cross a discontinuity unless  $b \mid q$ . We therefore have the exact identity

$$Q_{N_k}(\theta_k) = Q_{N_k}(a/b) + \tilde{\delta}_k N_k + \mathbf{1}_{\{\zeta=-1\}} \sum_{\substack{q \leq N_k \\ b \mid q}} \frac{1}{q}.$$

Periodicity gives

$$Q_N(a/b) = \frac{b-1}{2b} \log N + O_b(1), \quad \sum_{\substack{q \leq N \\ b \mid q}} q^{-1} = \frac{1}{b} \log N + O_b(1).$$

Because  $|\tilde{\delta}_k N_k| \leq \kappa$ , it follows that

$$\sum_{q \leq N_k} \frac{\overline{B}_1(q\theta_k)}{q} = -\frac{\zeta}{2b} \log N_k + O_b(1). \quad (28.8)$$

It remains to control  $N_k < q \leq k$ , if this interval is nonempty. For every fixed nonzero integer  $h$ , if  $b \nmid h$ , then  $\|h\theta_k\|$  stays bounded away from zero. If  $b \mid h$ , then eventually  $\|h\theta_k\| = |h\tilde{\delta}_k|$  and  $(N_k + 1)|\tilde{\delta}_k| > \kappa$ . Hence (28.7) bounds every nonconstant Fourier term over the high interval by a constant depending on  $h, b, \kappa$ , independently of  $k$ . Apply the polynomial bounds again to obtain

$$\left| \sum_{q=N_k+1}^k \frac{\overline{B}_1(q\theta_k)}{q} \right| \leq \varepsilon (H_k - H_{N_k}) + O_{\varepsilon, b, \kappa}(1).$$

Consequently this high sum divided by  $\log k$  tends to zero, by letting  $\varepsilon \downarrow 0$ . Finally  $\log N_k / \log k \rightarrow \tilde{\alpha}$ , so (28.8) proves (28.6). The exactly rational case follows directly from the periodicity computation.  $\square$

[Note of the batch-98 write: manuscript 08 writes the lower summation limit in (28.7) as  $M$ ; it is printed as  $M'$  here because  $M$  is the zero-row count of Part I.]

### 28.3 Proofs of the arithmetic and sharp uniform results

*Proof of Theorem 28.2.* If  $t_0$  is irrational, every  $dt_0$  is irrational. By Lemma 28.5,  $Q_k(dt_k)/\log k \rightarrow 1/2$  for every fixed  $d$ . Since  $0 \leq Q_k \leq H_k$  and  $\sum d^{-2} < \infty$ , dominated convergence in (28.5) gives

$$\frac{S_k(t_k)}{k^2 \log k} \longrightarrow \frac{1}{2\zeta(2)}.$$

If  $t_0 = r/m$  is reduced, then  $dt_0$  has reduced denominator  $m/(d, m)$ . The sign of  $d\delta_k$  is  $\varsigma$ , and replacing  $\delta_k$  by  $d\delta_k$  does not change  $\alpha$ . Thus dominated convergence gives

$$\frac{\mathcal{S}_k(t_k)}{k^2 \log k} \longrightarrow \frac{1}{2\zeta(2)} - \frac{\varsigma\alpha}{2m} \sum_{d \geq 1} \frac{\mu(d)(d, m)}{d^2}.$$

The absolutely convergent Euler product evaluates to

$$\sum_{d \geq 1} \frac{\mu(d)(d, m)}{d^2} = \prod_{p|m} (1 - 1/p) \prod_{p \nmid m} (1 - 1/p^2) = \frac{1}{\zeta(2)} \prod_{p|m} \frac{p}{p+1} = \frac{m}{\zeta(2)\psi(m)}.$$

The same calculation with the exact rational case of Lemma 28.5 treats  $\delta_k = 0$  identically. Combining these limits with (28.1), (28.3) and (28.4) proves all statements.  $\square$

*Proof of Theorem 28.1.* The right-continuous formulas follow from Theorem 28.2. The root multiset is invariant under  $t \mapsto 1 - t$ , so  $\mathcal{F}_k(t-) = 1 - \mathcal{F}_k(1 - t)$  for  $0 < t < 1$ . Apply the right-continuous formula at  $1 - t$ , whose reduced denominator is again  $m$ , to obtain the left formula. Alternatively it follows by counting the multiples of  $m$  in the exact fractional-part identity.  $\square$

*Proof of Corollary 28.3.* For the lower bound, fix any irrational  $t \in (0, 1/2)$  and use Theorem 28.1; then let this fixed  $t$  decrease to zero.

For the upper bound it suffices to examine an arbitrary subsequence of angles  $t_k \in [0, 1]$  at which the scaled absolute discrepancies approach their limsup. Terms with  $t_k = 1$  have discrepancy zero and may be discarded. After taking a subsequence, compactness gives  $t_k \rightarrow t_0$ . If  $t_0$  is irrational, the absolute limit from Theorem 28.2 is strictly below  $3/(2\zeta(2))$ . If  $t_0 = r/m \in (0, 1)$  is reduced, pass to a further subsequence on which either  $t_k = t_0$  or  $\delta_k$  has fixed sign and the truncated exponent has a limit  $\alpha \in [0, 1]$ . The absolute limit is bounded by

$$\frac{3}{\zeta(2)} \left( \left| \frac{r}{m} - \frac{1}{2} \right| + \frac{1}{2\psi(m)} \right) \leq \frac{3}{\zeta(2)} \left( \frac{1}{2} - \frac{1}{m} + \frac{1}{2m} \right) < \frac{3}{2\zeta(2)}.$$

At  $t_0 = 0$ , the possible bracket in (28.2) is  $-1/2 + \alpha/2 \in [-1/2, 0]$ ; exact  $t_k = 0$  gives zero. At  $t_0 = 1$ , it is  $1/2 - \alpha/2 \in [0, 1/2]$ . These exhaust all subsequences and prove the upper bound. The coordinate map to the vertical line is strictly increasing, so it preserves Kolmogorov distance.  $\square$

## 28.4 Interpretation and useful examples

At  $t = 1/2$ , the right-continuous correction is  $3/(2\zeta(2)\psi(2)) = 1/(2\zeta(2))$ , and the left correction is its negative. For  $t_k = t_0 + k^{-\alpha}$  with rational  $t_0 \in (0, 1)$  and  $0 < \alpha < 1$ , the rational correction is  $\alpha$  times its exact right-continuous size. Approaching from below changes its sign. If  $t_k = k^{-\alpha}$  with  $0 < \alpha < 1$ , then

$$\frac{k}{\log k} (\mathcal{F}_k(t_k) - t_k) \longrightarrow -\frac{3(1 - \alpha)}{2\zeta(2)}.$$

If  $t_k = 1/\log k$ , the limit is  $-3/(2\zeta(2))$ , explicitly realizing the sharp constant along a moving endpoint sequence. The sharp constant therefore reflects a logarithmic range of angles approaching an endpoint, rather than a single fixed rational atom.

## 29 The arithmetic law at the extreme scale

The Cauchy law has unbounded support. In contrast, the largest imaginary coordinate of a finite polynomial is approximately  $k/(2\pi)$ . To describe this range one must both rescale the coordinate and compensate for the small proportion of zeros there.

### 29.1 An exact tail count

For  $y > 0$ , put

$$\Omega(y) = \frac{\pi}{\arctan(1/(2y))} > 2, \quad Z_k(y) = D_k \nu_k^z((y, \infty)). \quad (29.1)$$

A positive ordinate in a  $q$ -grid is  $\frac{1}{2} \cot(\pi r/q)$ ,  $1 \leq r < q/2$ . The condition that it exceed  $y$  is exactly  $q > \Omega(y)r$ . For real  $K > 0$  and  $Q \geq 0$ , define

$$\mathcal{P}(K, Q) = \#\{(a, q) \in \mathbb{Z}_{>0}^2 : (a, q) = 1, a + q \leq K, q > Q\}.$$

Using the representation  $i = ga$ ,  $j = g(a + q)$ , we obtain the exact formula

$$Z_k(y) = 2 \sum_{g, r \geq 1} g \mathcal{P}(k/g, \Omega(y)r). \quad (29.2)$$

Every nonzero term satisfies  $gr < k/\Omega(y)$ , so this sum is finite. The strict inequality is important at a threshold that is itself a zero.

**Lemma 29.1** (Primitive points in a truncated triangle). *Uniformly for  $K \geq 1$ ,  $Q \geq 0$ ,*

$$\mathcal{P}(K, Q) = \frac{(K - Q)_+^2}{2\zeta(2)} + O(K \log(2K)), \quad (29.3)$$

*with an absolute implied constant.*

*Proof.* If  $Q \geq K$ , both the count and the main term vanish. Otherwise let  $\mathcal{P}^*(K, Q)$  be the count without coprimality. Summing the possible positive values of  $a$  for each  $q > Q$ , and comparing the linear function  $K - v$  with its integral, gives

$$\mathcal{P}^*(K, Q) = \frac{1}{2}(K - Q)^2 + O(K + 1)$$

uniformly in  $Q$ . Möbius inversion gives

$$\mathcal{P}(K, Q) = \sum_{d \leq K} \mu(d) \mathcal{P}^*(K/d, Q/d).$$

Its main term is  $\frac{1}{2}(K - Q)^2 \sum_{d \leq K} \mu(d)/d^2$ , and its error is  $O(\sum_{d \leq K} (K/d + 1)) = O(K \log(2K))$ . Extending the main sum to infinity costs only  $O(K)$ .  $\square$

### 29.2 The edge tail, its density, and its cutoff

**Theorem 29.2** (Quantitative edge law). *Let  $\Xi$  be defined in (24.8). For every fixed  $\varepsilon > 0$ ,*

$$\sup_{x \geq \varepsilon} |k \nu_k^z((kx, \infty)) - \Xi(x)| = O_\varepsilon(\log k/k). \quad (29.4)$$

*The same assertion holds for the two-sided tail with limit  $2\Xi(x)$ .*

*Proof.* Uniformly for  $x \geq \varepsilon$ ,

$$\Omega(kx) = 2\pi kx + \frac{\pi}{6kx} + O_\varepsilon(k^{-3}), \quad \Omega(kx) > 2\pi kx.$$

Thus all nonzero terms in (29.2) belong to the fixed finite set  $gr < 1/(2\pi\varepsilon)$ . Apply Lemma 29.1; replacing  $\Omega(kx)$  by  $2\pi kx$  in the squared areas costs only  $O_\varepsilon(1)$ . We obtain

$$\begin{aligned} Z_k(kx) &= \frac{1}{\zeta(2)} \sum_{g,r \geq 1} g \left( \frac{k}{g} - 2\pi kxr \right)_+^2 + O_\varepsilon(k \log k) \\ &= \frac{k^2}{\zeta(2)} \sum_{g,r \geq 1} \frac{1}{g} (1 - 2\pi xgr)_+^2 + O_\varepsilon(k \log k) \\ &= \frac{k^2}{\zeta(2)} \sum_{n \geq 1} \sigma_{-1}(n) (1 - 2\pi xn)_+^2 + O_\varepsilon(k \log k). \end{aligned} \quad (29.5)$$

At a breakpoint  $2\pi xgr = 1$ , the actual term is zero:  $\Omega(kx) > 2\pi kx$  and a contributing  $q$  would have to be both larger than  $k/g$  and smaller than  $k/g$ . The area term is zero as well. Hence the proof does not exclude breakpoints. Divide (29.5) by  $D_k = k^3/3 + O(k^2 \log k)$ , and multiply by  $k$ . Symmetry proves the two-sided statement.  $\square$

On  $(0, \infty)$ ,  $\Xi$  is decreasing, convex, continuously differentiable, and piecewise quadratic. Its negative derivative is

$$-\Xi'(x) = \frac{12\pi}{\zeta(2)} \sum_{n \geq 1} \sigma_1(n) (1 - 2\pi nx)_+, \quad \sigma_1(n) = \sum_{d|n} d. \quad (29.6)$$

Thus the locally finite measures  $\nu_k^e(A) = k \nu_k^z(kA)$  on  $(0, \infty)$  converge vaguely to  $-\Xi'(x) dx$ . This follows first on intervals bounded away from zero by subtracting the tail limits, and then for compactly supported continuous functions by step-function approximation. The limit is not a probability measure: its mass diverges toward the origin.

At  $x_n = 1/(2\pi n)$ , the jump in the second derivative is

$$\Xi''(x_n^-) - \Xi''(x_n^+) = \frac{24\pi^2}{\zeta(2)} n^2 \sigma_{-1}(n). \quad (29.7)$$

Divisor sums are therefore recoverable from the slope changes of the edge density. On the outermost interval,

$$\frac{1}{4\pi} \leq x \leq \frac{1}{2\pi}, \quad \Xi(x) = \frac{3}{\zeta(2)} (1 - 2\pi x)^2.$$

**Proposition 29.3** (Exact extreme zeros). *For  $k \geq 4$ , the two zeros of largest absolute imaginary coordinate are*

$$-\frac{1}{2} \pm \frac{i}{2} \cot \frac{\pi}{k-1}, \quad (29.8)$$

*each of algebraic multiplicity exactly two. In particular,*

$$\max |\operatorname{Im} u| < \frac{k}{2\pi}, \quad \frac{\max |\operatorname{Im} u|}{k} \longrightarrow \frac{1}{2\pi}.$$

*For  $k = 3$ , the only reduced zero is  $-1/2$ , with multiplicity two.*

*Proof.* Every grid has  $q = (j - i)/g_{ij} \leq k - 1$ . The least positive angle entering a positive cotangent is  $\pi/(k - 1)$ , attained only at  $(i, j) = (1, k)$ ,  $r = 1$ . That pair has  $g_{ij} = 1$ ; no other grid can produce this same extreme angle. The bound follows from  $\cot z < 1/z$  for  $0 < z < \pi/2$ .  $\square$

The edge theorem describes a collection of order  $k^2$  zeros with multiplicity, out of a total of order  $k^3$ . It does not describe a limiting process for the finitely many largest individual roots. The latter question is finer than (29.4).

## 30 Matching the edge to the Cauchy tail

### 30.1 A divisor-sum expansion with its constant

**Theorem 30.1.** As  $x \downarrow 0$ ,

$$\Xi(x) = \frac{1}{2\pi x} - \frac{3}{2\zeta(2)} \log \frac{1}{x} + \frac{3}{2\zeta(2)} \left( \frac{3}{2} - \gamma_E \right) + O\left(x \log \frac{1}{x}\right). \quad (30.1)$$

*Proof.* Put

$$\mathcal{H}(t) = \sum_{n \geq 1} \sigma_{-1}(n) (1 - tn)_+^2, \quad X = 1/t, \quad B_3(v) = v^3 - \frac{3}{2}v^2 + \frac{1}{2}v.$$

Regroup the divisors as  $n = d\ell$ . The elementary finite-sum identity

$$\sum_{\ell=1}^{\lfloor v \rfloor} (v - \ell)^2 = \frac{B_3(v) - B_3(\{v\})}{3}$$

then gives the exact equality

$$\mathcal{H}(1/X) = \frac{X}{3} \sum_{d \leq X} \frac{1}{d^2} - \frac{1}{2} \sum_{d \leq X} \frac{1}{d} + \frac{\lfloor X \rfloor}{6X} - \frac{1}{3X^2} \sum_{d \leq X} d B_3(\{X/d\}). \quad (30.2)$$

Define  $\Upsilon(v) = v B_3(\{1/v\})$  for  $v > 0$ , with  $\Upsilon(0) = 0$ . The periodic extension of  $B_3$  and its first derivative agree at integer endpoints. Thus  $\Upsilon$  is continuous,  $|\Upsilon(v)| \leq Cv$ , and away from zero  $|\Upsilon'(v)| \leq C(1 + 1/v)$ . Splitting off  $[0, 1/X]$  and bounding the remaining Riemann-sum error by the integral of  $|\Upsilon'|/X$  proves

$$\frac{1}{X^2} \sum_{d \leq X} d B_3(\{X/d\}) = \mathcal{I}_3 + O(\log X/X), \quad \mathcal{I}_3 = \int_0^1 \Upsilon(v) dv. \quad (30.3)$$

Here

$$\mathcal{I}_3 = \int_1^\infty \frac{B_3(\{v\})}{v^3} dv = \frac{3}{2} \log(2\pi) - \frac{11}{4}.$$

For clarity, the constant can be evaluated without a contour argument. Two integrations by parts using the periodic Bernoulli polynomials give

$$\mathcal{I}_3 = \frac{1}{4} + 3 \int_1^\infty \frac{\{v\} - 1/2}{v} dv.$$

The last integral converges by the zero mean of its periodic numerator. Its integral through  $N + 1$  is

$$N - (N + 1/2) \log(N + 1) + \log(N!),$$

which tends to  $\frac{1}{2} \log(2\pi) - 1$  by Stirling's formula.

Finally,

$$\frac{X}{3} \sum_{d \leq X} d^{-2} = \frac{\zeta(2)X}{3} - \frac{1}{3} + O(X^{-1}), \quad \sum_{d \leq X} d^{-1} = \log X + \gamma_E + O(X^{-1}), \quad \frac{\lfloor X \rfloor}{6X} = \frac{1}{6} + O(X^{-1}).$$

Insert these and (30.3) into (30.2). The result is

$$\mathcal{H}(t) = \frac{\zeta(2)}{3t} - \frac{1}{2} \log \frac{1}{t} + \frac{3}{4} - \frac{\gamma_E}{2} - \frac{1}{2} \log(2\pi) + O\left(t \log \frac{1}{t}\right). \quad (30.4)$$

Since  $\Xi(x) = 3\mathcal{H}(2\pi x)/\zeta(2)$ , the  $\log(2\pi)$  terms cancel, giving (30.1).  $\square$

[Note of the batch-98 write: in the evaluation of  $\mathcal{I}_3$ ,  $N$  is a running integer of that display only, not the diagonal index of Part I.]

The leading term matches the Cauchy tail. The next term has exactly the scale that appears in the global discrepancy theorem. The remainder in (30.1) has not been differentiated; the density formula (29.6) follows directly from the exact finite sum.

### 30.2 One formula covering the entire intermediate range

**Theorem 30.2** (Uniform transition formula). *For  $y > 0$ , set  $\mathfrak{m} = k/\Omega(y)$ . If  $\mathfrak{m} \leq 1$ , both  $Z_k(y)$  and  $\mathcal{H}(1/\mathfrak{m})$  vanish. If  $\mathfrak{m} \geq 1$ , then*

$$Z_k(y) = \frac{k^2}{\zeta(2)} \mathcal{H}(1/\mathfrak{m}) + O(k\mathfrak{m} \log(2\mathfrak{m}) \log(2k)), \quad (30.5)$$

with an absolute implied constant. For sufficiently large  $k$ , its normalized form is

$$\nu_k^z((y, \infty)) = \frac{1}{k} \Xi\left(\frac{\Omega(y)}{2\pi k}\right) + O\left(\frac{\log(2\mathfrak{m}) \log(2k)}{\Omega(y)k}\right), \quad \mathfrak{m} \geq 1. \quad (30.6)$$

*Proof.* Apply Lemma 29.1 to (29.2) without approximating  $\Omega$ . The squared areas regroup exactly into  $k^2 \mathcal{H}(\Omega/k)/\zeta(2)$ . Only  $gr < \mathfrak{m}$  contribute. Their total error is at most

$$Ck \sum_{g \leq \mathfrak{m}} \left\lfloor \frac{\mathfrak{m}}{g} \right\rfloor \log(2k/g) \leq C' k \mathfrak{m} \log(2\mathfrak{m}) \log(2k).$$

This proves (30.5). Also  $\mathcal{H}(1/\mathfrak{m}) = O(\mathfrak{m})$  for  $\mathfrak{m} \geq 1$ , for example by summing its divisor expression and using  $\sum d^{-2} < \infty$ . The normalization  $D_k = k^3/3 + O(k^2 \log k)$  gives (30.6); its extra normalization error is absorbed because  $\log(2\mathfrak{m})$  is bounded below by  $\log 2$ .  $\square$

**Corollary 30.3** (All mesoscopic Cauchy tails). *If  $y_k \rightarrow \infty$  and  $y_k = o(k)$ , then*

$$\nu_k^z((y_k, \infty)) \sim \frac{1}{2\pi y_k}, \quad \nu_k^z(\{|Y| > y_k\}) \sim \frac{1}{\pi y_k}.$$

*Proof.* Now  $\Omega(y_k) \sim 2\pi y_k$ , so  $\mathfrak{m}_k = k/\Omega(y_k) \rightarrow \infty$ . Equations (30.4) and (30.6) imply

$$\nu_k^z((y_k, \infty)) = \frac{1}{\Omega(y_k)} \left[ 1 + O\left(\frac{\log(2\mathfrak{m}_k)}{\mathfrak{m}_k} + \frac{\log(2\mathfrak{m}_k) \log(2k)}{k}\right) \right].$$

Both errors tend to zero; note that  $\mathfrak{m}_k < k/2$ . Symmetry gives the second formula.  $\square$

The exact effective argument in (30.6) is  $\Omega(y)/(2\pi k)$ . Replacing it by  $y/k$  is justified for growing  $y$ , but would lose the correct Cauchy coordinate for bounded  $y$ . The mesoscopic result places no extra restriction such as  $y = o(k/\log k)$ .

## 31 The transition in absolute moments

The zeros are symmetric in their imaginary coordinate, so every signed odd moment is zero. Absolute moments reveal the heavy tail. For  $p > 0$ , retain the notation  $M_p(k) = \int |Y|^p d\nu_k^z(Y)$ .

### 31.1 A uniform bound at every height

Before applying its multiplicity, a  $q$ -grid has at most  $2q/\Omega(y)$  points with  $|Y| > y$ . As  $\sum_{i < j} g_{ij}q_{ij} = c$ , we have the useful exact bound

$$\nu_k^z(\{|Y| > y\}) \leq \min\left(1, \frac{4c}{D_k \Omega(y)}\right) \leq \min\left(1, \frac{2c}{\pi D_k y}\right). \quad (31.1)$$

The right-hand side is bounded by  $\min(1, C/y)$  for an absolute constant and all  $k \geq 3$ . The finitely many small  $k$ 's can be included after increasing the constant, since  $2c/D_k \rightarrow 1$ .

### 31.2 Orders below and above one

*Proof of (24.10).* For a nonnegative random variable, the layer-cake identity gives

$$M_p(k) = p \int_0^\infty y^{p-1} \nu_k^z(\{|Y| > y\}) dy.$$

When  $0 < p < 1$ , (31.1) supplies an integrable majorant,  $py^{p-1} \min(1, C/y)$ . Weak convergence to the continuous Cauchy law gives convergence of the tails at every positive  $y$ . Dominated convergence proves moment convergence. Finally, the change of variables  $z = 2y$  evaluates the limit:

$$\int_{\mathbb{R}} |y|^p \frac{2 dy}{\pi(1+4y^2)} = \frac{2^{1-p}}{\pi} \int_0^\infty \frac{z^p}{1+z^2} dz = \frac{2^{-p}}{\cos(\pi p/2)}.$$

□

*Proof of (24.12).* The exact cutoff in Proposition 29.3 and the layer-cake identity yield

$$k^{1-p} M_p(k) = p \int_0^{1/(2\pi)} x^{p-1} k \nu_k^z(\{|Y| > kx\}) dx.$$

By (31.1), the integrand is at most a constant times  $x^{p-2}$ . This is integrable at zero exactly when  $p > 1$ . The edge theorem and dominated convergence therefore give

$$\begin{aligned} \lim_{k \rightarrow \infty} k^{1-p} M_p(k) &= \frac{6p}{\zeta(2)} \sum_{n \geq 1} \sigma_{-1}(n) \int_0^{1/(2\pi n)} x^{p-1} (1 - 2\pi n x)^2 dx \\ &= \frac{12}{(2\pi)^p (p+1)(p+2)\zeta(2)} \sum_{n \geq 1} \frac{\sigma_{-1}(n)}{n^p} \\ &= \frac{12\zeta(p)\zeta(p+1)}{(2\pi)^p (p+1)(p+2)\zeta(2)}. \end{aligned}$$

Tonelli's theorem justifies the nonnegative interchange. The last Dirichlet-series identity follows directly by writing  $n = d\ell$ :

$$\sum_{n \geq 1} \frac{\sigma_{-1}(n)}{n^p} = \sum_{d, \ell \geq 1} \frac{1}{d^{p+1} \ell^p} = \zeta(p+1)\zeta(p).$$

□

This computation shows why cutting off a continuous Cauchy density at its largest zero would give the wrong higher-moment constants. The entire divisor-sum edge profile contributes. For instance,

$$M_2(k) \sim \frac{\zeta(3)}{4\pi^2} k. \quad (31.2)$$

### 31.3 The critical moment, with its constant term

**Lemma 31.1.** *Uniformly for integers  $q \geq 1$ ,*

$$S_1(q) := \frac{1}{2} \sum_{\ell=1}^{q-1} \left| \cot \frac{\pi \ell}{q} \right| = \frac{q}{\pi} (\log q + \gamma_E - \log \pi) + O(1). \quad (31.3)$$

*Proof.* By symmetry,

$$S_1(q) = \sum_{\ell=1}^{\lfloor (q-1)/2 \rfloor} \cot(\pi \ell / q).$$

The function  $h(v) = \cot(\pi v) - 1/(\pi v)$  extends smoothly across zero on  $[0, 1/2]$ , and

$$\int_0^{1/2} h(v) dv = \frac{1}{\pi} \log \frac{2}{\pi}.$$

Subtract  $q/(\pi \ell)$  in the sum. A Riemann-sum error  $O(1)$ , together with the harmonic-number expansion, gives

$$S_1(q) = \frac{q}{\pi} (\log(q/2) + \gamma_E + \log(2/\pi)) + O(1).$$

The finitely many smallest values of  $q$  are absorbed in the constant.  $\square$

Define the two weighted sums

$$\mathcal{S}_k^{\text{gap}} = \sum_{i < j} (j - i) \log(j - i), \quad \mathcal{S}_k^{\text{gcd}} = \sum_{i < j} (j - i) \log g_{ij}.$$

For the first, grouping by  $h = j - i$  and using a Riemann sum gives

$$\mathcal{S}_k^{\text{gap}} = \sum_{h=1}^{k-1} h(k-h) \log h = c \left( \log k - \frac{5}{6} \right) + O(k^2 \log k). \quad (31.4)$$

The constant follows from  $\int_0^1 v(1-v) \log v dv = -5/36$  and  $\int_0^1 v(1-v) dv = 1/6$ . The integrable endpoint derivative also gives the displayed Riemann-sum error.

For the second, let  $\Lambda_{\text{vM}}(n)$  be the von Mangoldt function:  $\Lambda_{\text{vM}}(p^a) = \log p$  for prime powers, and zero otherwise. The elementary identity  $\log n = \sum_{d|n} \Lambda_{\text{vM}}(d)$  gives the exact finite formula

$$\mathcal{S}_k^{\text{gcd}} = \sum_{d \leq k} \Lambda_{\text{vM}}(d) d \binom{\lfloor k/d \rfloor + 1}{3}. \quad (31.5)$$

Since

$$d \binom{\lfloor k/d \rfloor + 1}{3} = \frac{k^3}{6d^2} + O(k^2/d),$$

we obtain

$$\mathcal{S}_k^{\text{gcd}} = -c \frac{\zeta'(2)}{\zeta(2)} + O(k^2 \log^2 k). \quad (31.6)$$

For completeness, no prime number theorem is needed for this error:  $\Lambda_{\text{vM}}(d) \leq \log d$  implies

$$\sum_{d \leq k} \frac{\Lambda_{\text{vM}}(d)}{d} = O(\log^2 k), \quad \sum_{d > k} \frac{\Lambda_{\text{vM}}(d)}{d^2} = O(\log k/k).$$

Logarithmic differentiation of the absolutely convergent Euler product at 2 gives  $\sum_{d \geq 1} \Lambda_{\text{vM}}(d)/d^2 = -\zeta'(2)/\zeta(2)$ .

*Proof of (24.11).* Apply Lemma 31.1 to each grid in the exact product. Since  $gq = j - i$ ,

$$M_1(k) = \frac{2}{\pi D_k} \{ \mathcal{S}_k^{\text{gap}} - \mathcal{S}_k^{\text{gcd}} + (\gamma_E - \log \pi) c \} + O(G_k/D_k).$$

Insert (31.4) and (31.6). Because

$$D_k = 2(c - G_k), \quad G_k = O(k^2 \log k),$$

the normalization correction multiplied by  $\log k$  is  $O(\log^2 k/k)$ . This proves

$$M_1(k) = \frac{\log k + \gamma_E - \log \pi - 5/6 + \zeta'(2)/\zeta(2)}{\pi} + O(\log^2 k/k).$$

□

The constant after  $(\log k)/\pi$  is approximately

$$\frac{\gamma_E - \log \pi - 5/6 + \zeta'(2)/\zeta(2)}{\pi} = -0.6273278444052116 \dots \quad (31.7)$$

In particular, a good numerical fit to the logarithmic coefficient alone can conceal a substantial finite constant.

### 31.4 An exact second moment and a shift-dependent example

The finite identity

$$\sum_{\ell=1}^{q-1} \cot^2 \frac{\pi \ell}{q} = \frac{(q-1)(q-2)}{3}$$

can be checked by logarithmically differentiating the finite sine product twice:  $\sum_{\ell=0}^{q-1} \csc^2(z + \pi \ell/q) = q^2 \csc^2(qz)$ . Letting  $z \rightarrow 0$  and removing the common  $z^{-2}$  term gives the sum of squared cosecants, and subtracting  $q - 1$  gives the displayed cotangent sum. Thus

$$M_2(k) = \frac{1}{6D_k} \left\{ \sum_{i < j} \frac{(j-i)^2}{g_{ij}} - 3c + 2G_k \right\}. \quad (31.8)$$

This identity supplies an exact rational check of (31.2), independent of numerical root summation.

The positive absolute imaginary moments of the full zero measure have the exact relation

$$\int |\operatorname{Im} u|^p d\mu_{k,s}^z(u) = \frac{D_k}{\deg H_{k,s}} M_p(k). \quad (31.9)$$

For  $p > 1$  and  $s \gg k^2$ , this gives

$$\int |\operatorname{Im} u|^p d\mu_{k,s}^z(u) \sim \frac{\mathcal{M}_p}{3} \frac{k^{p+1}}{s}.$$

Hence  $\mu_{k,s}^z \implies \delta_{-1}$  can coexist with divergent  $p$ -th moments when  $k^2 \ll s \ll k^{p+1}$ . If  $s/k^{p+1} \rightarrow \varkappa_p \in (0, \infty)$ , the moment tends to  $\mathcal{M}_p/(3\varkappa_p)$ . This is a concrete illustration of why weak convergence and moment convergence must be kept distinct.

## 32 Manuscript 08's checks and numerical illustrations

Manuscript 08's program `code/07-zero-laws-reproduce.py` (delivered as `code/reproduce.py`) regenerates the data and all four figures without accessing the network. The proofs above do not rely on these computations. The purpose of the checks is to test the enumerative normalization, detect multiplicity or factor-of-two errors, and display the rate at which the asymptotic regimes become visible.

### 32.1 What is checked exactly

The program constructs the counts allowing zero rows,  $A(n, j; u)$  of Section 2, directly from the column recurrence

$$A(n, j; u) = u \sum_{h=1}^n \binom{h+j-1}{j-1} A(n-h, j; u) \quad (n \geq 1, j \geq 1),$$

which is (2.2), uses inclusion–exclusion to obtain  $T_u(n, k)$ , and takes the determinant before comparing it with (24.15). Thus the determinant entries are not generated from the product being checked.

| Verification                                                                                          | Completed checks |
|-------------------------------------------------------------------------------------------------------|------------------|
| Complete polynomial determinant identities in $\mathbb{Q}[u]$ , $1 \leq k \leq 4$ , $1 \leq s \leq 3$ | 12               |
| Rational specializations, including exceptional and central roots                                     | 96               |
| Exact nonreal algebraic-root specializations                                                          | 3                |
| Degree and primitive-multiplicity identities                                                          | 91               |
| Independent root merging and exact CDF comparisons                                                    | 18               |
| Independent direct and aggregated tail counts                                                         | 20               |
| Finite cotangent identities at 70-digit precision                                                     | 78               |
| CDF maxima over complete supports, through $k = 2048$                                                 | 18               |
| Independent high-precision absolute first-moment checks                                               | 2                |

Table 13: Manuscript 08's checks. All listed checks passed. High-precision and tail checks are distinguished from exact polynomial and rational computations.

The empirical CDF maximum is found by checking every support angle and both its one-sided values. These values are exact rational numbers, rather than maxima sampled from a plotting grid. The second and fourth moments are also computed by exact finite cotangent identities. In edge plots, the root masses are integer counts, but comparison with transcendental thresholds uses floating-point arithmetic. Independent high-precision tail checks use thresholds away from atoms.

[Note of the batch-98 write: the program's record `data/07-zero-laws-verification_summary.txt` also lists four checks of PDF structure (trailer, cross-reference offsets and page count of each figure), which the table does not count. The shipped `data/07-zero-laws-pdf_validation.json` is a different record: it reports that the four figure PDFs were opened and rendered by PyMuPDF, a tool that is not a dependency of the program and is not shipped, so that record cannot be regenerated from the package.]

### 32.2 Bulk convergence and arithmetic concentrations

The computed Kolmogorov distances illustrate a useful distinction: the theorem determines the exact limiting constant, but moderate sizes do not approximate that constant especially closely. Write  $\Delta_k^K = d_K(\nu_k^Z, \mathcal{C})$ .

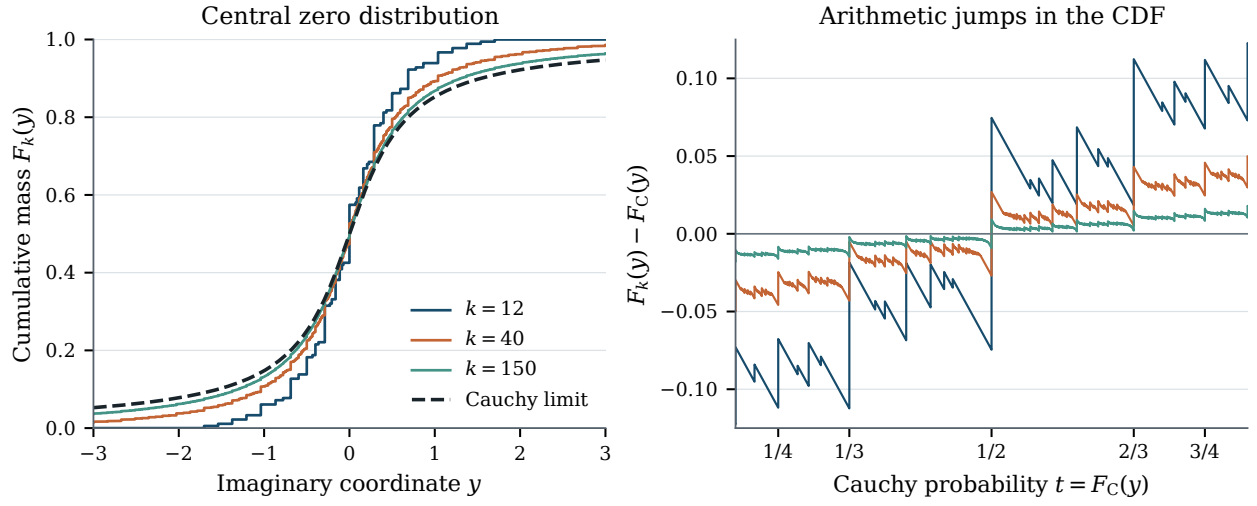


Figure 3: The reduced zero distribution for  $k = 12, 40, 150$  (manuscript 08). Left: CDFs of the imaginary coordinates and the Cauchy limit. Right: discrepancy in the uniform angular coordinate  $t = F_C(y)$ . The arithmetic jumps are visible even when the CDFs are close on the original scale. [The figure's  $F_k(y)$  is the distribution function of  $\nu_k^z$ , that is  $\mathcal{F}_k(F_C(y))$ , and its  $F_C$  is  $F_C$ .]

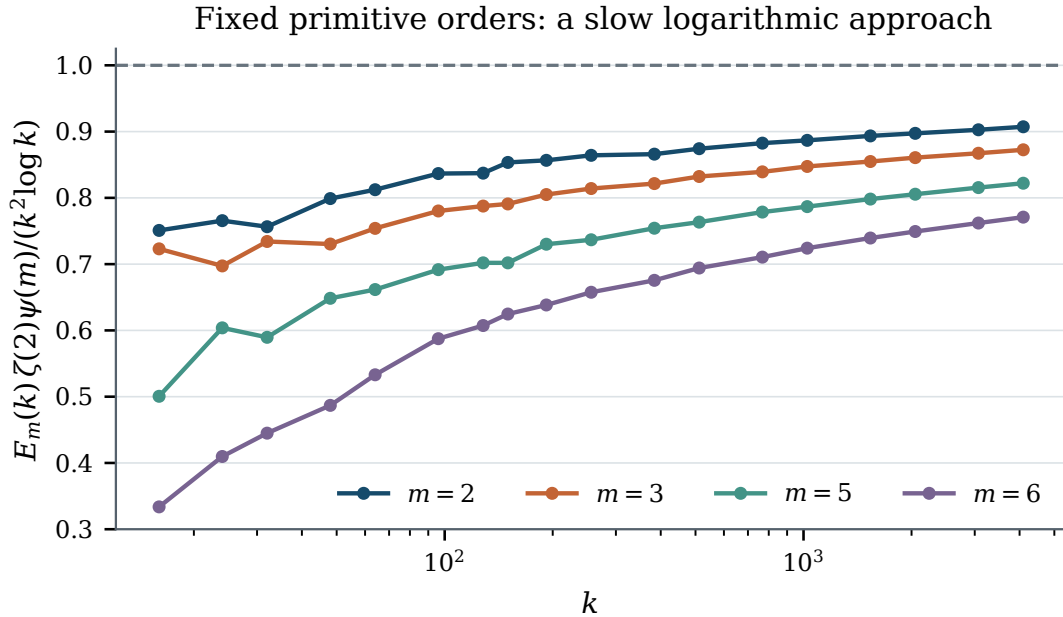


Figure 4: Exact primitive-order multiplicities divided by the leading terms  $k^2 \log k / (\zeta(2) \psi(m))$ , for  $m = 2, 3, 5, 6$  (manuscript 08). The convergence is slow because the proved error has relative order  $1/\log k$  at each fixed order. [The figure's  $E_m(k)$  is  $E_m(k)$ .]

| $k$   | $\Delta_k^K$                 | $k\Delta_k^K/\log k$ | One maximizing left limit |
|-------|------------------------------|----------------------|---------------------------|
| 64    | 0.0350281076162              | 0.539038690          | 1/7                       |
| 128   | 0.0208961596153              | 0.551255513          | 1/7                       |
| 256   | 0.0122991367143              | 0.567804913          | 1/7                       |
| 512   | 0.00708644151164             | 0.581607767          | 1/11                      |
| 1024  | 0.00406672748617             | 0.600785672          | 1/11                      |
| 2048  | 0.00230151896685             | 0.618196247          | 1/11                      |
| Limit | $9/\pi^2 = 0.911890653\dots$ |                      |                           |

Table 14: Complete-support exact discrepancies, displayed in decimals (manuscript 08). These data neither determine the limiting constant by themselves nor suggest that the maximum always occurs at the order-two atom.

The movement from an order-seven to an order-eleven endpoint is compatible with the moving-angle proof of the sharp constant. It is finite evidence, not a theorem predicting the denominator of every maximizing endpoint (Research question 9).

### 32.3 The extreme tail and the moment transition

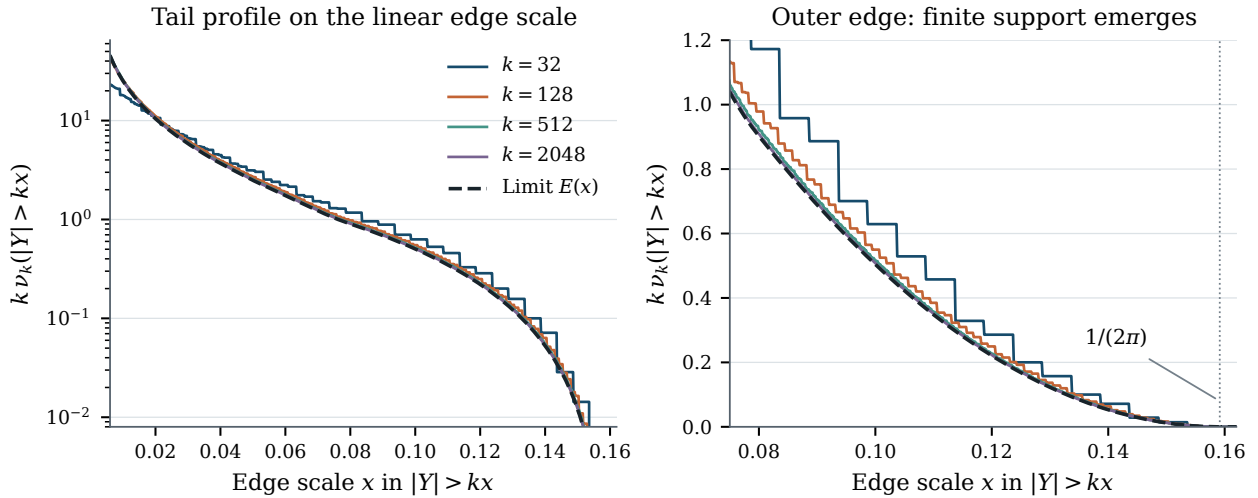


Figure 5: The two-sided extreme tail  $k\nu_k^Z(\{|Y| > kx\})$  for  $k = 32, 128, 512, 2048$ , compared with  $2\Xi(x)$  (manuscript 08). The enlarged outer region displays the finite endpoint  $1/(2\pi)$ . The limiting curve comes from the divisor-sum formula, not from a fitted numerical density. [The figure's  $\nu_k$  is  $\nu_k^Z$ , and its “Limit  $E(x)$ ” is the two-sided limit  $2\Xi(x)$ .]

For example, at  $k = 4096$  the computed values are

$$k^{-1}M_2(k) = 0.0305025370246502, \quad \frac{k^{-1}M_2(k)}{\zeta(3)/(4\pi^2)} = 1.00177611516,$$

and

$$k^{-3}M_4(k) = 0.000175765262189301, \quad \frac{k^{-3}M_4(k)}{\zeta(5)/(600\pi^2)} = 1.00377307676.$$

The constants here are consequences of (24.13); the exact finite moments provide independent numerical checks.

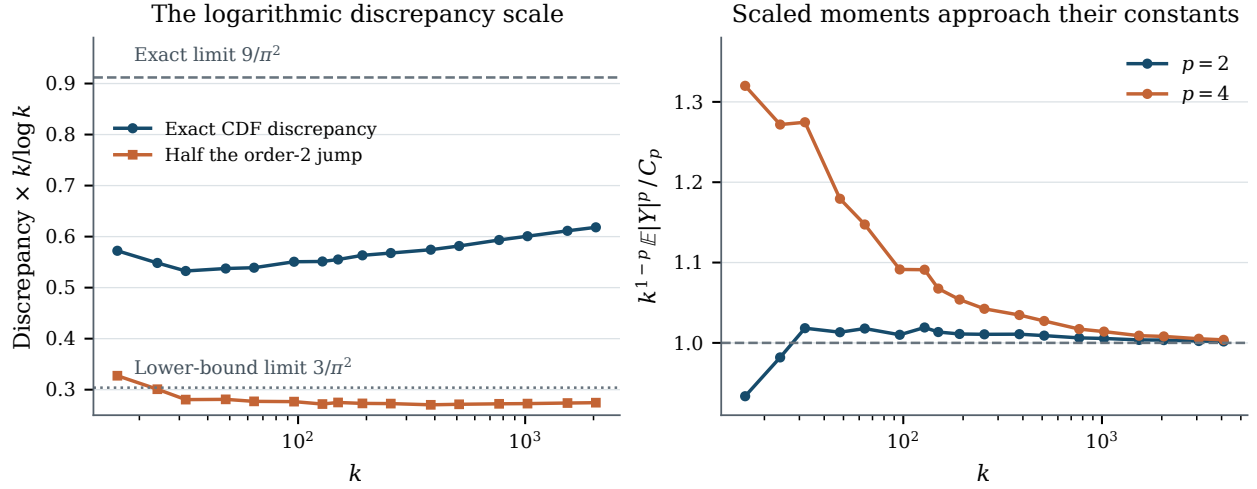


Figure 6: Left: discrepancy on the  $(\log k)/k$  scale, its order-two half-jump lower bound, and the proved limit  $9/\pi^2$ . Right: normalized second and fourth moments. Moment convergence in these plots is substantially faster than convergence of the scaled global discrepancy (manuscript 08; an observation, not a theorem). [The figure’s  $k^{1-p}\mathbb{E}|Y|^p/C_p$  is  $k^{1-p}\mathcal{M}_p(k)/\mathcal{M}_p$ ; its lower-bound limit  $3/\pi^2 = 1/(2\zeta(2))$  is that of half the central atom (25.7).]

| $k$   | $\mathcal{M}_1(k) - (\log k)/\pi$ |
|-------|-----------------------------------|
| 64    | −0.505972605                      |
| 256   | −0.573868502                      |
| 1024  | −0.605429214                      |
| 2048  | −0.613841428                      |
| 4096  | −0.619330934                      |
| Limit | −0.6273278444052116...            |

Table 15: The critical first-moment constant (manuscript 08). These entries use extended-precision cotangent summation; two cases are independently checked at 70-digit precision. [See Remark 32.1: “extended precision” is the platform’s `long double`.]

At  $k = 4096$ , the residual after subtracting the theorem’s logarithm and constant, multiplied by  $k/\log^2 k$ , is about 0.473444. This is consistent with the error scale in (24.11); it is not an empirical determination of an optimal remainder constant.

*Remark 32.1* (write; “extended precision”). [Note of the batch-98 write.] The program sums the cotangents for Table 15 and the edge-tail grid in NumPy’s `longdouble`. That type is an extended format on typical x86 Linux builds, but on Windows it is the ordinary 64-bit double; the package does not record the platform it was produced on. At the intake’s reruns on Windows, 16 of the 20 data rows of `data/07-zero-laws-first_absolute_moment.csv` and 20 of the 4,804 data rows of `data/07-zero-laws-edge_tail_grid.csv` differ from the shipped files in the last one or two of their seventeen printed digits; every other CSV file was byte-identical. The nine-digit values of Table 15 and the plotted curves are unaffected (the four PNG figures were pixel-identical), and the two 70-digit cross-checks do not depend on the type. The caption’s “extended precision” therefore depends on the platform, and the summary’s sentence “exact CSV values are reproducible” holds for the exact files, not for these two floating-point ones.

## 32.4 Reproduction and data

Manuscript 08’s package contains its L<sup>A</sup>T<sub>E</sub>X source and compiled PDF, the four vector figures and their PNG versions, the verification program, and the exact and numerical data used in its tables and plots. Its stated commands are `python3 code/reproduce.py` and a `latexmk` build of its own `.tex` file; its environment was SymPy 1.14.0, mpmath 1.3.0, NumPy and Matplotlib 3.10.8, and its finite checks and their scope are recorded in `data/07-zero-laws-verification_summary.json` and its text companion.

[Note of the batch-98 write: in this report the program, its data and its figures carry the prefix `07-zero-laws-`, the requirements file is `data/07-zero-laws-requirements.txt`, and its build script `code/07-zero-laws-build.sh` builds the delivered `.tex`, which is not shipped; the article here is a build of the merged text. The program writes unprefixed files into `data/` and `figures/` under the directory above its own, so it must be rerun on a copy with the delivered layout, as the report README describes, never in the report directory. At the intake’s rerun on such a copy (Python 3.13.5, the pinned SymPy, mpmath, NumPy 2.3.5 and Matplotlib), it exited with status 0 after about 45 s; nine CSV files were byte-identical, the text summary equal apart from line endings, the JSON summary equal apart from the recorded Python version, the four PNG figures pixel-identical and the four PDF figures identical apart from their creation dates; for the two remaining CSV files see Remark 32.1.]

## 33 Further questions and research

[Added 5 October 2026, batch 98.] This section collects every unproved claim of manuscript 08: its nine research questions, with its sketches; its explicit candidate for the correction of the fractional moments; its speculation about the last zeros; and its three numerical observations. Manuscript 08 states no claim that this write found wrong. Its question 8 is Part I’s Research question 5 and is printed there (see the paragraph on it below); its question 9 overlaps Section 10’s “practical order of attack” and is printed here with a note. Manuscript 08’s questions 1–7 and 9 are Research questions 9–16.

The original zero-distribution question is now answered for both normalizations, with fixed-order multiplicities, extreme tails, and a complete leading moment classification. The following questions

ask for finer information beyond those theorems. They are proposed research directions for this family, not claims that every formulation has been checked against all literature.

**Research question 9** (The next term in the global discrepancy; manuscript 08). Determine a second asymptotic term for  $\Delta_k^K$  beyond  $(9/\pi^2)(\log k)/k$ , and describe the denominators of maximizing support angles.

The sharp-constant proof uses compactness and harmonic averaging, and does not give a quantitative error in that constant. Table 14 suggests why such an error would matter in practice. A first target is a two-sided effective bound with a smaller error than  $o((\log k)/k)$ . A more ambitious target is an arithmetic description of the maximizing order and its transitions as  $k$  grows. The fixed rational endpoint formulas alone cannot settle a maximization over growing denominators.

[Note of the batch-98 write.] *Evidence, not proof:* in Table 14 the maximizing left limit moves from  $1/7$  (at  $k = 64, 128, 256$ ) to  $1/11$  (at  $k = 512, 1024, 2048$ ), and  $k\Delta_k^K/\log k$  is still  $0.618$  at  $k = 2048$  against the limit  $0.912$ . Manuscript 08 calls the movement “finite evidence, not a theorem”. The moving-endpoint example of Section 28, which realizes the sharp constant along  $t_k = 1/\log k$ , suggests that the maximizing angles drift to the endpoints with growing denominators; that is a heuristic, not a statement proved here.

**Research question 10** (Cyclotomic orders that grow with  $k$ ; manuscript 08). For which growing ranges of  $m = m_k$  can (24.6) be strengthened to a useful uniform relative asymptotic?

The present absolute error guarantees a relative equivalent for  $\psi(m) = o(\log k)$ , but says little when  $\psi(m)$  is comparable to or larger than  $\log k$ . The exact expression (25.4) retains more information: when  $m$  is a fixed fraction of  $k$ , only a bounded number of unreduced denominators are multiples of  $m$ . An analysis of that transition should keep the short coprimality intervals rather than replace them by a global density.

**Research question 11** (Local spacing with multiplicity weights; manuscript 08). What are the local spacing and pair-correlation laws of the distinct zeros, and how do their multiplicity weights change those laws?

The distinct angles are the reduced fractions of denominator at most  $k - 1$ , so their unweighted spacing belongs to the classical Farey setting. The present weights  $E_m(k)$  impose a different arithmetic sampling. A natural first comparison is between a uniformly selected distinct zero and a zero selected with its algebraic multiplicity. A satisfactory result should state the coordinate used for spacing, since the cotangent map strongly stretches the ends.

**Research question 12** (The last finitely many zeros; manuscript 08). Describe the successive extreme ordinates and their multiplicities at a resolution finer than  $Y/k$ .

Proposition 29.3 gives the largest ordinate exactly. The edge theorem controls order  $k^2$  roots, counted with multiplicity, and does not resolve a fixed extreme rank. The next denominators near  $k - 1$ , together with their short coprimality counts, are an explicit finite starting point. This problem may exhibit dependence on arithmetic features of  $k$  that are averaged out by  $\Xi$ . [Note of the batch-98 write: that last sentence is manuscript 08’s speculation; it is not proved or refuted here.]

**Research question 13** (Subleading fractional moments; manuscript 08). For  $0 < p < 1$ , determine the first correction to  $M_p(k) \rightarrow 2^{-p}/\cos(\pi p/2)$ , uniformly on useful ranges of  $p$ .

The edge expansion suggests a term of order  $k^{p-1}$ . One concrete candidate is the analytic continuation in  $p$  of the coefficient  $\mathcal{M}_p$  in (24.13), using the continued zeta function. This candidate is not proved here. A proof must subtract the Cauchy tail before integrating the edge profile and control the small- $x$  region uniformly. Merely continuing the formula established for  $p > 1$  does not justify the asymptotic statement for  $p < 1$ .

[Note of the batch-98 write.] The candidate, stated as a claim, is: for fixed  $0 < p < 1$ ,  $\mathcal{M}_p(k) = 2^{-p}/\cos(\pi p/2) + \mathcal{M}_p k^{p-1} + o(k^{p-1})$  with  $\mathcal{M}_p$  continued analytically; it is open. What is missing is the subtraction described above and a bound on the bulk error, which by Theorem 27.1 is of order  $(\log k)/k$  for smooth compactly supported statistics, smaller than  $k^{p-1}$  for every  $0 < p < 1$ ; but  $|Y|^p$  is neither compactly supported nor smooth at the origin, so that theorem does not apply to it directly. (Added after the independent check of 7 October 2026; numerical evidence, not proof.) At  $p = \frac{1}{2}$  the continued constant is  $\mathcal{M}_{1/2} = -2.9608$ , and exact multiplicities give  $(M_{1/2}(k) - 1)k^{1/2} = -2.395, -2.516, -2.573, -2.614, -2.694$  at  $k = 500, 1024, 1500, 2048, 4096$ ; the differences from  $\mathcal{M}_{1/2}$  are 2.04, 2.05, 2.05, 2.06, 2.05 times  $(\log k)/k^{1/2}$ , as a further term of order  $(\log k)/k$  in  $M_{1/2}(k)$  would make them. The data are consistent with the candidate at  $p = \frac{1}{2}$ ; they do not test other  $p$ .

**Research question 14** (An expansion through the critical moment order; manuscript 08). Can the three moment regimes be replaced by one uniform formula when  $p = 1 + \lambda/\log k$  and  $\lambda$  remains bounded?

Such a formula should recover the logarithm and the constant in (24.11) when  $\lambda \rightarrow 0$ , while matching the two noncritical regimes in an overlap region. The pole of  $\zeta(p)$  in  $\mathcal{M}_p$  identifies the cancellation that a uniform derivation must handle. This is a more precise objective than substituting a varying  $p$  into a fixed- $p$  theorem. [Here  $\lambda$  is a local parameter, not the ratio  $n/k$  of Part I.]

**Research question 15** (Smaller Hankel minors and other row models; manuscript 08). Which submaximal Hankel determinants, or variants of the column enumeration, retain a vertical-line zero law?

The exact resultant product is decisive for the maximal determinant. Smaller minors contain sums of products, and their zeros need not inherit its geometry. A first project is an exact classification for small deficits in the matrix size, supported by symbolic examples and then by a structural identity. Bounded entries and alternative column weights are separate variants whose spectra must first be established. [Note of the batch-98 write: this continues the last sentence of Part I's Research question 5, which asks which minors other than the maximal ones have a definite sign, and the variants of Part I's Research question 8 and of Part II's paragraph on weighted rows and restricted alphabets (Section 20).]

**Manuscript 08's question 8: a combinatorial model for the positive product.** Manuscript 08's eighth question asks whether the coefficients of (24.15) can be explained directly by counted objects or a sign-reversing involution. It says itself that the question “was already present in the repository report”: it is Part I's Research question 5, “A combinatorial explanation of determinant positivity” (Section 10), and is not repeated here. Manuscript 08's addition to it, which arithmetic factors a model would have to explain, is printed there in a dated note. The zero analysis does not answer it.

**Research question 16** (Effective bounds and formal certificates; manuscript 08). Develop explicit constants and a proof-producing implementation for the algebraic and arithmetic parts of the theory.

The determinant, cyclotomic multiplicities, exact degree identities, and full-support CDF comparisons are finite objects. They offer a practical first formalization stage. The next stage is the primitive-lattice error with an explicit constant, followed by the CDF bound (26.2). The sharp constant requires the harmonic approximation argument; the numerical verification suite does not currently formalize that analytic step. Effective constants would also permit a principled switch between exact data and asymptotic evaluation. [Note of the batch-98 write: the first stage extends the first target named by both earlier manuscripts, the reduced denominator, the resultants and the maximal Hankel product (Section 10, “A practical order of attack”, and Section 20). Nothing of Part III is formalized, and this paragraph is a proposal, not a plan of the repository.]

**Observations that are not theorems.** Manuscript 08 records three numerical observations, kept here as such: (i) the drift of the maximizing endpoint from  $1/7$  to  $1/11$  (Table 14; Research question 9); (ii) the residual 0.473444 of the first moment at  $k = 4096$  after scaling by  $k/\log^2 k$ , “not an empirical determination of an optimal remainder constant” (Section 32); whether  $O(\log^2 k/k)$  in (24.11) is the true order is open; and (iii) “moment convergence in these plots is substantially faster than convergence of the scaled global discrepancy” (Figure 6), a finite comparison with no rate proved for either side beyond the theorems above.

## Part IV

# Packed matrices on the total-weight $2N$ diagonal: fixed-order asymptotics, controlled inversion, and joint marked limits

## 34 Provenance, scope and notation of Part IV

[Added 6 October 2026, batch 108.] Part IV was added in the write of batch 108. It is built from Report 169 of a session bundle of research reports, which studies the diagonal  $a_N = T(2N, N)$  of this report, OEIS A261784 [3]. Report 169 was written without sight of this report, and several of its theorems are theorems of Parts I and II proved by a different route: its expansion of  $a_N$  to every fixed order is Corollaries 7.4 and 17.1, its column mean and variance are Corollary 7.5 at  $n/k = 2$ , and its Lambert- $W$  inversion and integer brackets are Propositions 18.1 and 18.2. These are printed as marked second routes, with remarks of the write that identify them (Table 16); Report 169’s own proofs are printed because its marked theorem reuses them. New relative to Parts I–III are the two further terms of the inversion, an explicit pole-tail bound, a Morse/Lagrange finite formula for the coefficients, a complex-uniform expansion with two markers (cells with entry at least 2, and columns), the Poisson limit of the number of such cells jointly with the column count, and the transfer to entries at least 3 and to the total excess. Everything of Report 169 is printed below, in the letters of Table 17. Bracketed notes marked “[Note of the batch-108 write]” and remarks marked “[write]” are the write’s; all other text of Part IV is Report 169’s.

### 34.1 The manuscript

Report 169 is the archive `Packed_Matrices_Asymptotics_Inverses_and_Joint_Limits_Source.zip` (571,859 bytes; 27 files in an inner directory `report169/`), which arrived in commit 60f54ea06

and was placed in this report by commit 602e5bd0f (batch 108). Its article is titled *Report 169: Packed matrices on the total-weight  $2n$  diagonal*, subtitled *Fixed-order asymptotics, controlled inversion, and joint marked limits* (16 pages; main file `source/report169.tex`, 686 lines, 60 labels; dated 3 October 2026). Its author line and PDF author field are empty; it names no person, tool or addressee. It has no pin: it names no ProveIt commit, cites nothing in the repository and none of Parts I–III, which were written on the same day. Its literature list overlaps this report’s (Munarini, Poneti and Rinaldi’s Proposition 29, Louchard, Kotěšovec’s leading formula). It is not an edition of any earlier source of this report: 0.66 % of its word 8-grams occur in this report, and 0.13 % of this report’s in it.

**Status and non-claims of Report 169, kept.** Its statements concern fixed-order asymptotics: no convergence of the coefficient series, no optimal truncation and no effective numerical threshold are claimed;  $O$ -constants and onsets are existence statements. It proves no local limit theorem for the column count; the coupling of the total excess with the repeated-cell count says nothing about unbounded moments. Its decimal values are not interval-certified. It credits the leading equivalent to Kotěšovec (OEIS, 18 February 2017, updated 20 April 2024) and the exact transform to Munarini, Poneti and Rinaldi; it names Cerbai–Claesson, Cameron–Prelberg–Stark, Louchard and Greenhill–McKay as precedents; it makes “no claim of historical priority or global novelty”; and it says that Kotěšovec’s mathematics page and the A261784 history were not available to its review. Exact computations and independent checking “do not amount to formal proof-assistant verification or certified interval arithmetic” (Section 43).

### 34.2 What Part IV adds

Table 16: Report 169 against Parts I–III. “Second route” means the same statement proved here by a different argument; the remark named identifies it.

| Report 169 (Part IV)              | Parts I–III                                               | Status here                                                                                 |
|-----------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Transform (36.1)                  | (2.4), credited to [4, Proposition 29]                    | the same identity; a dummy-column derivation                                                |
| Lemma 36.1                        | (7.5)                                                     | the same exponent, with an explicit sum over the poles (Remark 36.2)                        |
| Theorem 35.1, $\beta_1$ in (35.4) | Corollary 7.4 (first order), Corollary 17.1 (every order) | second route; (35.4) is the same rational function as Corollary 7.4’s $\beta$ (Remark 35.2) |
| (38.3), (38.5), (38.6)            | (7.16), (17.4)                                            | a second finite prescription, on the $Q$ -plane contour; $\beta_2, \beta_3$ agree           |
| Proposition 39.1                  | Proposition 18.1 (first correction)                       | first step a second route; the $x_0^{-2}$ and $x_0^{-3}$ terms are new (Remark 39.3)        |
| Theorem 39.2                      | Proposition 18.2                                          | the same statement, second route                                                            |
| Theorem 40.1                      | (7.25): real column weight, first order                   | new: complex column weight, a marker for cells $\geq 2$ , every order (Remark 40.2)         |
| Theorem 41.1, (41.2)              | Corollary 7.5 at $n/k = 2$                                | second route, identical formulas (Remark 41.2)                                              |
| Theorem 41.1, (41.3)              | —                                                         | new: Poisson( $Lr$ ) limit of the repeated cells, independent of the Gaussian column count  |

| Report 169 (Part IV)        | Parts I–III | Status here |
|-----------------------------|-------------|-------------|
| (41.5) and the total excess | —           | new         |

Two questions of Part II are touched; neither is answered (Section 20). The repeated-cell count is a statistic of the matrices themselves whose generating function carries the factor  $e^{Lr(\omega-1/2)}$ , and so bears on “A combinatorial realization of the latent defect”; it does not construct the decorated model that paragraph asks for (Remark 41.3). Theorem 41.1 answers, on the single ray  $n/k = 2$ , an analogue of “Joint laws with the number of columns” in which the repeated-cell count replaces the defect. Dated notes at both paragraphs say so.

### 34.3 Notation of Part IV

Report 169 writes  $n$  for the number of rows,  $\rho$  for  $\log 2$ ,  $t$  for the saddle,  $u$  and  $v$  for its two markers, and uses many letters that Parts I–III use for other objects. Part IV is printed in the letters of Parts I and II:  $N$  rows and total  $2N$  as in Section 7.5,  $L = \log 2$ , the saddle  $r$  of (7.18), the column weight  $u$  of Part I, the corrections  $\beta_j$  of Corollary 17.1, and the inverse core  $\Lambda$ ,  $\varpi$ ,  $x_0$  of Section 18. Table 17 lists every rename with the false reading the delivered letter would have in this report. The renames are changes of letter only; no normalization is changed.

Table 17: Report 169’s symbols and the letters used for them in Part IV. “False reading” gives what the Report-169 letter means elsewhere in this report.

| Notion                                       | Part IV                                           | Rep. 169                    | False reading if copied                                         |
|----------------------------------------------|---------------------------------------------------|-----------------------------|-----------------------------------------------------------------|
| Number of rows; total                        | $N$ ; $2N$                                        | $n$ ; $2n$                  | $n$ is the total mass                                           |
| Total in the triangle                        | $n$ , in $T(n, k)$                                | $N$ , in $T(N, k)$          | $N$ is the diagonal index                                       |
| A261784                                      | $a_N$                                             | $a_n$                       | —                                                               |
| $\log 2$                                     | $L$                                               | $\rho$                      | $\rho = L/\sqrt{L^2 + 8}$ is the pole-bound ratio               |
| Saddle $2 + W_0(-2e^{-2})$                   | $r$                                               | $t$                         | $t$ is a local variable                                         |
| Its half (contour radius)                    | $q = r/2$                                         | $q$                         | $q = 1/(1 + u_0)$ in Section 6; $q$ indexes $F_q$               |
| $\mathcal{D}_Q^2 \Phi(q) = 2(r - 1)/(2 - r)$ | $b_Q$                                             | $b$                         | $b = 2(r - 1)$ at $\lambda = 2$ (Corollary 7.4)                 |
| Growth constant; amplitude                   | $d_*$ , $c_*$                                     | $d$ , $C$                   | $d$ is a Hankel size; $C$ counts columns                        |
| Poisson mean of the repeated cells           | $Lr = 2v$                                         | $\lambda = \rho t$          | $\lambda = n/k$ ; $v = Lr/2$                                    |
| Column variance per row                      | $2\sigma_\infty^2 = \frac{1-L}{2L^2}$             | $\sigma^2$                  | $\sigma_\infty^2$ of (6.3) is per unit of total                 |
| Diagonal corrections                         | $\beta_j$                                         | $C_j$                       | $C_J$ is a bracket constant                                     |
| Logarithmic coefficients                     | $\beta_1, \beta_2, \beta_3$                       | $h_1, h_2, h_3$             | $h = n/k$ ; $\beta_1$ is (18.2)                                 |
| Ordered Bell (Fubini) numbers; weighted      | $F_j$ ; $F_j(u)$                                  | $B_j$ ; $B_j(v)$            | $B$ , $B(u)$ are pole-bound constants                           |
| Bernoulli numbers                            | $B_{2m}$                                          | $B_{2r}$                    | as above; $r$ is the saddle                                     |
| Fubini poles                                 | $L^{(\ell)} = L + 2\pi i\ell$ ;<br>$L_u^{(\ell)}$ | $R_\ell$ ; $R_\ell(v)$      | $R_p(w)$ are kernel polynomials                                 |
| $\log(1 + 1/u)$                              | $L_u$                                             | $R(v)$                      | —                                                               |
| Pole contributions                           | $H_N(\cdot)$ , $H_{N,\ell}$                       | $H_n(\cdot)$ , $H_{n,\ell}$ | $H_{n,k}(z)$ is the kernel, $H_{k,s}(u)$ the Hankel determinant |

| Notion                                            | Part IV                                                         | Rep. 169                               | False reading if copied                                     |
|---------------------------------------------------|-----------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------|
| Pole ratio $L/\sqrt{L^2 + 4\pi^2}$                | $\vartheta_0$                                                   | $\kappa$                               | $\kappa_j$ are cumulants                                    |
| Marked pole-tail base; bound on $ L_u $           | $\vartheta_*$ ; $L^* = 0.71$                                    | $\beta_*$ ; $r_*$                      | $\beta_j$ are corrections                                   |
| Kernel; marked kernel                             | $\mathfrak{K}_N(Q)$ ;<br>$\mathfrak{K}_N(Q, \omega, u)$         | $K_n(Q)$ ; $K_n$                       | $K$ is a compact set; $K_n$ also counts columns in Rep. 169 |
| Kernel coefficients                               | $\mathfrak{k}_j$                                                | $k_j$                                  | $k$ is the number of rows                                   |
| Euler operator in $Q$                             | $\mathcal{D}_Q$                                                 | $D$                                    | $D$ is overloaded in Part I;<br>$\mathcal{D} = r \, d/dr$   |
| Amplitude $(1 - Q)^{-1/2} e^{LQ}$                 | $\mathfrak{A}(Q)$                                               | $A(Q)$                                 | $A(n, k)$ is a count                                        |
| Logarithmic corrections; their exponential        | $\mathfrak{h}_j(Q)$ ; $\mathfrak{p}_j(Q)$                       | $H_j(Q)$ ; $P_j(Q)$                    | $P_j(t)$ , $P_j(w)$ are other polynomials                   |
| Morse-coordinate auxiliaries                      | $\mathfrak{f}(x)$ , $\mathfrak{S}(x)$ , $\mathfrak{F}_j(x)$     | $f(x)$ , $S(x)$ ,<br>$F_j(x)$          | $f(z) = e^z - 1$ ; $F_k(z; u)$ is a generating function     |
| Gaussian-lemma amplitude; outputs                 | $\mathfrak{a}_N(\theta, p)$ ; $\mathfrak{c}_j(p)$               | $a_n(\theta, p)$ ; $\gamma_j(p)$       | $a_N$ is A261784; $\gamma$ in Corollary 8.2                 |
| Gaussian-lemma Taylor variable; phase             | $\varrho$ ; $\phi$                                              | $r$ ; $f$                              | $r$ is the saddle                                           |
| Marker of cells with entry $\geq 2$               | $\omega$                                                        | $u$                                    | $u$ is the column weight; $\omega_j$ are roots of unity     |
| Marker of columns                                 | $u$                                                             | $v$                                    | $v = Lr/2$                                                  |
| Marker of cells with entry $\geq 3$               | $\omega'$                                                       | $s$                                    | $s$ is the window parameter and the Hankel shift            |
| Marked count                                      | $\mathsf{T}_N(\omega, u)$                                       | $A_n(u, v)$                            | $A(n, k)$ allows zero rows                                  |
| Marked leading term                               | $\mathcal{V}_N(\omega, u)$                                      | $L_n(u, v)$                            | $L = \log 2$                                                |
| Marked constants                                  | $d_*(u)$ , $c_*(\omega, u)$ ,<br>$\beta_j(\omega, u)$           | $d(v)$ , $C(u, v)$ ,<br>$C_j(u, v)$    | —                                                           |
| Marker polydisc                                   | $\mathcal{U}$                                                   | $\mathcal{P}$                          | $\mathcal{P}(K, Q)$ in Part III                             |
| Inverse of $w(\cdot, \omega)$                     | $\mathbf{z}(w, \omega)$ , $\mathbf{z}_3$                        | $\psi$ , $\psi_3$                      | $\psi$ is Dedekind's function                               |
| Column weight factor $u/(1 + u)$                  | $p_u$                                                           | $r$                                    | $r$ is the saddle                                           |
| Cells $\geq 2$ ; $\geq 3$ ; columns; total excess | $\mathcal{J}_N$ ; $\mathcal{J}_N^{(3)}$ ; $C$ ; $\mathcal{E}_N$ | $J_n$ ; $L_n^{(3)}$ ; $K_n$ ;<br>$E_n$ | $J_q$ is a block count, $J$ an order;<br>$E_j$ , $E_n(w)$   |
| Target value; inverse core; $1 + W$               | $Y$ ; $\Lambda$ , $\varpi$ , $x_0$ ;<br>$1 + \varpi$            | $x$ ; $Y$ , $W$ , $v$ ; $L$            | as in Section 18; $L = \log 2$                              |
| Core; truncated logarithm; its root               | $\mathfrak{g}_0$ ; $\mathfrak{g}_{J'}$ ; $\tilde{x}_{J'}(Y)$    | $F$ ; $G_H$ ; $s_H(x)$                 | $F_k$ , $G_k$ (gcd sums)                                    |
| Smooth comparator; its root                       | $\mathcal{A}^{[J]}$ ; $x_J(Y)$                                  | $f_M$ ; $v_M$                          | as in (18.1)                                                |
| Integer threshold; bracket                        | $\mathcal{N}(Y)$ ; $\epsilon_J$ , $C_J$                         | $N(x)$ ; $\delta_M$ ,<br>$K_M$         | as in (18.7)                                                |
| Truncation order                                  | $J$                                                             | $M$                                    | $M$ is the zero-row count                                   |
| Nonempty-column count                             | $A(n, k)$                                                       | $c_{N,k}$                              | $c_k$ is the fixed- $k$ amplitude                           |

Local letters whose meaning is fixed where they occur, and which do not refer to the objects of Parts I–III with the same name: the coefficient variables  $z$ ,  $w = -\log(1 - z)$  (Sections 36–37) or  $w = \log g(z, \omega)$  (Section 40),  $x$ ,  $y$ ,  $s$ ; the contour variable  $Q$  (not the Stirling index  $Q$  of Corollary 7.5); the generic pole argument  $R$ ; the phase functions  $I$ ,  $\Phi$ ,  $h$  of Section 37 ( $\Phi$  is unindexed, unlike the cyclotomic  $\Phi_m$  of Part III);  $F$ ,  $G$  in the proof of Theorem 40.1;  $\varphi_1$ ,  $\varphi_2$ ,  $\theta$ ,  $\tau$  in the proof of Theorem 41.1; and  $\epsilon$ ,  $\delta$ ,  $\delta_0$ ,  $p$ ,  $c$ ,  $c_1$  and  $\text{const}$  (Report 169's  $C_k$ ,  $C_M$ ) as small or generic constants. The cell series  $g(z, \omega)$  is not the function  $g$  of Section 7. Report 169 uses  $\epsilon$  both for  $N^{-1/2}$  and for  $1/N$ ; each use is stated where it occurs.

### 35 The model, attribution, and results

A *packed matrix* here has entries in  $\mathbb{Z}_{\geq 0}$  and has no zero row or zero column. Both line orders are retained; repeated lines are allowed. Let  $T(n, k)$  count such matrices of total  $n$  with exactly  $k$  rows. We study

$$a_N = T(2N, N) \quad (N \geq 1), \quad a_0 = 1.$$

There are at most  $2N$  columns, so each set being counted is finite. Reading columns as weakly increasing words on the ordered row alphabet identifies this definition with A261784. In particular,

$$a_0, a_1, a_2, a_3, a_4, a_5 = 1, 2, 66, 5546, 893490, 235804122.$$

The sequence is neither a binary-matrix count nor a lonesum or triangular-matrix count. No row or column factorial is divided out. [Note of the batch-108 write: these packed matrices are the matrix compositions of Part I,  $T(n, k)$  is the triangle A261781 of Section 1, and  $a_N$  is the diagonal of Section 7.5.]

Set

$$L = \log 2, \quad r = 2 + W_0(-2e^{-2}), \quad \mathbf{q} = r/2, \quad b_Q = \frac{2\mathbf{q} - 1}{1 - \mathbf{q}}. \quad (35.1)$$

Here  $W_0$  is the principal real Lambert branch. Equivalently  $\mathbf{q}$  is the unique positive solution of  $-\log(1 - \mathbf{q}) = 2\mathbf{q}$  in  $(0, 1)$ ; it lies in  $(1/2, 1)$ . The zero solution is excluded. Define

$$d_* = \frac{4}{L^2 r(2 - r)}, \quad c_* = \frac{e^{Lr/2}}{4\pi L \sqrt{r - 1}}, \quad Lr, \quad 2\sigma_\infty^2 = \frac{1 - L}{2L^2}. \quad (35.2)$$

For orientation only,  $r \approx 1.5936242600$ ,  $d_* \approx 12.8556891737$ ,  $c_* \approx 0.2588649231$ , and  $Lr \approx 1.1046161627$ . [Note of the batch-108 write:  $r, d_*, c_*$  are (7.18), (7.19) and (7.20) ( $e^{Lr/2} = 2^{r/2}$ );  $Lr = 2v$  with  $v = Lr/2$  of Section 7.5; and  $2\sigma_\infty^2$  is twice the constant of (6.3), because it is measured per row, and there are two units of total per row.]

**Theorem 35.1** (All fixed orders; second route to Corollaries 7.4 and 17.1). *For every fixed integer  $J \geq 0$  there are  $\beta_j \in \mathbb{Q}(L, r)$ , with  $\beta_0 = 1$ , such that*

$$a_N = c_* d_*^N \frac{(N!)^2}{N} \left( \sum_{j=0}^J \frac{\beta_j}{N^j} + O_J(N^{-J-1}) \right). \quad (35.3)$$

*The first correction is*

$$\begin{aligned} \beta_1 = \frac{r}{48(r-1)^3} & (2L^2 r^4 - 9L^2 r^3 + 12L^2 r^2 - 5L^2 r \\ & - 6Lr^3 + 6Lr^2 + 12Lr - 12L \\ & - 2r^3 - 3r^2 + 24r - 24). \end{aligned} \quad (35.4)$$

*Section 38 gives finite exact representations for  $\beta_2, \beta_3$  and a prescription for each fixed coefficient.*

The  $O$ -constant and the threshold in  $N$  may depend on  $J$ . This theorem asserts neither convergence of the coefficient series nor validity of optimal truncation. It supplies no explicit numerical threshold for a prescribed error.

*Remark 35.2* (write; the identification with Parts I and II). Theorem 35.1 is Corollary 7.4 at first order and Corollary 17.1 at every order: the constants  $d_*$ ,  $c_*$  are the same, and the coefficients agree. Indeed, two expansions of the same sequence in powers of  $1/N$  with the same leading factor have the same coefficients (subtract them and multiply by  $N^j$  successively), so Theorem 35.1 and Corollary 17.1 together force Report 169's coefficients to be the  $\beta_j$  of Corollary 17.1; this is why the same letter is used. For  $\beta_1$  more is true: the right side of (35.4) and the expression for  $\beta$  in Corollary 7.4, with  $b = 2(r - 1)$ ,  $\kappa_3$ ,  $\kappa_4$  and  $v = Lr/2$  substituted, are the same rational function of two *independent* indeterminates  $L$  and  $r$  (checked with SymPy at the write), so no relation between  $L$  and  $r$  is used; (35.4) is that expression expanded. The intake evaluated it to  $-0.5218067015564826638891441$ , equal to  $\beta$  in every digit, and the write found Report 169's exact expressions for  $\beta_1$ ,  $\beta_2$ ,  $\beta_3$  (`data/08-packed-companion-coefficient-formulas.txt`) equal to the fifty-digit values of Section 17 to within  $10^{-50}$ . The route is different. Part I reduces the positive sum to the analytic amplitude  $H_{n,k}$  on rays and integrates on  $|z| = r$  (Section 7); Part II integrates the entire kernel  $W_n$ ; Report 169 uses the exact pole lattice of the Fubini numbers, the substitution  $z = 1 - e^{-LQ/N}$ , Euler–Maclaurin summation of  $\sum_\nu \log(1 - \nu Q/N)$  and a Gaussian lemma on the circle  $|Q| = q$  (Section 37), and specializes to  $\lambda = 2$  from the start, so no central-binomial correction enters. Its proof is printed in full because Section 40 reuses it with complex markers.

**What was already known.** The inspected OEIS entry [3] credits the equivalent  $a_N \sim c_* d_*^N (N!)^2/N$  to Kotěšovec, dated 18 February 2017 and updated 20 April 2024. It gives a Lambert- $W$  formula for  $d_*$  and a numerical amplitude matching (35.2). That leading equivalent is credited here, not claimed as a discovery. The exact transform used below is Proposition 29, equation (36), of Munarini–Poneti–Rinaldi [4]. Exact row/column marking is also part of the Burge-matrix enumeration of Cerbai–Claesson [19, 20]. Cameron–Prellberg–Stark [17, 18] provide closely related proportional-dimension binary saddles; Louchard [6] proves column Gaussian laws in a fixed-row-dimension composition model; Greenhill–McKay [22] analyze sparse specified-margin multiple entries. These precedents are essential context for the refinements and probability consequences, rather than instances of a new generic Gaussian or Poisson phenomenon. Section 43 explains the model distinctions and the limits of the literature comparison. [Note of the batch-108 write: Part I credits the same two results in Section 1 and Table 1. The Cerbai–Claesson, Cameron–Prellberg–Stark, Greenhill–McKay, Duchamp–Hivert–Thibon, Khera–Lundberg–Melczer and Hwang–Jin–Schlosser citations are Report 169's; the write did not reread them.]

### 36 An exact finite transform and pole reduction

Let  $F_j = \sum_{c=0}^j c! \{c \atop j\}$  be the ordered Bell numbers, with exponential generating function  $(2 - e^z)^{-1}$ . We use unsigned Stirling numbers of the first kind. The known identity is

$$T(n, k) = \frac{k!}{n!} \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \left\{ j \atop k \right\} F_j. \quad (36.1)$$

One direct route starts with  $m$  dummy columns: for the ordinary cell series  $(1 - z)^{-1}$  the no-zero-row series is  $((1 - z)^{-m} - 1)^k$ . Weighting dummy columns by  $2^{-m-1}$  erases all inserted zero columns, because  $\sum_{m \geq C} 2^{-m-1} \binom{m}{C} = 1$ . Expanding  $(e^{mw} - 1)^k$  at  $w = -\log(1 - z)$  and using

$$[z^n] w^j = \frac{j!}{n!} \begin{bmatrix} n \\ j \end{bmatrix}$$

gives (36.1). This also fixes the ordered-line normalization. [Note of the batch-108 write: (36.1) is (2.4); Part I derives it from  $F_q = \frac{1}{2} \sum_b b^q 2^{-b}$  and a binomial difference.]

For every  $j \geq 1$ , the exact pole identity is

$$F_j = \frac{j!}{2} \sum_{\ell \in \mathbb{Z}} (L + 2\pi i \ell)^{-j-1}. \quad (36.2)$$

Indeed  $(2 - e^z)^{-1} = (1 + \coth((L - z)/2))/4$ ; the differentiated partial fractions of  $\coth$  converge absolutely and locally uniformly away from the poles. Coefficient extraction for  $j \geq 1$  gives (36.2). The exceptional undifferentiated constant does not enter when  $N \geq 1$ . [Note of the batch-108 write: this is Theorem q2:thm:fubini of the transseries volume [11], quoted in the note after (7.5).]

Put  $L^{(\ell)} = L + 2\pi i \ell$ . The ordinary Stirling-column identity

$$\sum_{j \geq N} \left\{ \begin{matrix} j \\ N \end{matrix} \right\} x^j = \frac{x^N}{\prod_{\nu=1}^N (1 - \nu x)}$$

then gives

$$a_N = \sum_{\ell \in \mathbb{Z}} H_N(L^{(\ell)}), \quad H_N(R) = \frac{N!}{2R} [z^{2N}] \frac{(w/R)^N}{\prod_{\nu=1}^N (1 - \nu w/R)}. \quad (36.3)$$

Only  $N \leq j \leq 2N$  contribute to the coefficient. Thus this is a finite coefficient calculation with an absolutely convergent pole sum, not an assumption about convergence of a factorially divergent generating series in  $N$ .

**Lemma 36.1** (Exponential principal-pole reduction). *With  $\vartheta_0 = L/\sqrt{L^2 + 4\pi^2} < 0.110$ ,*

$$\frac{|a_N - H_N(L)|}{H_N(L)} \leq \sum_{\ell \neq 0} \left( \frac{L}{|L^{(\ell)}|} \right)^{N+1} = O(\vartheta_0^N). \quad (36.4)$$

*Proof.* Each finite-transform summand for  $H_N(L)$  is nonnegative. Replacing  $L$  by  $L^{(\ell)}$  changes its factor  $L^{-j-1}$  to  $(L^{(\ell)})^{-j-1}$ ; since  $j \geq N$ , termwise absolute values give the inequality. If  $r_\ell = L/|L^{(\ell)}|$ , then  $r_\ell \leq \vartheta_0$  for  $\ell \neq 0$  and

$$\sum_{\ell \neq 0} r_\ell^{N+1} \leq \vartheta_0^{N-1} \sum_{\ell \neq 0} r_\ell^2 \quad (N \geq 1).$$

The last sum converges. In particular all nonreal poles are controlled together.  $\square$

*Remark 36.2* (write). Part I's estimate (7.5) replaces  $F_q$  by its first pole with relative error  $O(\vartheta^q)$  for every fixed  $\vartheta > \vartheta_0$ , by Cauchy's estimate; Lemma 36.1 uses the exact identity (36.2) to give an explicit bound by the pole sum itself, with the same exponential rate. ( $r_\ell$  here is a local ratio, not  $r_k = 1 - a_k$  of Section 1.)

## 37 The saddle and a complete fixed-order remainder argument

Make the exact substitution  $w = LQ/N$ ,  $z = 1 - e^{-w}$  in the residue for  $H_N(L)$ . It gives

$$H_N(L) = \frac{N! N^N}{2L^{2N+1}} \frac{1}{2\pi i} \oint_{|Q|=q} Q^{-N} \prod_{\nu=1}^N (1 - \nu Q/N)^{-1} \mathfrak{K}_N(Q) \frac{dQ}{Q}, \quad (37.1)$$

where

$$\mathfrak{K}_N(Q) = \left( \frac{w}{1 - e^{-w}} \right)^{2N} \frac{w}{e^w - 1}. \quad (37.2)$$

The product poles have  $Q = N/\nu \geq 1 > \mathbf{q}$ . For sufficiently large  $N$  the inverse map is analytic on the relevant shrinking  $w$ -disk, has winding one, and creates no other singularity inside the contour. This is a residue change of coordinate.

### 37.1 Uniform Euler–Maclaurin expansion

Let

$$I(Q) = 1 + \frac{(1-Q)\log(1-Q)}{Q}, \quad \Phi(Q) = I(Q) - \log Q, \quad \mathcal{D}_Q = Q \frac{d}{dQ}. \quad (37.3)$$

Fix  $\delta_0 > 0$  with  $\mathbf{q} + \delta_0 < 1$ . For  $f_Q(x) = -\log(1-Qx)$ , all fixed  $x$  derivatives are bounded uniformly on  $|Q| \leq \mathbf{q} + \delta_0$ ,  $0 \leq x \leq 1$ , since  $|1-Qx| \geq 1-\mathbf{q}-\delta_0$ . The right-endpoint Euler–Maclaurin formula, using one term beyond the last retained term to bound its remainder, gives for each fixed  $p \geq 0$

$$\begin{aligned} \sum_{\nu=1}^N f_Q(\nu/N) &= NI(Q) - \frac{1}{2} \log(1-Q) \\ &\quad + \sum_{m=1}^p \frac{B_{2m} Q^{2m-1} \{(1-Q)^{-(2m-1)} - 1\}}{2m(2m-1)N^{2m-1}} + O(N^{-2p-1}). \end{aligned} \quad (37.4)$$

Here  $B_{2m}$  are Bernoulli numbers. The error is uniform on that disk. Formula (37.4) always means a finite truncation with its stated remainder.

Put  $h(Q) = -\log(1-Q)/Q$ . Direct differentiation gives

$$\mathcal{D}_Q \Phi = h - 2, \quad \mathcal{D}_Q h = (1-Q)^{-1} - h.$$

Therefore  $\mathcal{D}_Q \Phi(\mathbf{q}) = 0$  and  $\mathcal{D}_Q^2 \Phi(\mathbf{q}) = b_Q > 0$ . Expanding (37.2) uniformly on the disk gives  $\log \mathfrak{K}_N = LQ + O(N^{-1})$ , and to any fixed order it gives analytic coefficient functions with uniform remainders. The leading amplitude is

$$\mathfrak{A}(Q) = (1-Q)^{-1/2} e^{LQ}. \quad (37.5)$$

### 37.2 The entire contour, not only a local saddle

For  $a = \nu \mathbf{q}/N$ ,

$$\frac{|1 - ae^{i\theta}|^2}{(1-a)^2} = 1 + \frac{4a \sin^2(\theta/2)}{(1-a)^2}.$$

For a positive fraction of the indices  $\nu$ ,  $a \in [\mathbf{q}/2, \mathbf{q}]$ . On this compact interval the logarithm of the right side is at least a positive constant times  $\sin^2(\theta/2)$ . Consequently

$$\left| \frac{\prod_{\nu=1}^N (1 - \nu \mathbf{q} e^{i\theta}/N)^{-1}}{\prod_{\nu=1}^N (1 - \nu \mathbf{q}/N)^{-1}} \right| \leq e^{-cN \sin^2(\theta/2)} \quad (37.6)$$

for some  $c > 0$ , uniformly on the full circle. The kernel is uniformly bounded there. Thus  $|\theta| \geq N^{-2/5}$  contributes  $O(e^{N\Phi(\mathbf{q})} e^{-c_1 N^{1/5}})$  to the integral. Dividing by the eventual main scale  $e^{N\Phi(\mathbf{q})} N^{-1/2}$  introduces a factor  $\sqrt{N}$ , which is absorbed into a smaller positive exponential constant. There is a unique dominant arc. [Note of the batch-108 write: the index  $\nu$  of the products is Report 169's  $r$ , which is the saddle here.]

### 37.3 A Gaussian lemma without logarithmic loss

We give the remainder step explicitly, in a form also valid for compact complex marker sets. Write a local integral after removal of its constant exponential factor as

$$\int e^{N\phi(\theta)} \mathbf{a}_N(\theta, p) d\theta, \quad \phi(0) = \phi'(0) = 0, \quad \phi''(0) = -b_Q < 0,$$

where  $p$  lies in a compact parameter set. Suppose the phase is analytic in a fixed neighborhood and the amplitude has, uniformly there,

$$\mathbf{a}_N(\theta, p) = \sum_{j=0}^J \mathbf{a}_j(\theta, p) N^{-j} + O(N^{-J-1}),$$

with bounded fixed derivatives of the analytic  $\mathbf{a}_j$ . These hypotheses follow from (37.4) and the kernel expansion; choose sufficiently many Euler–Maclaurin terms for the desired  $J$ .

Set  $\epsilon = N^{-1/2}$ ,  $\theta = \epsilon y$ , and

$$h_*(z) = \frac{\phi(z) + b_Q z^2/2}{z^2}, \quad h_*(0) = 0, \quad H(\varrho, y) = y^2 h_*(\varrho y).$$

Choose  $\delta > 0$  so that  $|h_*(z)| \leq b_Q/8$  for  $|z| \leq \delta$ . On the central interval  $|y| \leq N^{1/10}$ , for  $0 \leq \varrho \leq \epsilon$  and large  $N$ ,

$$|H(\varrho, y)| \leq b_Q y^2/8, \quad |\partial_{\varrho}^m H(\varrho, y)| \leq \text{const}_m |y|^{m+2} \quad (m \geq 1).$$

Apply Taylor's theorem in the real variable  $\varrho$  through degree  $2J+1$  to

$$e^{H(\varrho, y)} \sum_{j=0}^J \varrho^{2j} \mathbf{a}_j(\varrho y, p).$$

The product and chain rules bound every required derivative by a fixed polynomial in  $|y|$  times  $e^{b_Q y^2/8}$ . After multiplication by  $e^{-b_Q y^2/2}$  the Taylor remainder is bounded by

$$\text{const}_J \epsilon^{2J+2} (1 + |y|^{e_J}) e^{-3b_Q y^2/8}. \quad (37.7)$$

The omitted amplitude remainder has the same type of integrable Gaussian bound. The coefficient of  $\epsilon^k$  is a polynomial in  $y$  of parity  $k$ , so odd degrees integrate to zero on the symmetric interval. Extending the retained polynomial–Gaussian integrals to  $\mathbb{R}$  costs an exponentially small error. The integral of (37.7) is  $O(\epsilon^{2J+2})$  with no logarithmic or interval-length loss. Restoring  $d\theta = \epsilon dy$  proves

$$\int_{|\theta| \leq N^{-2/5}} e^{N\phi(\theta)} \mathbf{a}_N(\theta, p) d\theta = N^{-1/2} \left( \sum_{j=0}^J \mathbf{c}_j(p) N^{-j} + O(N^{-J-1}) \right).$$

All constants are uniform over the compact parameter set. If the leading saddle amplitude is bounded away from zero, division gives the corresponding relative expansion.

### 37.4 Normalization

The leading integral in (37.1) is

$$\frac{e^{N\Phi(\mathbf{q})} \mathfrak{A}(\mathbf{q})}{\sqrt{2\pi N b_Q}}.$$

Use

$$\frac{N^N}{N!} = \frac{e^N}{\sqrt{2\pi N}} \exp \left\{ -\frac{1}{12N} + \frac{1}{360N^3} + \cdots \right\},$$

again as a fixed-order expansion, together with  $I(\mathbf{q}) = 2\mathbf{q} - 1$ ,  $1 - \mathbf{q} = e^{-2\mathbf{q}}$ , and  $(1 - \mathbf{q})b_Q = 2\mathbf{q} - 1$ . The result is exactly (35.2). Lemma 36.1 and the Gaussian remainder prove Theorem 35.1.

### 38 Exact correction coefficients

This section specifies the coefficients by finite algebra, without numerical fitting. Let  $\mathfrak{h}_j(Q)$  denote the coefficient of  $s^j$  in the following formal logarithmic correction:

$$\begin{aligned} & \sum_{m \geq 1} \frac{B_{2m} Q^{2m-1} \{(1-Q)^{-(2m-1)} - 1\}}{2m(2m-1)} s^{2m-1} - \frac{LQ}{2} s \\ & - \sum_{m \geq 1} \frac{B_{2m} (LQ)^{2m}}{2m(2m)!} (2s^{2m-1} + s^{2m}) - \sum_{m \geq 1} \frac{B_{2m}}{2m(2m-1)} s^{2m-1}. \end{aligned} \quad (38.1)$$

The lines come from Euler–Maclaurin, the odd and even kernel terms, and inverse Stirling normalization, respectively. For a fixed degree only finitely many terms enter. Define

$$\mathfrak{p}_j(Q) = [s^j] \exp \left( \sum_{i \geq 1} \mathfrak{h}_i(Q) s^i \right).$$

In particular,

$$\begin{aligned} \mathfrak{h}_1 &= \frac{Q^2}{12(1-Q)} - \frac{LQ}{2} - \frac{(LQ)^2}{12} - \frac{1}{12}, \\ \mathfrak{h}_2 &= -\frac{(LQ)^2}{24}, \quad \mathfrak{h}_3 = -\frac{Q^3 \{(1-Q)^{-3} - 1\}}{360} + \frac{(LQ)^4}{1440} + \frac{1}{360}, \\ \mathfrak{p}_0 &= 1, \quad \mathfrak{p}_1 = \mathfrak{h}_1, \quad \mathfrak{p}_2 = \mathfrak{h}_2 + \frac{1}{2} \mathfrak{h}_1^2, \quad \mathfrak{p}_3 = \mathfrak{h}_3 + \mathfrak{h}_1 \mathfrak{h}_2 + \frac{1}{6} \mathfrak{h}_1^3. \end{aligned} \quad (38.2)$$

#### 38.1 A Gaussian prescription at every fixed order

For a standard normal  $U$ ,  $\beta_m$  is the coefficient of  $\epsilon^{2m}$  in

$$\begin{aligned} & \mathbb{E} \left[ \frac{\mathfrak{A}(\mathfrak{q} e^{i\epsilon U / \sqrt{b_Q}})}{\mathfrak{A}(\mathfrak{q})} \sum_{j \geq 0} \mathfrak{p}_j(\mathfrak{q} e^{i\epsilon U / \sqrt{b_Q}}) \epsilon^{2j} \right. \\ & \left. \times \exp \left\{ \sum_{k \geq 3} \frac{(\mathcal{D}_Q^k \Phi)(\mathfrak{q})}{k!} \left( \frac{iU}{\sqrt{b_Q}} \right)^k \epsilon^{k-2} \right\} \right]. \end{aligned} \quad (38.3)$$

Only finitely many derivatives and Gaussian moments can contribute. Differentiating via  $\mathcal{D}_Q h = (1-Q)^{-1} - h$  and then putting  $h(\mathfrak{q}) = 2$  eliminates all logarithms at the saddle. Odd Gaussian moments vanish and even ones are  $(2i-1)!!$ ; therefore  $\beta_m \in \mathbb{Q}(L, \mathfrak{q}) = \mathbb{Q}(L, r)$ . [Note of the batch-108 write: here  $\epsilon = N^{-1/2}$ . As with (7.16), the expression is expanded and truncated in  $\epsilon$  first and the Gaussian expectation is applied to each polynomial coefficient (Section 21, “Formal Gaussian algebra”).]

#### 38.2 An alternative finite coefficient representation

The following form makes  $\beta_2$  and  $\beta_3$  compact and is convenient for exact calculation. Put

$$\mathfrak{f}(x) = \Phi(\mathfrak{q}(1+x)) - \Phi(\mathfrak{q}), \quad \mathfrak{G}(x) = \frac{2\mathfrak{f}(x)}{b_Q x^2}, \quad \mathfrak{F}_j(x) = \frac{\mathfrak{A}(\mathfrak{q}(1+x)) \mathfrak{p}_j(\mathfrak{q}(1+x))}{\mathfrak{A}(\mathfrak{q})(1+x)}. \quad (38.4)$$

The apparent singularity in  $\mathfrak{S}$  is removable and  $\mathfrak{S}(0) = 1$ . Then

$$\beta_m = \sum_{j=0}^m \frac{(-1)^{m-j} (2(m-j)-1)!!}{b_Q^{m-j}} [x^{2(m-j)}] \mathfrak{F}_j(x) \mathfrak{S}(x)^{-(m-j)-1/2}, \quad (38.5)$$

with  $(-1)!! = 1$ . In particular the promised exact finite expressions are

$$\begin{aligned} \beta_2 &= \mathfrak{p}_2(\mathfrak{q}) - \frac{1}{b_Q} [x^2] \mathfrak{F}_1 \mathfrak{S}^{-3/2} + \frac{3}{b_Q^2} [x^4] \mathfrak{F}_0 \mathfrak{S}^{-5/2}, \\ \beta_3 &= \mathfrak{p}_3(\mathfrak{q}) - \frac{1}{b_Q} [x^2] \mathfrak{F}_2 \mathfrak{S}^{-3/2} + \frac{3}{b_Q^2} [x^4] \mathfrak{F}_1 \mathfrak{S}^{-5/2} - \frac{15}{b_Q^3} [x^6] \mathfrak{F}_0 \mathfrak{S}^{-7/2}. \end{aligned} \quad (38.6)$$

All symbols in these finite representations are explicit analytic series defined in (37.3), (37.5), (38.2), and (38.4). Expand only through  $x^6$ , using  $\log(1 - \mathfrak{q}) = -2\mathfrak{q}$ . The companion contains the resulting exact rational expressions as well as an exact calculator. [Note of the batch-108 write: shipped as `data/08-packed-companion-frozen-coefficients_C0_C3.json` and `data/08-packed-companion-coefficient-formulas.txt`, in the variables  $q = \mathfrak{q}$  and  $R = L$ , and `code/08-packed-companion-packed_matrix.py`.]

For completeness, the analytic Morse coordinate  $y = x\sqrt{\mathfrak{S}(x)}$  satisfies  $\mathfrak{f}(x) = b_Q y^2/2$ . Residue substitution gives

$$[y^{2k}] \mathfrak{F}_j(x(y)) x'(y) = [x^{2k}] \mathfrak{F}_j(x) \mathfrak{S}(x)^{-k-1/2}.$$

The even Gaussian moments on the local steepest-descent contour contribute  $(-1)^k (2k-1)!! b_Q^{-k}$ ; summing the amplitude orders proves (38.5). This also gives an algebraically independent way to check (38.3).

Uncertified decimal illustrations, deliberately shorter than working precision, are

$$\beta_1 \approx -0.521806701556483, \quad \beta_2 \approx 0.128288828088706, \quad \beta_3 \approx -0.002443538422849.$$

The exact formulas, rather than the rounded values, define the coefficients. [Note of the batch-108 write: they agree in every printed digit with Corollary 7.4, (9.1) and (17.2). Equation (38.5) is a third finite prescription beside (7.16) and (17.4).]

### 39 Lambert- $W$ inversion and integer thresholds

Taking logarithms in Theorem 35.1 and using Stirling's series gives

$$\log a_N = 2N \log N + (\log d_* - 2)N + \log(2\pi c_*) + \frac{\tilde{\beta}_1}{N} + \frac{\tilde{\beta}_2}{N^2} + \frac{\tilde{\beta}_3}{N^3} + O(N^{-4}), \quad (39.1)$$

where

$$\tilde{\beta}_1 = \beta_1 + \frac{1}{6}, \quad \tilde{\beta}_2 = \beta_2 - \frac{1}{2}\beta_1^2, \quad \tilde{\beta}_3 = \beta_3 - \beta_1\beta_2 + \frac{1}{3}\beta_1^3 - \frac{1}{180}. \quad (39.2)$$

The  $\log N$  term cancels exactly. Omitting  $\log(2\pi c_*)$ ,  $1/6$ , or  $-1/180$  would change the stated inverse corrections. [Note of the batch-108 write:  $\tilde{\beta}_1$  is (18.2), and (39.1) is (18.3) with two more terms;  $2N \log N + (\log d_* - 2)N = 2N \log(\alpha N)$  with  $\alpha = \sqrt{d_*}/e$ . The write evaluates  $\tilde{\beta}_2 = -0.0078522888059223\dots$  and  $\tilde{\beta}_3 = 0.0115833114844424\dots$  from Report 169's exact expressions.]

For large positive  $Y$ , define

$$\Lambda = \frac{1}{2} \log \frac{Y}{2\pi c_*}, \quad \varpi = W_0 \left( \frac{\sqrt{d_*}}{e} \Lambda \right), \quad x_0 = \frac{\Lambda}{\varpi}. \quad (39.3)$$

The leading logarithmic core  $\mathfrak{g}_0(x) = 2x \log x + (\log d_* - 2)x + \log(2\pi c_*)$  satisfies  $\mathfrak{g}_0(x_0) = \log Y$  and  $\mathfrak{g}'_0(x_0) = 2(1 + \varpi)$ . [Note of the batch-108 write:  $\Lambda, \varpi, x_0$  are those of (18.4); Report 169 calls them  $Y, W, v$ , calls the target value  $x$ , and writes  $L$  for  $1 + \varpi$ .]

**Proposition 39.1** (Controlled smooth expansion). *For a fixed integer  $J' \geq 3$ , let  $\tilde{x}_{J'}(Y)$  be the eventual real root of*

$$\mathfrak{g}_{J'}(x) = \mathfrak{g}_0(x) + \sum_{j=1}^{J'} \tilde{\beta}_j x^{-j} = \log Y,$$

where the  $\tilde{\beta}_j$  are the coefficients of the fixed-order logarithmic expansion and  $\mathfrak{g}_0$  is the exact core above. Writing  $x = \tilde{x}_{J'}(Y)$ , one has

$$\begin{aligned} x = x_0 - \frac{\tilde{\beta}_1}{2x_0(1 + \varpi)} - \frac{\tilde{\beta}_2}{2x_0^2(1 + \varpi)} \\ - \frac{1}{x_0^3} \left( \frac{\tilde{\beta}_3}{2(1 + \varpi)} + \frac{\tilde{\beta}_1^2}{4(1 + \varpi)^2} + \frac{\tilde{\beta}_1^2}{8(1 + \varpi)^3} \right) + O\left(\frac{x_0^{-4}}{1 + \varpi}\right). \end{aligned} \quad (39.4)$$

In particular stopping after the  $x_0^{-2}$  term gives error  $O(x_0^{-3}/(1 + \varpi))$ . Higher fixed orders are obtained by finite substitution or Newton recursion; their coefficients are rational functions of  $1 + \varpi$ , not just constants times  $(1 + \varpi)^{-1}$ .

*Proof.* Substitute  $x = x_0 + \delta$  in  $\mathfrak{g}_0(x) + \tilde{\beta}_1/x + \tilde{\beta}_2/x^2 + \tilde{\beta}_3/x^3 + \dots = \log Y$ . Here  $\mathfrak{g}''_0(x_0) = 2/x_0$ . The successive leading shifts are  $-\tilde{\beta}_1/(2x_0(1 + \varpi))$  and  $-\tilde{\beta}_2/(2x_0^2(1 + \varpi))$ . At order  $x_0^{-3}$ , the residual is

$$\frac{1}{x_0^3} \left( \tilde{\beta}_3 + \frac{\tilde{\beta}_1^2}{2(1 + \varpi)} + \frac{\tilde{\beta}_1^2}{4(1 + \varpi)^2} \right),$$

from the unshifted  $\tilde{\beta}_3$ , the shifted  $\tilde{\beta}_1/x$ , and the quadratic core term. Dividing by  $2(1 + \varpi)$  gives (39.4). After substitution the remaining residual is  $O(x_0^{-4})$ , while the derivative is  $2(1 + \varpi) + O(x_0^{-2})$  in the relevant neighborhood. The mean-value theorem gives the claimed root error. The same residual-over-derivative argument controls any fixed Newton/substitution order.  $\square$

This smooth expansion alone is not an exact ceiling formula for the discrete sequence. To make the relation precise, for fixed  $J$  set

$$\mathcal{A}^{[J]}(x) = c_* d_*^x \frac{\Gamma(x+1)^2}{x} \sum_{j=0}^J \beta_j x^{-j} \quad (39.5)$$

on its eventually positive increasing real branch. Write  $x_J(Y) = (\mathcal{A}^{[J]})^{-1}(Y)$ . [Note of the batch-108 write: this is the smooth comparator (18.1) and its root  $x_J(Y)$  of Section 18.]

**Theorem 39.2** (Safe integer inverse; second route to Proposition 18.2). *Let  $\mathcal{N}(Y) = \min\{N \geq 0 : a_N \geq Y\}$ . For every fixed  $J$  there is  $C_J > 0$  such that, for all sufficiently large  $Y$ ,*

$$\lceil x_J - \epsilon_J \rceil \leq \mathcal{N}(Y) \leq \lceil x_J + \epsilon_J \rceil, \quad \epsilon_J = C_J \frac{x_J^{-J-1}}{\log x_J}. \quad (39.6)$$

*The constant and the threshold are existence statements, not numerically certified bounds.*

*Proof.* The logarithmic derivative of (39.5) is  $2 \log x + \log d_* + O(x^{-1})$ , hence is positive for large  $x$ . Theorem 35.1 implies

$$|\log a_N - \log \mathcal{A}^{[J]}(N)| \leq \Theta_J N^{-J-1}$$

for some  $\Theta_J$  and every sufficiently large integer  $N$ . Also  $a_{N+1}/a_N \sim d_* N(N+1) > 1$ , so the exact counts are eventually strictly increasing.

Choose  $C_J$  sufficiently large relative to  $\Theta_J$ . On  $|x - x_J| \leq 2$ , the mean-value theorem and uniform comparability of  $x$  and  $x_J$  show that the change in  $\log \mathcal{A}^{[J]}$  over a distance  $\epsilon_J$  exceeds the error bound with a strict margin. Apply this directly to the last integer below  $x_J - \epsilon_J$  and the first integer at or above  $x_J + \epsilon_J$ : the former has  $a_N < Y$ , and the latter has  $a_N \geq Y$ . Eventual monotonicity handles all more distant integers; the finitely many earlier values are below  $Y$  when  $Y$  is large. This proves both inequalities, including equality thresholds.  $\square$

Away from the shrinking neighborhoods of integer boundaries the two ceilings coincide. Near a threshold the interval can leave two adjacent possibilities; claiming a universal exact ceiling of a finite asymptotic expansion would discard that genuine uncertainty. Formula (39.4) approximates these smooth roots to the indicated order when the logarithmic truncation in (39.5) is taken sufficiently far.

*Remark 39.3* (write; relation to Section 18). (i) Through its  $x_0^{-1}$  term, (39.4) is Proposition 18.1: the same core, the same  $\Lambda$ ,  $\varpi$ ,  $x_0$ , and the correction  $-\tilde{\beta}_1/(2x_0(1+\varpi))$ ; its error  $O(x_0^{-2}/\log x_0)$  there is  $O(x_0^{-2}/(1+\varpi))$  here, since  $1+\varpi = \log(\alpha x_0) + 1$  grows like  $\log x_0$ . The  $x_0^{-2}$  and  $x_0^{-3}$  terms are new; the write rederived the  $x_0^{-3}$  coefficient by matching powers of  $x_0^{-1}$  in  $\mathfrak{g}_0(x_0 + \delta) - \mathfrak{g}_0(x_0) = 2(1+\varpi)\delta + \delta^2/x_0 + O(\delta^3/x_0^2)$  against  $-\sum_j \tilde{\beta}_j(x_0 + \delta)^{-j}$ . (ii) Proposition 39.1 concerns the root  $\tilde{x}_{J'}$  of the truncated logarithm, and Section 18 the root  $x_J$  of  $\mathcal{A}^{[J]}$ ; for  $J, J' \geq 3$  the two satisfy the same expansion (39.4). Indeed, by Stirling's series at real  $x$ ,  $2 \log \Gamma(x+1) - \log x = 2x \log x - 2x + \log(2\pi) + \frac{1}{6x} - \frac{1}{180x^3} + O(x^{-5})$ , and  $\log(\sum_{j \leq J} \beta_j x^{-j}) = \beta_1 x^{-1} + (\beta_2 - \frac{1}{2}\beta_1^2)x^{-2} + (\beta_3 - \beta_1\beta_2 + \frac{1}{3}\beta_1^3)x^{-3} + O(x^{-4})$  for  $J \geq 3$ ; hence  $\log \mathcal{A}^{[J]}(x) = \mathfrak{g}_3(x) + O(x^{-4})$ , and  $\mathfrak{g}_{J'} = \mathfrak{g}_3 + O(x^{-4})$  as well. Both functions have derivative  $2(1+\varpi) + O(x_0^{-1})$  on a fixed neighbourhood of  $x_0$ , so their roots differ by  $O(x_0^{-4}/(1+\varpi))$  by the mean-value theorem. (iii) Theorem 39.2 is Proposition 18.2 with the same comparator and the same bracket. The two proofs differ only in monotonicity: Report 169 uses the eventual growth  $a_{N+1}/a_N \sim d_* N(N+1)$ , Section 18 an injection that gives  $a_N < a_{N+1}$  for every  $N$ . The non-claim after Theorem 39.2 is the one after Proposition 18.2. Proposition 18.1 is an instance of the transseries volume's perturbed inversion [11], as the note at the head of Section 18 says; Proposition 39.1 carries the same substitution two steps further, and no novelty is claimed for the method.

## 40 A complex-uniform two-marker theorem

For a uniformly chosen counted matrix let  $\mathcal{J}_N$  be the number of cells with entry at least 2, and  $C$  the number of columns. Define the finite polynomial

$$\mathsf{T}_N(\omega, u) = \sum_M \omega^{\mathcal{J}_N(M)} u^{C(M)}, \quad \mathsf{T}_N(1, 1) = a_N.$$

The cell function and its local logarithm are

$$g(z, \omega) = \frac{1 + (\omega - 1)z^2}{1 - z}, \quad w(z, \omega) = \log g(z, \omega),$$

with the logarithm defined by its power series at  $z = 0$ . Let

$$L_u = \text{Log}(1 + 1/u), \quad L_1 = L, \quad L_u^{(\ell)} = L_u + 2\pi i \ell.$$

All branches below continue from  $u = 1$ . [Note of the batch-108 write:  $\mathsf{T}_N(1, u) = T(2N, N; u)$ , the column-weighted count of Section 7.6, and  $L_u$  is Part I's  $\log(1 + 1/u)$ , continued to complex  $u$  by the principal branch.]

**Theorem 40.1** (Uniform marked expansion). *On the compact polydisc*

$$\mathcal{U} = \{|\omega - 1| \leq 2.1, \quad |u - 1| \leq 0.02\}, \quad (40.1)$$

define

$$\begin{aligned} d_*(u) &= \frac{4}{L_u^2 r(2 - r)}, \\ c_*(\omega, u) &= \frac{\exp\{L_u r(\omega - 1/2)\}}{2\pi(1 + u)L_u \sqrt{r - 1}}, \quad \mathcal{V}_N(\omega, u) = c_*(\omega, u) d_*(u)^N \frac{(N!)^2}{N}. \end{aligned} \quad (40.2)$$

For every fixed  $J \geq 0$ , there are holomorphic coefficient functions  $\beta_j(\omega, u)$  on a neighborhood of  $\mathcal{U}$ , with  $\beta_0(\omega, u) = 1$  and  $\beta_j(1, 1) = \beta_j$ , such that

$$\frac{\mathsf{T}_N(\omega, u)}{\mathcal{V}_N(\omega, u)} = \sum_{j=0}^J \beta_j(\omega, u) N^{-j} + O_J(N^{-J-1}) \quad (40.3)$$

uniformly on  $\mathcal{U}$ . On every smaller compact polydisc the error admits termwise differentiation of any fixed marker order, preserving its  $N$ -order.

#### 40.1 Exact marked decomposition

With  $m$  dummy columns the no-zero-row series is  $(g(z, \omega)^m - 1)^N$ . Put  $p_u = u/(1 + u)$ . A genuine  $C$ -column matrix receives weight

$$\frac{1}{1 + u} \sum_{m \geq C} \binom{m}{C} p_u^m = u^C.$$

On  $|u - 1| \leq 0.02$ ,  $|p_u| \leq 1.02/1.98 < 1$ . Each fixed  $z$  coefficient grows at most polynomially with  $m$ , so this series and the coefficient operations converge uniformly. Expanding the exponential yields

$$\mathsf{T}_N(\omega, u) = N! \sum_{j=N}^{2N} \left\{ \begin{matrix} j \\ N \end{matrix} \right\} \frac{\mathsf{F}_j(u)}{j!} [z^{2N}] w(z, \omega)^j, \quad \mathsf{F}_j(u) = \sum_C C! \left\{ \begin{matrix} j \\ C \end{matrix} \right\} u^C. \quad (40.4)$$

The exponential generating function of  $\mathsf{F}_j(u)$  is  $(1 - u(e^z - 1))^{-1}$ . Its partial fractions give, for  $j \geq 1$ ,

$$\frac{\mathsf{F}_j(u)}{j!} = \frac{1}{1 + u} \sum_{\ell \in \mathbb{Z}} (L_u^{(\ell)})^{-j-1}.$$

Thus

$$\mathsf{T}_N = \sum_{\ell} \mathsf{H}_{N, \ell}, \quad \mathsf{H}_{N, \ell} = \frac{N!}{(1 + u)L_u^{(\ell)}} [z^{2N}] \frac{(w/L_u^{(\ell)})^N}{\prod_{\nu=1}^N (1 - \nu w/L_u^{(\ell)})}. \quad (40.5)$$

As before, all exchanges with the pole sum use the finite range  $N \leq j \leq 2N$ . [Note of the batch-108 write:  $\mathsf{F}_j(u)$  is Part I's weighted Fubini number  $\mathsf{F}_q(u)$  of the proof of Corollary 7.5.]

## 40.2 An explicit complex pole-tail estimate

Since  $dL_u/du = -1/(u(1+u))$ , integration along the segment from 1 to  $u$  gives

$$|L_u - L| \leq \frac{0.02}{0.98 \cdot 1.98} < 0.011.$$

In particular  $|L_u| < L^* = 0.71$ ,  $|1+u| \geq 1.98$ , and  $|L_u^{(\ell)}| \geq 2\pi|\ell| - L^*$ . Let  $w_0(z) = -\log(1-z)$ . Coefficientwise absolute values satisfy

$$|w(z, \omega)|_{\text{coeff}} \preceq 3w_0(\sqrt{2.1}z). \quad (40.6)$$

The  $w_0(z)$  part is bounded by  $w_0(\sqrt{2.1}z)$ , and the absolute coefficient of  $z^{2k}$  from  $\log(1+(\omega-1)z^2)$  is at most  $2.1^k/k$ , exactly that of  $2w_0(\sqrt{2.1}z)$ . Convolution gives

$$|[z^{2N}]w^j| \leq 2.1^N 3^j [z^{2N}]w_0^j.$$

Positivity is used only in the unmarked comparison core  $H_N(L)$ , not in a complex marked integral. Since  $3L/|L_u^{(\ell)}| < 1$  for  $\ell \neq 0$  and  $j \geq N$ ,

$$|H_{N,\ell}(\omega, u)| \leq \frac{2}{|1+u|} H_N(L) 2.1^N \frac{L}{|L_u^{(\ell)}|} \left( \frac{3L}{|L_u^{(\ell)}|} \right)^N. \quad (40.7)$$

The sum over  $\ell \neq 0$  is uniformly bounded by a constant times the  $|\ell| = 1$  exponential scale: after taking out that scale, the remaining ratios can be dominated by their third powers for  $N \geq 2$ , giving a convergent series.

The explicit  $c_*(\omega, u)$  is holomorphic and nonzero on a neighborhood of  $\mathcal{U}$ , so its modulus is bounded away from zero. The unmarked asymptotic already proved gives

$$\frac{H_N(L)}{|\mathcal{V}_N(\omega, u)|} \leq \text{const} \cdot \left( \frac{L^*}{L} \right)^{2N}.$$

Hence, uniformly on the entire complex polydisc,

$$\frac{|\sum_{\ell \neq 0} H_{N,\ell}(\omega, u)|}{|\mathcal{V}_N(\omega, u)|} = O(0.83^N), \quad (40.8)$$

because the derived exponential base is

$$\vartheta_* = \frac{3 \cdot 2.1 \cdot (L^*)^2}{L(2\pi - L^*)} < \frac{105861}{128110} < 0.83.$$

The first strict bound uses only  $L > 0.69$  and  $\pi > 3.14$ . It is therefore not dependent on floating-point certification. [Note of the batch-108 write: 105861/128110 is the value of the middle expression at  $L = 0.69$ ,  $\pi = 3.14$ ; with the true values  $\vartheta_* = 0.8221 \dots$ ]

## 40.3 Uniform analytic inverse and full-circle kernel

On  $|z| = 0.01$  and  $|\omega - 1| \leq 2.1$ ,

$$|w(z, \omega) - z| \leq \frac{0.01^2}{2(1 - 0.01)} + \frac{2.1 \cdot 0.01^2}{1 - 2.1 \cdot 0.01^2} < 0.000263,$$

$$|w_z(z, \omega) - 1| \leq \frac{0.01}{0.99} + \frac{2 \cdot 2.1 \cdot 0.01}{1 - 2.1 \cdot 0.01^2} < 0.053.$$

For a target  $|w| < 0.009$ , Rouché's theorem applied against  $z - w$  gives exactly one solution in  $|z| < 0.01$ , since  $|z - w| > 0.001$  on the boundary. Its derivative is nonzero. Uniqueness and the analytic implicit function theorem give a jointly holomorphic inverse  $z = z(w, \omega)$ , with

$$z(w, \omega) = w - (\omega - 1/2)w^2 + O(w^3) \quad (40.9)$$

uniformly on compact subsets. The strict bounds hold on a slightly larger open marker domain as well.

For  $N \geq 64$ ,  $|L_u Q/N| \leq 0.71q/64 < 0.009$  on the entire closed  $Q$ -disk. The last inequality can also be seen from  $q < 0.8$ , which follows from  $\log 5 > 1.6$ . Substitute  $w = L_u Q/N$  into the residue. The inverse has its unique zero at  $w = 0$  and the rational product has no pole before  $Q = 1$ . Thus

$$\begin{aligned} H_{N,0} &= \frac{N! N^N}{(1+u)L_u^{2N+1}} \frac{1}{2\pi i} \oint_{|Q|=q} Q^{-N} \prod_{\nu=1}^N (1 - \nu Q/N)^{-1} \mathfrak{K}_N(Q, \omega, u) \frac{dQ}{Q}, \\ \mathfrak{K}_N &= \left( \frac{w}{z(w, \omega)} \right)^{2N+1} z_w(w, \omega). \end{aligned} \quad (40.10)$$

Both inverse factors are analytic and nonzero near zero, with value one there. Let  $F = \log(w/z)$  and  $G = \log z_w$ , choosing their branches at zero. Then  $F = (\omega - 1/2)w + O(w^2)$  and  $G = O(w)$ . With  $\epsilon = 1/N$ , the expression

$$\exp\{(2/\epsilon + 1)F(L_u Q\epsilon, \omega) + G(L_u Q\epsilon, \omega)\}$$

has a removable singularity at  $\epsilon = 0$ . Taylor expansion on compact sets gives

$$\mathfrak{K}_N = e^{(2\omega-1)L_u Q} \left( \sum_{j=0}^J \mathfrak{k}_j(Q, \omega, u) N^{-j} + O(N^{-J-1}) \right), \quad \mathfrak{k}_0 = 1, \quad (40.11)$$

uniformly on the full circle and marker domain. In particular the kernel is uniformly bounded on that circle.

The phase and product are exactly those of Section 37. Their all-arc bound is unchanged, and the leading amplitude becomes

$$(1 - Q)^{-1/2} \exp\{(2\omega - 1)L_u Q\}.$$

It is nonzero at  $Q = q$ , uniformly bounded away from zero over the compact marker set. The Gaussian lemma therefore applies uniformly. Its normalization gives (40.2); adding (40.8) proves (40.3).

#### 40.4 Normalized residuals and their logarithm

After subtracting the finite expansion, the residual in (40.3) is holomorphic:  $\mathbb{T}_N$  is a polynomial,  $\mathbb{V}_N$  is holomorphic and nonzero, and all coefficient functions are holomorphic. On

$$|\omega - 1| \leq 2.05, \quad |u - 1| \leq 0.01, \quad (40.12)$$

Cauchy estimates using the positive margins to (40.1) preserve the  $O(N^{-J-1})$  bound for every fixed mixed derivative. Slightly smaller circles can be used at the margins.

There is a second point needed for cumulants. The  $J = 0$  estimate puts  $\mathbb{T}_N/\mathbb{V}_N$  in  $|z - 1| < 1/2$  uniformly for large  $N$ . Its analytic logarithm near 1 therefore exists and is  $O(N^{-1})$ . Fixed derivatives of this logarithm are also  $O(N^{-1})$  on (40.12). This supplies differentiated logarithmic, rather than merely pointwise, asymptotics.

*Remark 40.2* (write; relation to Section 7.6). At  $\omega = 1$  and real  $u$  near 1, Theorem 40.1 is the column-weighted expansion (7.25) at  $n = 2N$ ,  $k = N$ : both have the factor  $e^{L_u r/2}/((1+u)L_u^{2N+1})$  in their  $u$ -dependence, and the saddle  $\mathbf{q} = r/2$  does not move with  $u$ , because  $u$  enters only through the scale  $L_u$  of  $w$ . Part I proves that expansion at first order for  $u$  in a compact real interval around 1, on every compact ray; Theorem 40.1 proves it at every order for complex  $u$  in the small disc  $|u - 1| \leq 0.02$ , on the single ray  $n/k = 2$ , and adds the marker  $\omega$ . Two specializations lie inside  $\mathcal{U}$ : at  $\omega = 0$ ,  $\mathbb{T}_N(0, 1)$  counts packed 0-1 matrices with  $N$  rows and total  $2N$ , and  $\mathbb{T}_N(0, 1)/a_N \rightarrow e^{-Lr}$ ; at  $\omega = 1/2$  the exponential factor of  $c_*(\omega, 1)$  disappears (Remark 41.3). The write checked the first by exact counts:  $\mathbb{T}_N(0, 1)/a_N = 0.345668, 0.338424, 0.336044, 0.334861$  for  $N = 10, 20, 30, 40$ , against  $e^{-Lr} = 0.331338$ , with  $N(e^{Lr}\mathbb{T}_N(0, 1)/a_N - 1) = 0.4325, 0.4277, 0.4261, 0.4253$ , as an  $O(1/N)$  correction requires.

## 41 Columns, repeated cells, and total excess

Dividing (40.3) by its value at  $(1, 1)$  gives

$$\mathbb{E}[\omega^{\mathcal{J}_N u^C}] = \frac{2L}{(1+u)L_u} e^{r\{L_u(\omega-1/2)-L/2\}} \left(\frac{L}{L_u}\right)^{2N} (1 + O(N^{-1})), \quad (41.1)$$

with a holomorphic differentiable relative remainder on inner compact polydiscs.

**Theorem 41.1** (Moment corrections and joint limit). *The column count satisfies*

$$\begin{aligned} \mathbb{E}C &= \frac{N}{L} - \frac{1}{2} + \frac{1}{2L} - \frac{r}{4} + O(N^{-1}), \\ \text{Var } C &= 2N\sigma_\infty^2 - \frac{1}{4} - \frac{1}{4L} + \frac{1}{4L^2} + \frac{r}{8} + O(N^{-1}). \end{aligned} \quad (41.2)$$

Furthermore

$$\left(\mathcal{J}_N, \frac{C - N/L}{\sqrt{2\sigma_\infty^2 N}}\right) \xrightarrow{d} (P, Z), \quad P \sim \text{Poisson}(Lr), \quad Z \sim N(0, 1), \quad (41.3)$$

where  $P$  and  $Z$  are independent. Exact centering by  $\mathbb{E}C$  and scaling by  $\sqrt{\text{Var } C}$  give the same limit.

*Proof.* Set  $s = \log u$  and  $L_{e^s} = \log(1 + e^{-s})$ . Then  $L_{e^s} = L$ ,  $\frac{d}{ds}L_{e^s} = -1/2$  and  $\frac{d^2}{ds^2}L_{e^s} = 1/4$  at  $s = 0$ . The exponential part has  $\varphi_1(s) = -2\log L_{e^s}$ , with

$$\varphi_1'(0) = 1/L, \quad \varphi_1''(0) = (1 - L)/(2L^2) = 2\sigma_\infty^2.$$

At  $\omega = 1$  the variable part of the log-amplitude is  $\varphi_2(s) = -\log(1 + e^s) - \log L_{e^s} + rL_{e^s}/2$ . Its first two derivatives are exactly the constant terms in (41.2). Differentiation in  $s$ , justified by Section 40.4, produces ordinary cumulants, so no factorial-to-ordinary variance adjustment is missing.

For the joint limit fix real  $\theta, \tau$  and substitute

$$\omega = e^{i\theta}, \quad u = \exp(i\tau/\sqrt{2\sigma_\infty^2 N}).$$

The entire  $\omega$  characteristic circle satisfies  $|\omega - 1| \leq 2 < 2.05$  and hence lies strictly inside the uniform domain, including  $\omega = -1$ . The marker  $u$  lies there for large  $N$ . The linear part of  $N(\varphi_1(s) - \varphi_1(0))$  cancels the centering phase; its quadratic part tends to  $-\tau^2/2$  and its remaining part is  $O(N^{-1/2})$ .

The amplitude tends to  $\exp\{Lr(e^{i\theta} - 1)\}$ , while its mixed  $L_u(\omega - 1/2)$  part differs from its limit by  $O(N^{-1/2})$ . Thus the joint characteristic function tends to

$$\exp\{Lr(e^{i\theta} - 1) - \tau^2/2\}.$$

Lévy's continuity theorem proves (41.3). Formula (41.2) justifies changing the centering and scaling.  $\square$

*Remark 41.2* (write; second route to Corollary 7.5 at  $n/k = 2$ ). At  $n = 2N$ ,  $k = N$ , (7.23) reads  $\mathbb{E}C = \frac{2N+1}{2L} - \frac{1}{2} - \frac{r}{4} + O(N^{-1})$  and (7.24) reads  $\text{Var } C = \frac{(2N+1)(1-L)}{4L^2} - \frac{1}{4} + \frac{r}{8} + O(N^{-1})$ . Expanding,  $\frac{2N+1}{2L} = \frac{N}{L} + \frac{1}{2L}$  and  $\frac{(2N+1)(1-L)}{4L^2} = 2N\sigma_\infty^2 + \frac{1}{4L^2} - \frac{1}{4L}$ , so the two are (41.2) term for term. The marginal Gaussian law in (41.3) is (7.22) at  $n = 2N$ :  $n/(2L) = N/L$  and  $n(1-L)/(4L^2) = 2\sigma_\infty^2 N$ . The routes differ: Part I uses the exact mixture over the Stirling index and real column weights, Theorem 41.1 the cumulants of the complex-uniform marked expansion. The joint limit with  $\mathcal{J}_N$  is new. It answers, on the ray  $n/k = 2$ , an analogue of Part II's question "Joint laws with the number of columns" (Section 20) in which a combinatorial statistic of the matrices, the number of cells with entry at least 2, replaces the defect; the two are asymptotically independent, Gaussian and Poisson. The question as posed, for the Stirling defect  $\Delta_{n,k}$  or a combinatorial realization of it on every compact ray, stays open. (Wording corrected after the independent check of 6 October 2026. The remark first said that the joint limit "settles one instance" of the question, and that the question stays open "for the Stirling defect  $\Delta_{n,k}$  or another combinatorial defect". The repeated-cell count is not a realization of the defect: its limit is Poisson( $2v$ ), the defect's Poisson( $v$ ) (Remark 41.3), so the answer is to an analogue, not to an instance.)

*Remark 41.3* (write; the repeated cells and the factor  $e^{Lr/2}$ ). By (40.2) at  $u = 1$ ,  $c_*(\omega, 1) = e^{Lr(\omega-1/2)}/(4\pi L\sqrt{r-1})$ . Hence  $c_* = e^{Lr/2}c_*(\frac{1}{2}, 1)$ : weighting each cell with entry at least 2 by  $\frac{1}{2}$  removes from the amplitude the factor  $e^{Lr/2} = e^v$  that Section 16 attributes to the Poisson cloud of the Stirling defect, and (41.1) gives  $\mathbb{E}[2^{-\mathcal{J}_N}] \rightarrow e^{-Lr/2}$ . The limiting law of  $\mathcal{J}_N$  is Poisson with mean  $Lr = 2v$ , twice the mean  $v$  of the defect  $\Delta_{2N,N}$  (Theorem 16.2).  $\mathcal{J}_N$  is a statistic of the matrices themselves;  $\Delta_{n,k}$  is defined by the transform weights (16.1). No exact relation between the two is proved or claimed here, no decorated construction inducing those weights is given, and the statement concerns the ray  $n/k = 2$  only. It bears on Part II's question "A combinatorial realization of the latent defect" (Section 20) without answering it.

#### 41.1 Entries at least three: an explicit smaller-order bound

Let  $\mathcal{J}_N^{(3)}$  count cells with entry at least 3. Its cell series and logarithm are

$$g_3(z, \omega') = \frac{1 + (\omega' - 1)z^3}{1 - z}, \quad w_3 = w_0(z) + \log(1 + (\omega' - 1)z^3).$$

For  $|\omega' - 1| \leq 1$  the coefficient majorant is  $|w_3|_{\text{coeff}} \leq 4w_0(z)$ : the correction at  $z^{3k}$  is at most  $1/k$ , the corresponding coefficient of  $3w_0(z)$ . With  $|u - 1| \leq 0.02$ , the normalized nonprincipal-pole base is now

$$\frac{4(L^*)^2}{L(2\pi - L^*)} < 1$$

(and is approximately 0.522). The inverse argument works with the same radii: on  $|z| = 0.01$  the extra logarithm is bounded by  $0.01^3/(1 - 0.01^3)$  and its derivative by  $3 \cdot 0.01^2/(1 - 0.01^3)$ .

The inverse series is

$$z_3(w, \omega') = w - \frac{1}{2}w^2 + \left(\frac{7}{6} - \omega'\right)w^3 + O(w^4).$$

Its quadratic term does not depend on  $\omega'$ , so neither does the leading saddle amplitude. More precisely, at  $u = 1$  the kernel comparison is

$$\log \mathfrak{K}_N(Q, \omega') - \log \mathfrak{K}_N(Q, 1) = \frac{2(\omega' - 1)L^2 Q^2}{N} + O(N^{-2}).$$

The uniform saddle expansion and analytic residual allow this perturbation to be evaluated at  $Q = \mathfrak{q}$  at order  $N^{-1}$ , giving

$$\mathbb{E}[\omega' \mathcal{J}_N^{(3)}] = 1 + \frac{2L^2 \mathfrak{q}^2 (\omega' - 1)}{N} + O(N^{-2}). \quad (41.4)$$

Cauchy differentiation therefore proves the sharper estimate

$$\mathbb{E} \mathcal{J}_N^{(3)} = \frac{(Lr)^2}{2N} + O(N^{-2}), \quad \mathbb{P}(\mathcal{J}_N^{(3)} \geq 1) = O(N^{-1}). \quad (41.5)$$

The probability statement uses Markov's inequality; it does not assert equality with the displayed expectation to that precision.

Let  $\mathcal{E}_N = 2N - \#\{\text{occupied cells}\}$  be total excess. On  $\mathcal{J}_N^{(3)} = 0$  every nonzero entry is 1 or 2, and  $\mathcal{E}_N = \mathcal{J}_N$ . Hence

$$\mathbb{P}(\mathcal{E}_N \neq \mathcal{J}_N) \leq \mathbb{P}(\mathcal{J}_N^{(3)} \geq 1) = O(N^{-1}).$$

The pairs  $(\mathcal{E}_N, (C - N/L)/\sqrt{2\sigma_\infty^2 N})$  and the pair in (41.3) agree with probability  $1 - O(N^{-1})$  under their natural coupling. They have the same independent Poisson/Gaussian distributional limit. This coupling alone makes no claim about unbounded moments of  $\mathcal{E}_N$  on the rare exceptional event.

## 42 Report 169's checks, numerical diagnostics, and reproducibility

The companion separates exact identities from approximate diagnostics. Its frozen OEIS prefix contains all 15 displayed terms, indexed 0 through 14, with source and retrieval date. The following two finite counting routes are independent:

1. The integer Stirling transform (36.1).
2. Nonempty-column recurrence followed by row inclusion-exclusion. For  $k \geq 1$ , let  $A(0, k) = 1$  and

$$A(n, k) = \sum_{i=1}^n \binom{i+k-1}{k-1} A(n-i, k).$$

These count ordered nonzero columns of height  $k$ , allowing zero rows. Thus

$$T(n, N) = \sum_{k=0}^N (-1)^{N-k} \binom{N}{k} A(n, k),$$

where  $A(n, 0) = 0$  for  $n > 0$  and  $A(0, 0) = 1$ .

Both routes reproduce the frozen prefix exactly. For marked checks, a direct enumeration of ordered nonzero column vectors with row coverage is compared with (40.4) at small  $N$ . For example,

$$\begin{aligned} T_1(\omega, u) &= \omega u + u^2, \\ T_2(\omega, u) &= \omega^2 u + 2\omega^2 u^2 + 2\omega u + 16\omega u^2 + 18\omega u^3 + u^2 + 12u^3 + 14u^4. \end{aligned}$$

These check both marker meanings, as well as their total values 2 and 66. [Note of the batch-108 write:  $A(n, k)$  is Part I's count with zero rows allowed, with Part I's conventions at  $n = 0$  and

$k = 0$ , and the second route is the inclusion–exclusion (2.3); Report 169 writes  $c_{N,k}$ . The write re-enumerated  $T_1$ ,  $T_2$  and  $T_3$  by brute force and found the displays above.]

Exact coefficient calculations are checked against the explicit  $\beta_1$  and the finite formula (38.5). Rational-expression files for  $\beta_0$  through  $\beta_3$  avoid using floating-point recognition as an algebraic proof. The scripts enforce bounded inputs, run offline, and use explicit exceptions rather than relying on Python assertions. The companion README gives commands for ordinary and optimized Python tests, optional computer algebra, and output regeneration.

Additional independently recomputed exact counts through  $N = 800$  support the following *uncertified* diagnostics. If  $\mathcal{R}_N = a_N/(c_* d_*^N (N!)^2/N)$ , then at  $N = 800$

$$\begin{aligned} N^2(\mathcal{R}_N - 1 - \beta_1/N) &\approx 0.1282857803, \\ N^3(\mathcal{R}_N - 1 - \beta_1/N - \beta_2/N^2) &\approx -0.0024381943, \\ N^4(\mathcal{R}_N - 1 - \beta_1/N - \beta_2/N^2 - \beta_3/N^3) &\approx 0.0042752840. \end{aligned}$$

At  $Y = a_{800}$  the Lambert core  $x_0$  has error approximately  $-2.7880 \cdot 10^{-5}$ ; the first corrected expression has error  $-7.6808 \cdot 10^{-10}$ , and the second has error  $2.4552 \cdot 10^{-12}$ . These values illustrate signs and scales, not global inequalities or certified thresholds. The proofs of the error orders are in Sections 37 and 39. [Note of the batch-108 write:  $\mathcal{R}_N$  is that of Section 9, and the first two lines continue the columns  $V_N$ ,  $W_N$  of Table 4 to  $N = 800$ . The errors are those of  $x_0$  and of (39.4) truncated after the  $x_0^{-1}$  and  $x_0^{-2}$  terms, measured against the integer 800. The  $N = 800$  count uses the Stirling route only (the companion caps inclusion–exclusion at  $N = 100$ ); the last line suggests a fourth coefficient near 0.0043, which is an uncertified observation, not a claim.]

The release includes this editable L<sup>A</sup>T<sub>E</sub>X source, a deterministic build script, the exact companion, frozen inputs, and public-source provenance. Build commands refuse to overwrite an existing output. PDF regeneration requires a compatible pdfL<sup>A</sup>T<sub>E</sub>X installation; byte-for-byte identity additionally depends on the same TeX engine, package versions, and font files. ZIP timestamps, ordering, and permissions are fixed. No external source download or online service is required to run the exact checks.

[Note of the batch-108 write: in this report Report 169’s L<sup>A</sup>T<sub>E</sub>X source and PDF are not shipped (the article here is a build of the merged text), and its companion, build and release scripts carry the prefix `08-packed-`; the report README maps the names and gives the rerun route on a copy with the delivered layout. On such copies the intake ran the companion’s unit tests (17 of 19 pass, under `python` and `python -0`; the two errors are POSIX output-safety tests that require `O_NOFOLLOW`, not mathematical checks), reproduced the four example JSON files byte for byte by calling the generator directly (its `generate` command refuses to write on Windows), and reran the optional SymPy identity check (output equal to the shipped one apart from line endings). The write reran the unit tests with the same result. Not rerun: the  $N = 800$  run, the PDF build and the archive build.]

## 43 Literature comparison and scope

The following comparison concerns the primary sources listed in the bibliography and inspected on 3 October 2026. It is a bounded literature review, not a priority certificate.

- **Exact model.** Munarini–Poneti–Rinaldi, Section 7 and Table 2, count precisely the nonzero-line model and give (36.1). Their fixed-row asymptotic in Proposition 13 concerns a different regime allowing zero rows. Cerbai–Claesson’s Theorem 4.8 in [20] gives a joint row/column

Stirling–Burge polynomial from which the row-fixed enumerator follows. Duchamp–Hivert–Thibon [21] provide the packed-matrix algebraic context and all-dimension series.

- **Existing leading equivalent.** A261784 records Kotěšovec’s leading formula and numerical amplitude. The separate two-page A120733 note [16] treats the sum over all row and column dimensions by a related Stirling-transform/defect mechanism. It does not fix  $N$  rows at total  $2N$ . The inspected A261784 representation linked no proof; that absence is not evidence that no proof exists elsewhere.
- **Adjacent saddle method.** Cameron–Prellberg–Stark [17], especially Theorem 7, treat binary incidence matrices with both dimensions proportional to weight. The unrestricted-entry setting replaces a falling-factorial structure by a rising one and changes the collision tilt. Their proportional saddle and Poisson collision mechanism are close precedents; they are not counts of the present unrestricted-entry diagonal. Their companion classification paper [18] uses other symmetry conventions.
- **Column statistics.** Louchard [6] fixes the column-vector dimension while total weight grows, allows empty rows, and derives Gaussian and local column laws. Here the nonempty row count itself grows as half the total. The present constants and uniform conditioning therefore require the argument given above.
- **Collisions and other matrix models.** Greenhill–McKay [22] analyze sparse integer tables with prescribed margins, including successive double-pair count ratios. Free margins and the present conditioning are different. Lonesum asymptotics [23] impose an extra binary condition; Fishburn asymptotics [24] impose upper-triangular square shape. Neither changes the definition of  $a_N$  here.

No reviewed source states this same combination of every-fixed-order diagonal corrections, controlled index inversion, and the complex-uniform marked consequences. This limited observation supports the usefulness of a consolidated refinement, not a worldwide novelty claim. The author’s separate mathematics page and the OEIS history endpoint were not available in the source review; their contents are not part of this comparison.

The proof establishes fixed-order error bounds with unspecified constants, holomorphic uniformity on stated marker domains, a joint weak limit, and shrinking integer inverse brackets. It does not establish a convergent expansion, optimal truncation, effective numerical thresholds, a local limit theorem for the growing column count, or an exact universal ceiling selector. Exact computations and independent mathematical checking do not amount to formal proof-assistant verification or certified interval arithmetic.

[Note of the batch-108 write: Report 169’s review did not include this report, whose Parts I and II were written on the same day. Two of the three items of that combination are in them: the every-fixed-order diagonal corrections (Theorem 7.1 at  $\lambda = 2$ , Corollary 17.1) and the controlled index inversion (Section 18: the Lambert- $W$  inverse with its first correction, Proposition 18.1, and integer brackets at every fixed order  $J$ , Proposition 18.2; the note first said “with integer brackets at first order”, corrected after the independent check of 6 October 2026); Part I also has the column mean and variance (Corollary 7.5). What Report 169 adds is listed in Table 16. A120733, Kotěšovec’s sum over all dimensions, is  $\sum_k T(n, k)$ ; the collection’s report [a260700-parabolic-double-cosets](#) [10] names it as the labelled analogue of its own sequence. The symmetric analogue of the present model, OEIS A138178, is treated in the collection’s report [a138178-symmetric-packed-matrices](#) [25] (batch 108), which uses the same devices (a geometric weighting that erases zero lines, the Fubini pole at  $\log(1 + 1/u)$ , complex-uniform marked expansions and Poisson laws of

repeated cells) for a different model, with an involution saddle and an expansion in powers of  $n^{-1/2}$ ; no theorem is shared.]

## 44 Further questions and research

[Added 6 October 2026, batch 108.] This section collects every unproved claim of Report 169: its six further questions, with its sketches. Report 169 states no claim that this write found wrong, and its uncertified diagnostics (Section 42) are labelled as such, not claimed. Its questions 1–5 are Research questions 17–21; the question counter of Part I continues, so no existing number changes. Its question 2 is re-scoped, because Part I answers its unmarked form; its question 6 is the paragraph “Beyond all algebraic orders” of Section 20 and is printed there (see the last paragraph below).

Several extensions are substantive and remain separate from the results proved here.

**Research question 17** (Effective remainders and threshold selection; Report 169). Make the Euler–Maclaurin, inverse-kernel, and Gaussian constants explicit enough to obtain usable numerical  $N$  thresholds, and combine these with exact local counts to settle integer inverse ambiguities near a threshold.

[Note of the batch-108 write: Part II’s “Effective certified computation and formalization” (Section 20) asks for explicit constants in the compact-ray bound and in (18.7); this is the same aim for Report 169’s route, with the exact counts as the tie-breaker. Open.]

**Research question 18** (A variable row-density regime; Report 169). Replace the total  $2N$  by  $n$  with  $N/n$  in a compact subinterval of  $(0, 1)$ . A density-dependent saddle should yield uniform coefficient functions; limits toward sparse or nearly unit row occupancy may require new transition scales.

[Note of the batch-108 write: for the unmarked count the first sentence is answered by Part I: Theorem 7.1 with (7.17) gives a uniform expansion to every fixed order for  $n/k$  in every compact subset of  $(1, \infty)$ , which is  $k/n$  in a compact subset of  $(0, 1)$ , with coefficients  $E_j(\lambda)$  that depend smoothly on  $\lambda$ ; and Corollary 7.5 gives the column mean and variance there. Its two limits,  $n/k \downarrow 1$  (nearly unit occupancy) and  $n/k \rightarrow \infty$  toward the coverage window (sparse occupancy), are Part I’s Research question 1. What stays open is the marked form on variable rays: Theorem 40.1 and Theorem 41.1, with the repeated-cell marker and complex column weight, uniformly for  $n/k$  in compact subsets of  $(1, \infty)$ .]

**Research question 19** (Local column probabilities and quantitative joint errors; Report 169). Derive a local central limit theorem and Edgeworth corrections for the column count  $C$ , jointly with fixed repeated-cell counts.

This requires characteristic-function control away from the small  $u$ -neighborhood, beyond the full  $\omega$ -circle control already proved. [Note of the batch-108 write: Part I’s Research question 3 asks the same in the coverage window, jointly with the zero-row count. Open.]

**Research question 20** (The hierarchy of large entries; Report 169). For cells at least  $r' \geq 3$ , determine leading constants and joint rare-event probabilities, including the interaction between triple entries and pairs of repeated cells.

Equation (41.5) supplies only the first moment at  $r' = 3$ . [Report 169 writes  $r$  for the threshold;  $r$  is the saddle here.]

**Research question 21** (Moments of total excess; Report 169). Prove uniform integrability and moment expansions for the total excess  $\mathcal{E}_N$ .

A direct excess marker could prove them. The high-probability equality  $\mathcal{E}_N = \mathcal{J}_N$  by itself is insufficient for those unbounded-moment conclusions.

**Report 169’s question 6: beyond fixed-order asymptotics.** Report 169’s sixth question asks for the growth of the  $\beta_j$ , possible Gevrey bounds, optimal truncation, and the relative role of the nonreal Fubini poles and other complex saddles; the exponentially small pole estimate of Lemma 36.1 identifies one scale, but does not supply a complete exponentially improved expansion. This is Part II’s paragraph “Beyond all algebraic orders” (Section 20), and is not repeated here; what Report 169 adds to it (the growth and Gevrey type of the coefficients, and saddles other than the Fubini sectors) is printed there in a dated note. It is open.

## A Provenance and editorial notes

[Added 3 October 2026, batch 85B.]

**Sources.** The report was built from two manuscripts of batch 85 of ProveIt’s incoming-reports intake, written independently on the same subject.

- *Manuscript 06* (Part I, the base): the archive `Matrix_Compositions_Uniform_Asymptotics.zip` (21 files), main file `matrix_compositions.tex` (2,078 lines, 100 labels), a 31-page PDF titled *Matrix Compositions: Exact Spectra, Hankel Products, and Uniform Asymptotics*, dated October 3, 2026 (Pacific time). Its author line and PDF metadata read “OpenAI ChatGPT”, with “Research prepared for Vladimir Reshetnikov”. Its pin is the search-index commit `ce37e13f4aa16819c87c3ccc611362d15758f2ac` (2026-10-03, 17:05) [7]. It arrived in commit `345284ac8`.
- *Manuscript 01* (Part II): the archive `OEIS_Matrix_Compositions_Research.zip` (26 files), main file `matrix_compositions.tex` (1,561 lines, 98 labels), a 22-page PDF titled *Matrix compositions: finite spectra, uniform asymptotics, and two Poisson limits*, dated 3 October 2026, author line “Research report prepared for Vladimir Reshetnikov”. Its pin is the tree revision `d68b65ea064084fb121df9bb431b21d086f7be81`; it also records the search-index commit `ce37e13f4` [9]. It arrived in commit `317c1ce2e`, with the other archives of batch 85; the placement of batch 85A (`713149ded`) held it back so that it could be merged with manuscript 06.

Both were placed together by commit `fa2f3e419`, which deleted both archives from `docs/incoming`; they survive in their arrival commits. Neither manuscript is an edition of the other: they share about 0.7% and 0.5% of their word 8-grams (bibliography entries and one phrase), and their pins, code and notation differ.

**Why manuscript 06 is the base.** On every theorem the two share, manuscript 06 has the weaker hypotheses or the more general statement: recurrence minimality for every column weight  $u > 0$  (manuscript 01:  $u = 1$ ); a pole bound uniform in  $k$ , also weighted (manuscript 01: a factor  $k - 1$ ); the coverage expansion of the whole generating function in complex  $w$  (manuscript 01:  $\mathbb{P}(M = 0)$  only); and the correct attribution of the recurrence’s existence to [4, Proposition 25]. It also has more new mathematics (the Hankel determinant product, the sharp total-variation rate,

the joint Gaussian–Poisson law, the column central limit theorem, the prescribed-column threshold). Manuscript 01 is more general only in the kernel lemma, and that lemma only serves compact rays, where both theorems coincide.

### Where the merge had to choose.

- Shared results are printed once, in manuscript 06’s form, with a bracketed note naming manuscript 01’s version (Table 5). Manuscript 01’s proofs that are the same argument are summarized in those notes, not reprinted; its genuinely different devices are recorded there (Bonferroni truncation for the coverage probability, the tail bound  $|\mathbf{W}_n(w)| \leq e^{|w|}$ , the direct uniqueness argument for the maximum on the circle) or printed in Part II where a new result needs them.
- Manuscript 01’s  $k$ -dependent pole bound is printed as Remark 4.2, because it is sharper for  $k \leq 4$ .
- Manuscript 01’s kernel theorem is printed in full in Part II as the general form of Lemma 7.2, and Remark 7.3 says so where the lemma stands. Manuscript 06’s lemma is kept unchanged.
- Manuscript 06’s moments of  $D$  in Corollary 7.5 are kept, with a pointer to the stronger Theorem 16.2.
- Manuscript 01’s closed form of  $\beta_1$  is printed as a note after Corollary 7.4, being the same expression; its fifty-digit constants and  $\beta_3$  are in Part II.
- Manuscript 01’s overstated claim about the order conjecture is narrowed (Section 11.2), its draft OEIS text is printed with the correction (Section 22), and its example (2.8) is credited to [4, Remark 26].
- Manuscript 01’s research question on joint column laws is re-scoped as partly answered by manuscript 06; manuscript 06’s Research question 2 is re-scoped as answered for the event  $M = 0$  by Theorem 13.1.
- Manuscript 01’s inversion section is printed with a note that it is an instance of the transseries volume’s apparatus; manuscript 01’s source manifest (its Appendix C) is replaced by this appendix and the report README, and its rebuild instructions by the README.
- Manuscript 01’s three tables (Tables 6, 7 and 8) are typeset, unchanged, from its shipped table bodies, the files `data/01-two-poisson-X.tex` with  $X$  one of `defect_table`, `error_table` and `coupon_table`. Its fourth body, `diagonal_table` (the same diagonal errors at more indices and digits), was not used by its article and is not used here.

### What the write changed in manuscript 06’s text.

- Every label gained the prefix `mxc`: (100 labels; every reference was updated); nothing was renamed otherwise and nothing was removed.
- Added: the second author and date lines of the title, the PDF metadata, the editorial note after the abstract, the part heading, Section 1.5 with Table 2, Remarks 2.3, 4.2 and 7.3, bracketed notes marked “[Added 3 October 2026, batch 85B]” in Sections 1, 2, 5, 7, 9 and 10, Part II, this appendix, and five bibliography entries [9, 10, 11, 12, 8].
- The figure path is now `figures/06-hankel-coupon_window.pdf`.

- No delivered sentence of manuscript 06 was changed or removed. Every non-claim is kept: the abstract’s last sentence; the scope table’s caption; the bounded audit; “The theorem controls the unrestricted count  $A(n, k)$  uniformly” and what it does not say; the moving-mean remark; the compactness paragraph after Corollary 7.5; the prescribed-column caveat; the Hankel caveat on total positivity; the numerics caveats; and the caveat on the research questions.

Manuscript 01’s text is restated in manuscript 06’s letters (Table 2), with some proofs condensed; none of its statements was strengthened, and every one of its non-claims is kept (Section 11, the remarks after Theorems 13.1 and 16.2, Remark 15.1, Sections 18, 19, 21 and 22).

**Overloaded letters in manuscript 06.** Manuscript 06 reuses some letters; none is renamed, and in each case the section determines the meaning.  $D$  is  $\binom{k+1}{2}$  in Section 2, the constant  $D_u = 16b/L_u^2$  in Section 4, the function  $D(\eta)$  in Section 5 and  $n - Q$  in Section 7.6.  $d$  is  $\binom{k+1}{2}$  and  $c$  is  $\binom{k+1}{3}$  in Section 3, beside the growth constants  $d_k, c_k$ .  $b$  is  $u/(1+u)$  in Sections 2–4, also  $u$  in Section 3, the variance  $\kappa_2$  in Section 7, and  $b_j$  are factorial moments.  $P_j(t)$  in Section 3 and  $P_j(w)$  in Section 7.4 are different polynomials;  $E_n(w)$  is a product,  $E_j(\lambda)$  a saddle coefficient and  $E_n = B\rho^{n-1}$  an error budget;  $r$  is  $r_k = 1 - a_k$ , the saddle, and a summation index;  $v$  is  $Lr/2$  in Section 7 but a tilting parameter in the proof of Theorem 6.2; and  $m$  indexes  $d_m$  and counts columns.

**Checks by the intake.** The placement reran both packages on copies: manuscript 06’s suite passed with its outputs byte-identical or equal apart from line endings and timings and its figure pixel-identical; manuscript 01’s suite passed likewise, and its checksum ledger matched 25/25. Manuscript 06’s diagonal constants and manuscript 01’s agree in every printed digit, and their coverage probabilities at  $k = 20, n = 60$  agree. The intake also verified Proposition 25 and Remark 26 in the published paper, the impulse of Remark 2.4 at  $k = 1, 2$ , the instances  $H_{3,s}(u)$ ,  $H_{2,1}(1) = 8$  and  $H_{3,1}(1) = 62208$ , the constants of Theorem 4.1 and Remark 4.2, the moments in Corollary 7.5, the increment expansion (2.11), the cumulant recursion at  $j = 3$ , and  $P_3$  numerically. These checks do not replace an independent review of the proofs.

**Part III (batch 98).** [Added 5 October 2026, batch 98.] Part III is built from manuscript 08 of batch 98, the archive `a261781_hankel_zero_laws.zip` (30 files, a 29-page PDF, main file of 2,041 lines with 110 labels), which arrived in commit 2172df76a and was placed in commit 0fba5167f. Its pin is d18416ec7 [15]; this report’s text did not change between the pin and the write. Its article, PDF and delivery README are not shipped; it has no checksum ledger. Its other 27 files are shipped byte-identical with the prefix `07-zero-laws-`, and they use the delivery names; its `requirements.txt` (shipped as `data/07-zero-laws-requirements.txt`) is byte for byte a generic file of an unrelated report in the Fabius tree. Manuscript 08 is a continuation written against this report, not an edition of either earlier manuscript: 1.25 % of its word 8-grams occur in this report, and 0.50 % of this report’s in it.

### Where the batch-98 write had to choose.

- Manuscript 08’s Section 2 re-derives Theorem 3.1 by the same argument; it is printed as a pointer (Remark 24.4), which keeps its two new small cases  $H_{1,s}, H_{2,s}$  and its remarks on the convention  $s \geq 1$ .
- Its question 8 is Part I’s Research question 5; it is printed once, there, with manuscript 08’s addition in a dated note, and a paragraph of Section 33 points to it. Its question 9 is printed with a note naming the formalization targets of Parts I and II. Its questions 1–7 and 9 are

Research questions 9–16; the question counter of Part I continues, so no existing number changes.

- Manuscript 08’s letters collide with Parts I and II in about thirty places (among them  $\nu_k$ ,  $M$ ,  $M_p$ ,  $D_k$ ,  $d_k$ ,  $c_k$ ,  $\Delta_k$ ,  $\gamma$ ,  $\Lambda$ ,  $L$ ,  $\rho_k$ ,  $\tau$  and  $\sigma$ ). Part III renames them, with sans-serif type for its colliding Latin objects and a superscript  $z$  on its zero measures, adopts Section 3’s unindexed  $c = \binom{k+1}{3}$  and writes  $\binom{k+1}{2}$  out; every rename is in Table 10. Two internal overloads of the manuscript are resolved by the same table:  $H_N$  (harmonic numbers) beside  $H_{k,s}$ , and  $\sigma$  (a sign) beside the divisor functions  $\sigma_{\pm 1}$ ; its lower summation limit  $M$  and running integer  $N$  are marked in notes. Its labels are printed as `mx:hz:` followed by the delivered name, including the `ar:` labels of its Section 6.
- Its figures are delivered files and keep its letters; the captions translate them. Its tables are retyped from its source with the renamed letters and unchanged numbers.
- Its Section 12 (source audit, attribution and package contents) is folded into Section 23, Section 32 and this appendix; its bibliography entries for A261781, A261784 and Munarini–Poneti–Rinaldi are this report’s [2, 3, 4], and three entries were added [13, 14, 15].
- Following the intake’s standing rule on unproved and wrong claims, every unproved claim of manuscript 08 stands in Section 33: its nine questions, its candidate for the correction of the fractional moments (Research question 13, stated there as a claim), its speculation that the last zeros depend on arithmetic features of  $k$  (Research question 12), and its three numerical observations. No claim of manuscript 08 was found wrong. Two imprecisions are recorded in remarks of the write: “extended precision” and “exact CSV values are reproducible” depend on the platform (Remark 32.1), and the uniformity in  $m$  of Proposition 25.2 is empty for  $m \geq k$  (Remark 25.3).

**What the batch-98 write changed.** It added to the preamble eight macros for Part III’s notation, a third title, author and date line and the PDF metadata; the batch-98 editorial note; bracketed notes marked “[Added 5 October 2026, batch 98]” in Sections 1 and 1.5, at the end of Section 3, in Research questions 5 and 7 and in “A practical order of attack” of Section 10; Part III (Sections 23–33); these four paragraphs; and three bibliography entries. All its labels carry the prefix `mx:hz:`. No existing label was renamed or removed, no existing number changed, and no delivered sentence of manuscripts 06 or 01 was changed or removed. Manuscript 08 is printed in full apart from its Section 2 and its Section 12, with renamed letters, notes of the write and light rewording where a sentence named its own package layout; none of its statements was strengthened, and every one of its non-claims is kept (Section 23).

**Checks by the intake for Part III.** The placement and the write each reran manuscript 08’s program on a copy with the delivered layout (Section 32): exit status 0, all counted checks passed, outputs as described there. The placement read every proof of Sections 25–31 and found no error; it re-derived by hand, among others, the Euler product for  $\psi$ , the grid and weight bounds of Theorem 27.1, the exact low-range identity of Lemma 28.5, the case analysis of Corollary 28.3, the jump formula (29.7), the Bernoulli identity of Theorem 30.1, the weights of Theorem 24.3,  $\mathcal{M}_2 = \zeta(3)/(4\pi^2)$ ,  $\mathcal{M}_4 = \zeta(5)/(600\pi^2)$  and the identity (31.8). An independent mpmath computation of its own (not shipped) found  $\mathcal{I}_3$  equal to  $\frac{3}{2} \log(2\pi) - \frac{11}{4}$  to  $10^{-15}$ ; at  $k = 500$  and 1500 it found  $D_k = 2(c - G_k)$  exactly,  $M_1(k) - (\log k)/\pi = -0.5923$  and  $-0.6107$  against the limit  $-0.62733$ ,  $(k/\log k)(\mathcal{F}_k(t) - t)$  at  $t = \sqrt{2} - 1$  equal to  $-0.127$  and  $-0.130$  against  $-0.156$ , and  $k\nu_k^z(kx, \infty)$  at

$x = 0.1$  equal to 0.2570 and 0.2545 against  $\Xi(0.1) = 0.2520$ : slow approaches consistent with the error terms proved. The write checked  $H_{1,s}$ ,  $H_{2,s}$ , the bound of Remark 25.3 and the translation of every renamed formula against the delivered source. These checks do not replace an independent review of the proofs.

**Independent check of the batch-98 write (7 October 2026).** An adversarial check made by the intake after the write (commit 324818ddb) and its reciprocal note (commit 639b038ab) re-derived every statement they added, with its own code. *Sources.* The archive (30 files; 29-page PDF with author “OpenAI ChatGPT” and “An OEIS-related research article”; 2,041 lines), the pin (a merge commit of 4 October 2026, 19:16 –0700; this report unchanged to the placement), the generic `requirements.txt` (the same blob as a file of the Fabius tree), the 8-gram shares 1.25 % and 0.50 % (word tokens), the decimals of the printed text (unchanged from the manuscript), [13] and [14] (23 pages) are as printed. *Statements.*  $H_{1,s}$ ,  $H_{2,s}$  ( $s \leq 3$ ) and  $H_{3,s}$  ( $s \leq 2$ ) were recomputed as determinants from the inclusion–exclusion generating function, and  $T_u(2, 2) = u + 2u^2$ ; the Part I note on  $E_m(k)$  (every order  $2 \leq m \leq k - 1$  occurs, through the pair  $(1, 1 + m)$ ), the masses  $3 \log k / (\zeta(2)\psi(m)k)$  of Research question 7’s note and Remark 25.3 were checked by hand. *Numbers.* The check’s own computation of the atoms ( $D_k = \sum \varphi(m)E_m(k) = 2(c - G_k)$  asserted for every  $k$  used) reproduces Table 14 (distances to  $10^{-12}$ ; maximizing angle  $1/7$  for  $k \leq 256$  and  $1/11$ , or its mirror  $10/11$ , above) and the entries  $k = 1024, 2048, 4096$  of Table 15, the limit  $-0.6273278444052115$ , and the placement’s values at  $k = 500$  and  $1500$  of  $M_1(k) - (\log k)/\pi$ ,  $(k/\log k)(\mathcal{F}_k(t) - t)$  and  $k\nu_k^z((kx, \infty))$ , with  $\Xi(0.05) = 1.2438$ ,  $\Xi(0.1) = 0.2520$  and  $\mathcal{I}_3 = \frac{3}{2} \log(2\pi) - \frac{1}{4}$  to fifteen digits. A rerun of manuscript 08’s program on a copy confirms Remark 32.1 exactly (16 of 20 and 20 of 4,804 rows; on Windows NumPy’s `longdouble` is the 64-bit double), the four PDF-structure checks of the text summary, and that the JSON summary records the Python version but not the platform. The source and the wording of the evidence note at Research question 9, of the fractional-moment candidate and of the last-zeros speculation are manuscript 08’s, as stated. No mathematical error was found. Corrected with the first wording kept: the reciprocal note after Theorem 16.2 (the defect of the sibling report is not “the number of missing blocks”, and the pair mechanism is named in a note of that report’s write, not in its source note). Added: numerical evidence for the fractional-moment candidate at  $p = \frac{1}{2}$  (Research question 13).

**Part IV (batch 108).** [Added 6 October 2026, batch 108.] Part IV is built from Report 169 of a session bundle of research reports, the archive `Packed_Matrices_Asymptotics_Inverses_and_Joint_Limits_Source.zip` (571,859 bytes; 27 files in an inner directory `report169/`, with a checksum ledger `SHA256SUMS` that matched 26 of 26 files at the intake), which arrived in commit 60f54ea06 and was placed in commit 602e5bd0f (batch 108). Its main file `source/report169.tex` has 686 lines and 60 labels, its PDF 16 pages; it is dated 3 October 2026, and its author line and PDF author field are empty. It has no pin and cites nothing in the repository. Its article, PDF, delivery README and checksum ledger are not shipped (they survive in the arrival commit). Its other 23 files are shipped byte-identical with the prefix `08-packed-`: the companion’s README at the report root, its four programs and the three release scripts in `code/`, and its data, frozen inputs, examples, symbolic and optional large-run records and source list in `data/`. It is not an edition of any earlier source: 0.66 % of its word 8-grams occur in this report, and 0.13 % of this report’s in it.

**Where the batch-108 write had to choose.**

- Report 169 proves again, by a different route, Corollaries 7.4 and 17.1, the moments of Corollary 7.5 at  $n/k = 2$ , and Propositions 18.1 and 18.2. Unlike Part III’s re-proof, these

are printed in full as marked second routes (Theorems 35.1, 39.2 and 41.1, Proposition 39.1), because the new marked theorem reuses their proofs; remarks of the write (Remarks 35.2, 39.3 and 41.2) identify each with its counterpart, and dated notes at the counterparts point to them.

- Report 169's letters are translated into those of Parts I and II (Table 17: about forty renames, among them  $n \rightarrow N$ ,  $\rho \rightarrow L$ ,  $t \rightarrow r$ ,  $C_j \rightarrow \beta_j$ ,  $h_j \rightarrow \tilde{\beta}_j$ , its markers  $u, v \rightarrow \omega, u$ , and the inverse core  $Y, W, v \rightarrow \Lambda, \varpi, x_0$ ); the product index  $r$  becomes  $\nu$  and the Bernoulli index  $m$ . Its labels are printed as `mx c:pm`: followed by the delivered name; the write added labels for the Part, its first and last sections, two of its unlabelled sections, its two tables, its research questions and the remarks of the write.
- Its six further questions: questions 1, 3, 4 and 5 are Research questions 17, 19, 20 and 21; question 2 is Research question 18, re-scoped because Part I answers its unmarked form, with a dated note at Part I's Research question 1; question 6 is printed once, as a dated note in Part II's paragraph "Beyond all algebraic orders", with a pointer paragraph in Section 44.
- Its bibliography entries for A261784, Munarini–Poneti–Rinaldi and Louchard are this report's [3, 4, 6]; nine entries were added [16, 17, 18, 19, 20, 21, 22, 23, 24], and one for the sibling report on symmetric packed matrices [25].
- Following the intake's standing rule on unproved and wrong claims, every unproved claim of Report 169 stands in Section 44; its uncertified  $N = 800$  diagnostics are kept with their label, and the fourth coefficient they suggest is recorded as an observation, not a claim. No claim of Report 169 was found wrong. Its statement that no reviewed source combines every-fixed-order diagonal corrections, controlled inversion and the marked consequences is true of the sources it reviewed but not of this repository, where Parts I and II have the first two; a note in Section 43 says so.

**What the batch-108 write changed.** It added a fourth title, author and date line and the PDF metadata; the batch-108 editorial note; bracketed notes marked "[Added 6 October 2026, batch 108]" in Sections 1 and 1.5, after Corollaries 7.4 and 7.5, in Research questions 1 and 3, after the fifty-digit constants of Section 17, at the end of Section 18, and in three paragraphs of Section 20; Part IV (Sections 34–44); these four paragraphs; and ten bibliography entries. All its labels carry the prefix `mx c:pm`:. No existing label was renamed or removed, no existing number changed, and no delivered sentence of manuscripts 06, 01 or 08 was changed or removed. Report 169 is printed in full, with renamed letters, notes of the write and light rewording where a sentence named its own package; none of its statements was strengthened, and every one of its non-claims is kept (Section 34).

**Checks by the intake for Part IV.** The placement verified the checksum ledger, compared  $\beta_1$  of (35.4) with  $\beta$  of Corollary 7.4 (25 digits) and the moments (41.2) with Corollary 7.5 (differences 0 and  $2 \cdot 10^{-40}$ ), computed  $\mathbb{E}\mathcal{J}_N = 1.0312, 1.0743, 1.0893$  at  $N = 20, 50, 100$  from the finite transform (so  $N(Lr - \mathbb{E}\mathcal{J}_N) = 1.47, 1.52, 1.53$ , consistent with  $Lr + O(1/N)$ ), checked  $\vartheta_0 = 0.10965$ , the base  $\vartheta_*$  and the inverse series of  $z_3$ , and reran the companion on copies (Section 42). The write checked (35.4) against  $\beta$  as rational functions of independent  $L$  and  $r$  (SymPy), Report 169's exact  $\beta_1, \beta_2, \beta_3$  against Part II's fifty-digit values ( $< 10^{-50}$ ), the coefficient of  $x_0^{-3}$  in (39.4) by hand and the whole expansion numerically (at  $\log Y = 10^3, 10^5, 10^7$  the error after the  $x_0^{-3}$  term times  $x_0^4(1 + \varpi)$  is  $3.6 \cdot 10^{-4}, 2.2 \cdot 10^{-4}, 1.5 \cdot 10^{-4}$ ), the displays  $T_1, T_2$  by brute force, the  $\omega = 0$  specialization by exact counts (Remark 40.2), the Euler–Maclaurin, kernel and Stirling terms of (38.2), the normalization

of Section 37, the majorant and the bounds of Section 40, and every renamed formula against the delivered source. These checks do not replace an independent review of the proofs.

**Independent check of the batch-108 write (6 October 2026).** An adversarial check made by the intake after the write (commit `ab4e30d32`) re-derived every identification and number of Part IV that the write added or relied on. (35.4) minus  $\beta$  of Corollary 7.4, with  $L$  and  $r$  independent symbols, simplifies to 0 (SymPy); Report 169’s exact  $\beta_1, \beta_2, \beta_3$ , evaluated at 80 digits, differ from Part II’s printed fifty-decimal truncations by  $6.5 \cdot 10^{-51}$ ,  $9.7 \cdot 10^{-51}$  and  $1.4 \cdot 10^{-51}$ . Independently of every route, exact values of  $a_N$  for  $N \leq 150$ , computed by its own row-and-column inclusion–exclusion (equal to the transform for  $N \leq 8$ ), give  $N^4(a_N/(c_* d_*^N(N!)^2/N) - 1 - \beta_1/N - \beta_2/N^2 - \beta_3/N^3) = 0.004376, 0.004322, 0.004304$  at  $N = 50, 100, 150$ : bounded, so  $\beta_1, \beta_2, \beta_3$  are right. The column mean and variance agree with Corollary 7.5 term for term, and exact column distributions show both errors to be  $O(1/N)$ . The coefficient of  $x_0^{-3}$  in (39.4) was re-derived by hand, and Remark 39.3 (ii) confirmed; at  $\log Y = 10^3, 10^5, 10^7$  the error after the  $x_0^{-3}$  term of the root of  $\log \mathcal{A}^{[3]}$ , times  $x_0^4(1 + \varpi)$ , is  $-0.00383, -0.00366, -0.00360$  (the write’s figures above use another comparator; both are consistent with the stated  $O(x_0^{-4}/(1 + \varpi))$ ).  $Lr = 1.10461616271868 \dots = 2v$ , the exact ratios of binary packed to all packed matrices,  $\mathbb{E}\mathcal{J}_N$  and the second factorial moment of  $\mathcal{J}_N$  for  $N \leq 40$  are consistent with Poisson( $Lr$ ) at rate  $O(1/N)$ ; Remark 41.3’s algebra was confirmed. All 288 earlier labels are present, the 79 new ones carry the prefix `mxc:pm:`, and the numbering before Part IV and of the appendix is unchanged. No mathematical error was found. Two wordings were corrected; the first wordings are quoted in the dated note of Section 20, in Remark 41.2 and in the note of Section 43: Part IV “settles one instance” of Part II’s “Joint laws with the number of columns” (abstract, front matter, the dated note in Section 20 and Remark 41.2) is now “answers an analogue”, since the repeated-cell count, whose limit is Poisson( $2v$ ), is not a realization of the defect, whose limit is Poisson( $v$ ); and the note of Section 43 credited Part II with “integer brackets at first order”, whereas Proposition 18.2 gives them at every fixed order  $J$  (only the explicit inverse of Proposition 18.1 is first order).

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